



A Tunable Bandpass Filter for Superconducting Quantum Computation

Connor Casey

Wang Lab | Department of Physics

11 May 2026





Scope of Talk

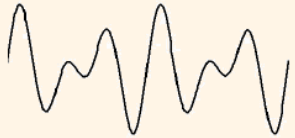
1. Research Motivation & Background
2. Literature Review & Objective
3. Geometries Available
4. Simulating the Proposed Design
5. Design Challenges for Lower Frequencies
6. The Actual Design

I. Research Motivation & Background



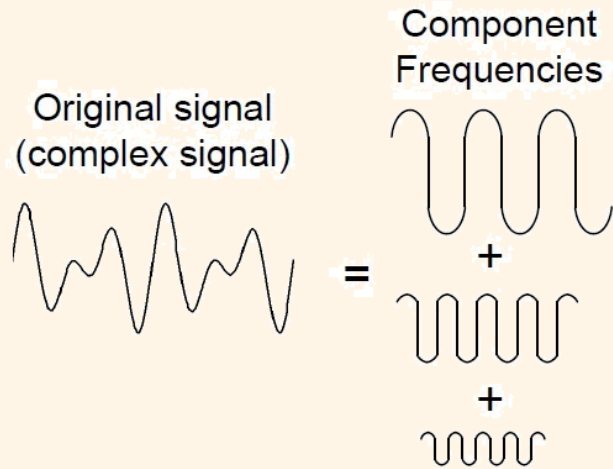
What is a Filter & Why We Need?

Original signal
(complex signal)



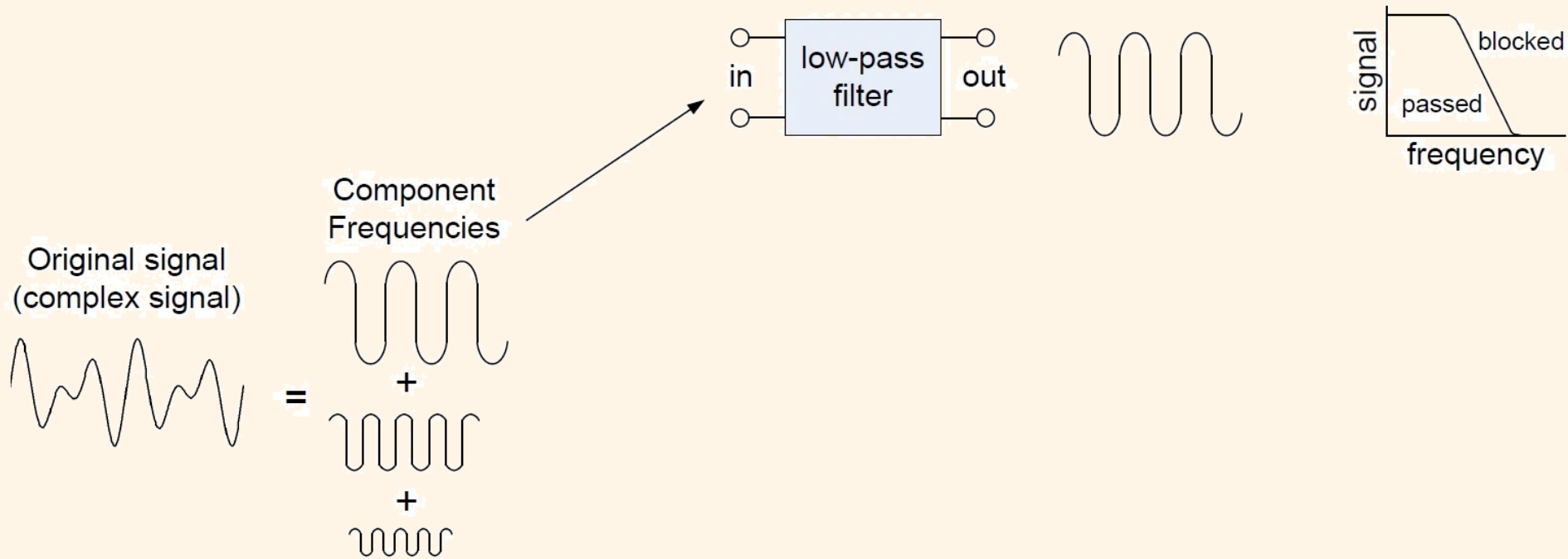


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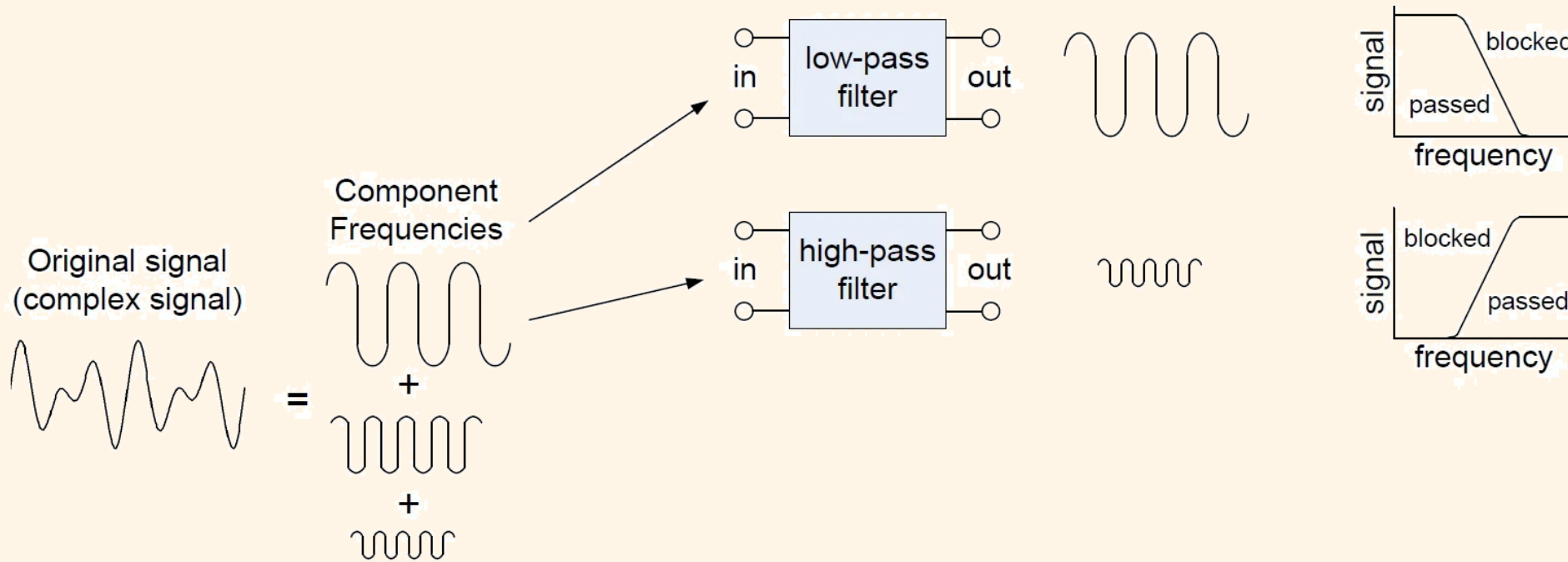


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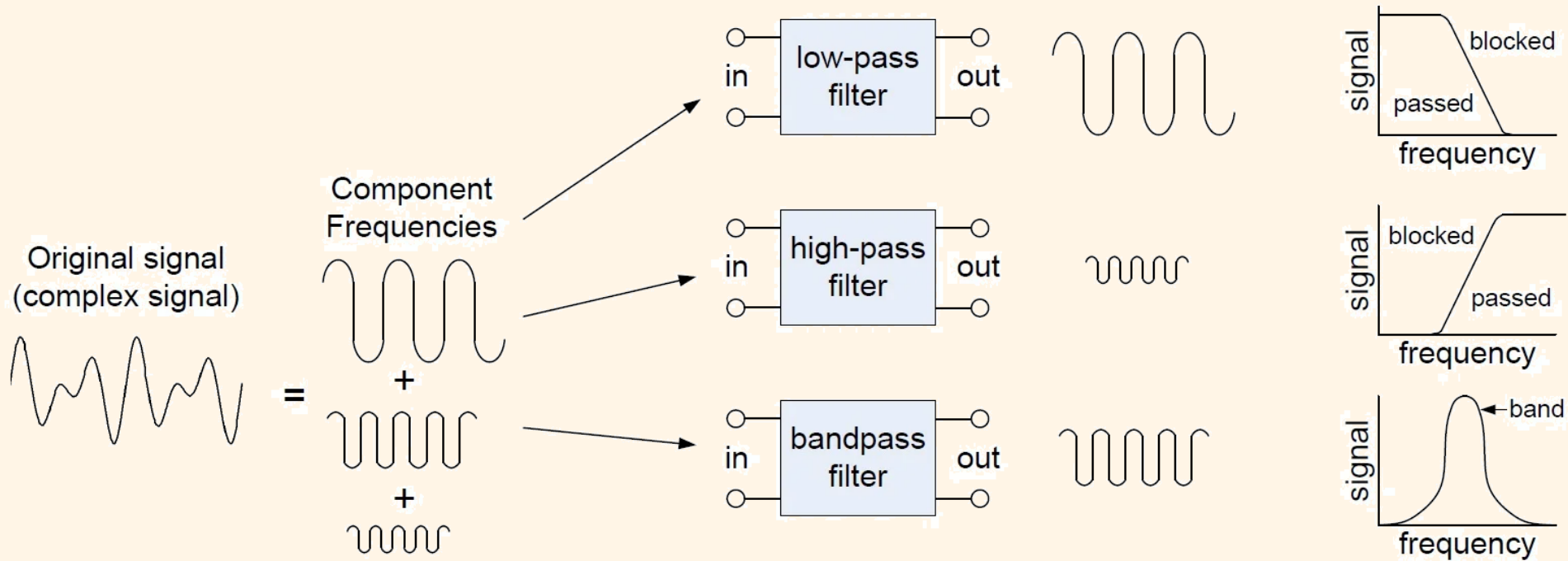


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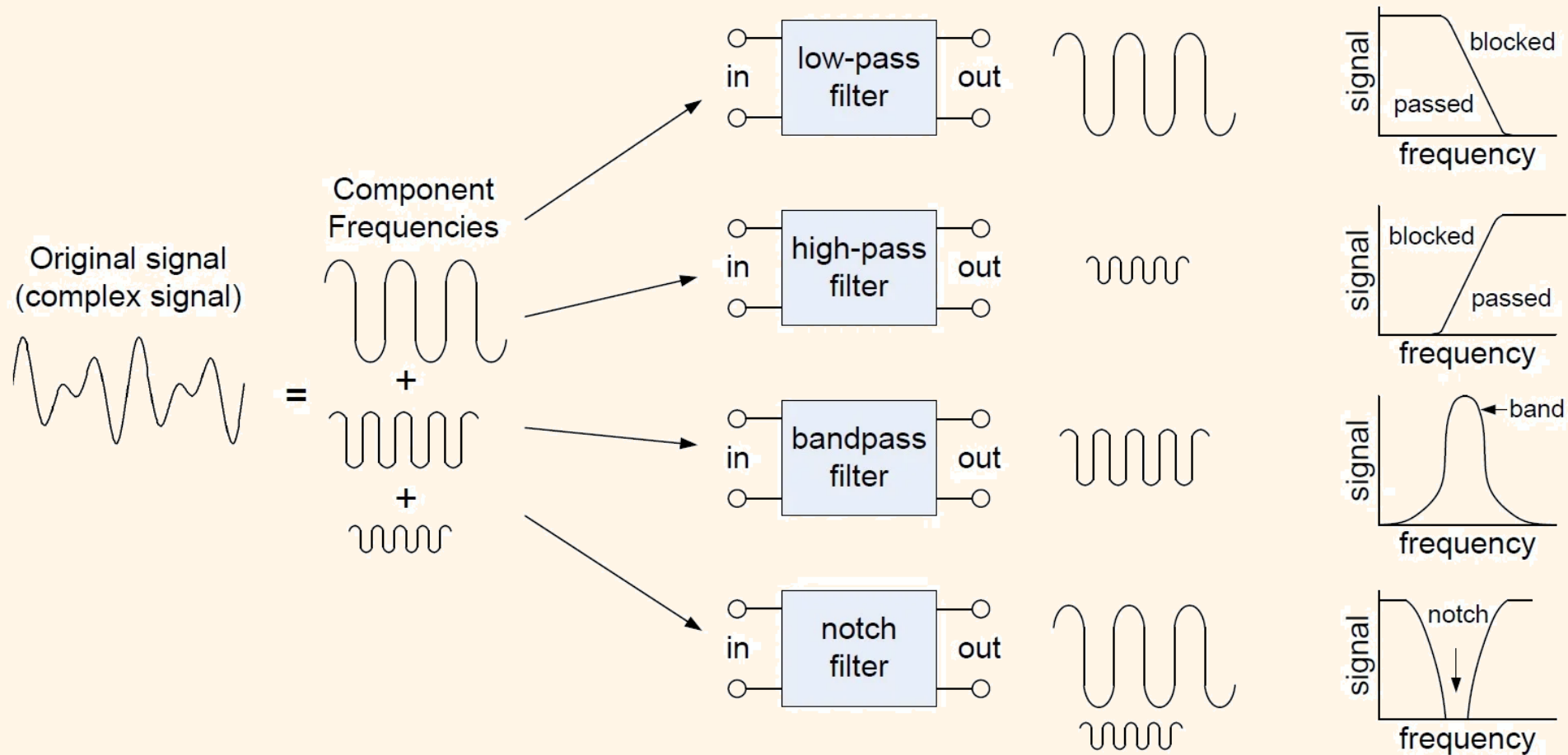


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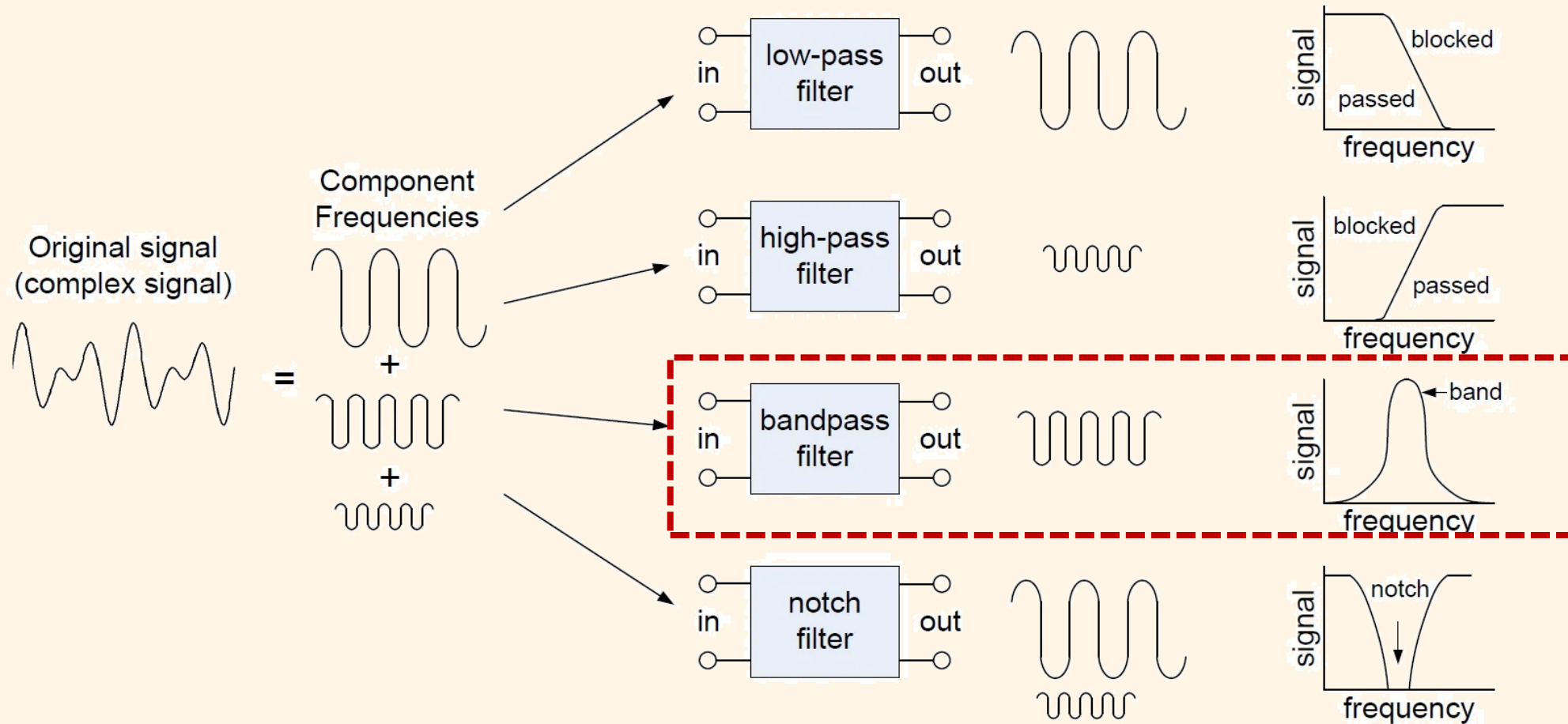


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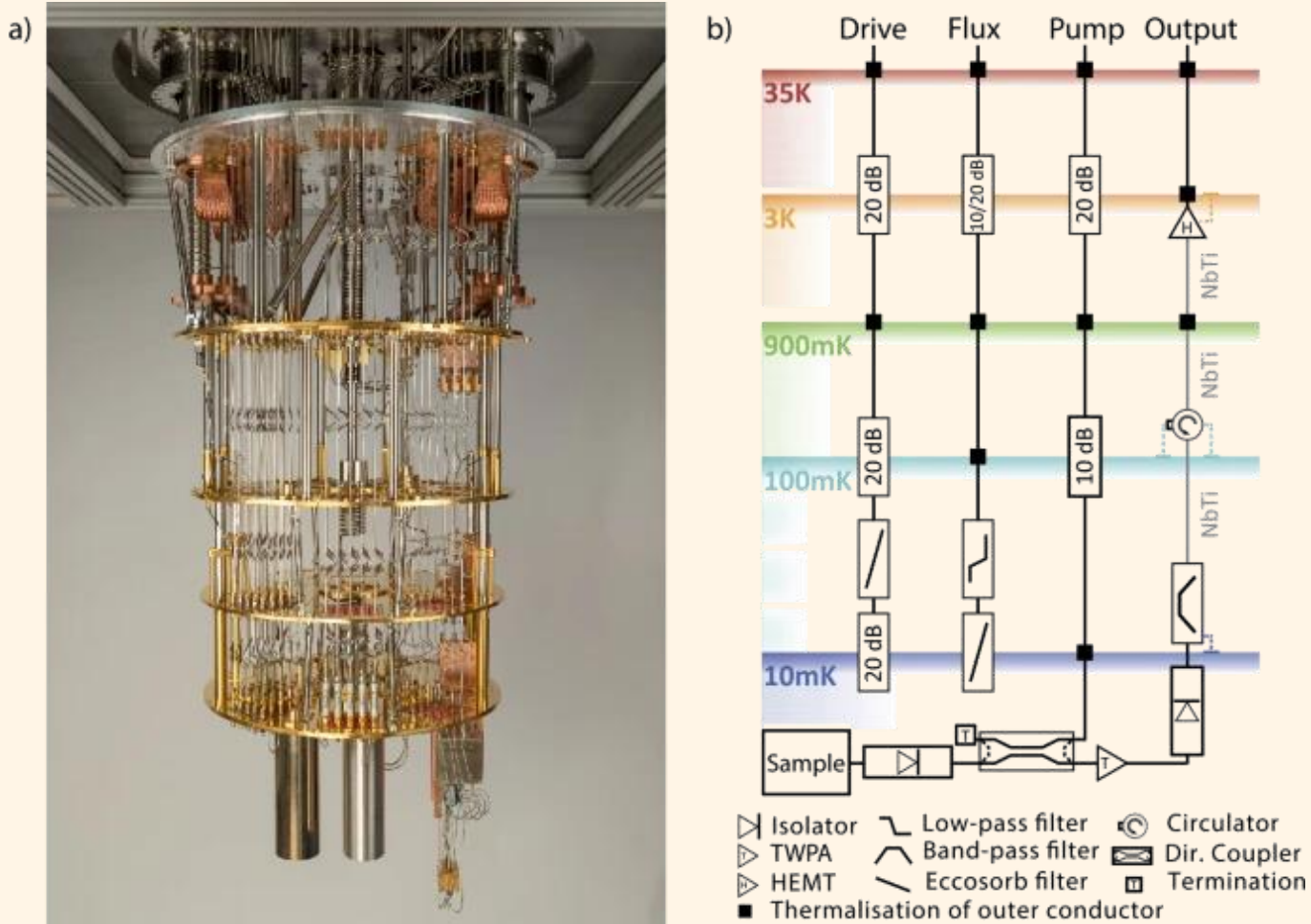


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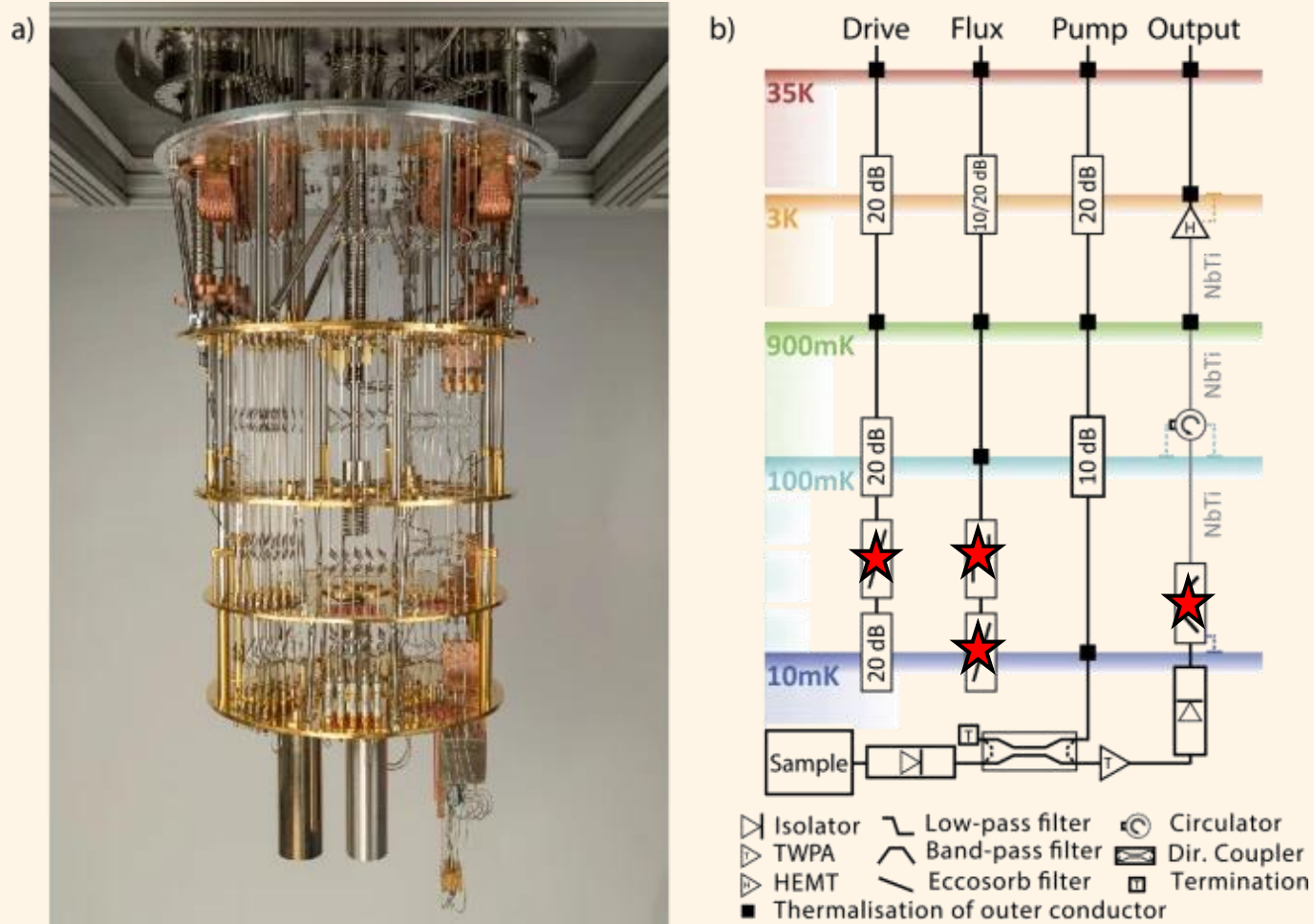
Filters in Superconducting Setups



S. Krinner et al. Engineering Cryogenic Setups for 100-qubit Scale Superconducting Circuit Systems. *EPJ Quantum Technology*. 2019.



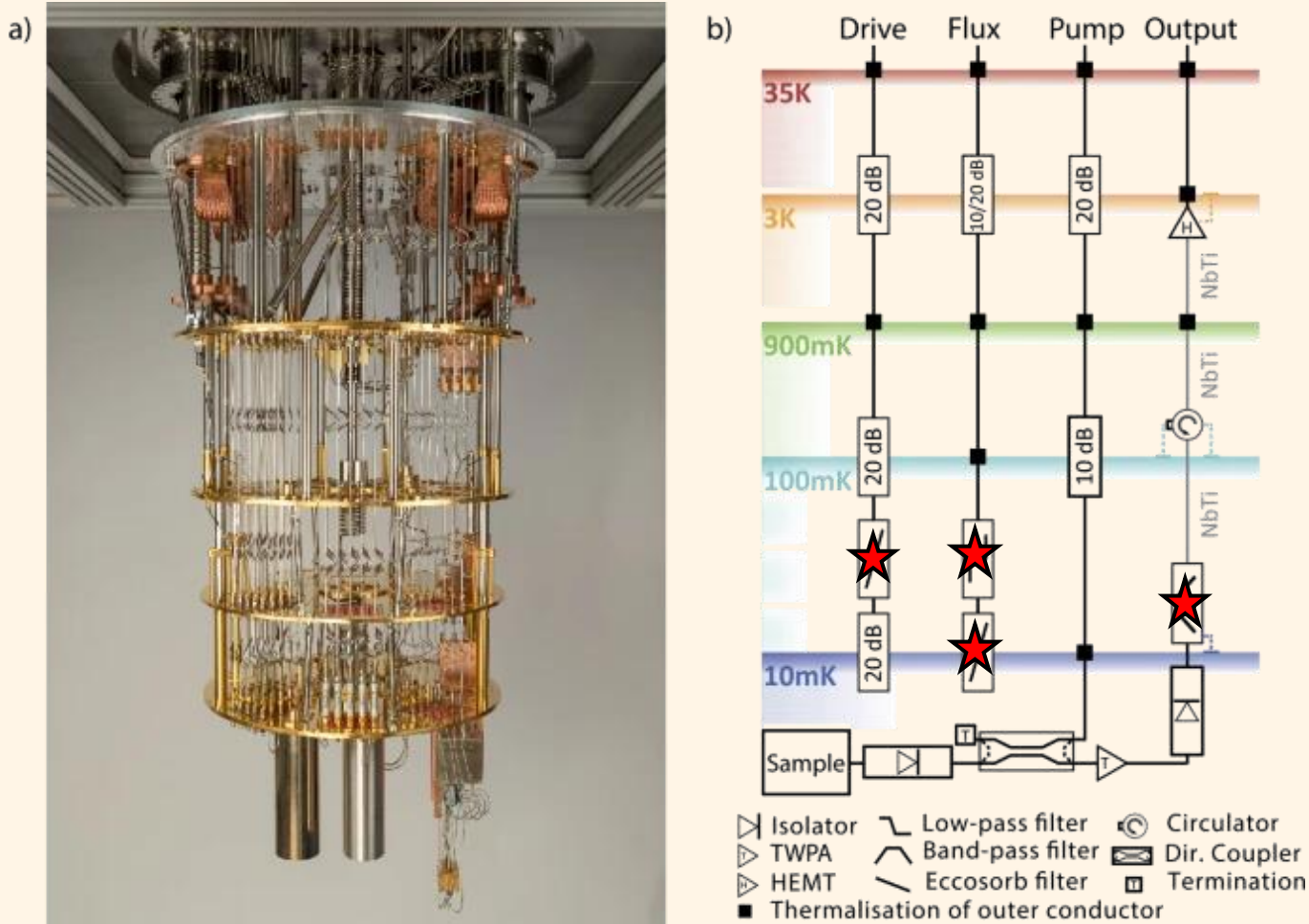
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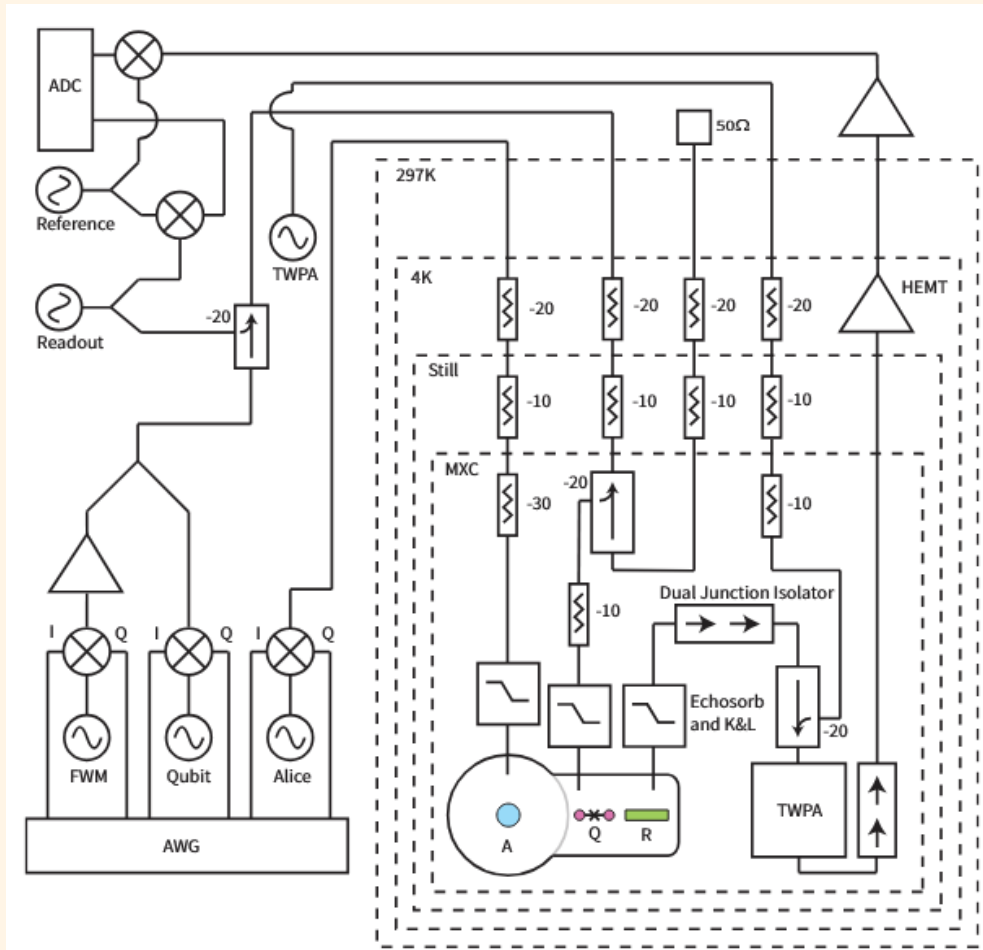
Filters in Superconducting Setups



As the number of qubits continues to increase and/or as experiments require more drive lines (as discussed later in PReSPA), the number of filters will also need to increase accordingly!



Where Does a BP Filter Go?

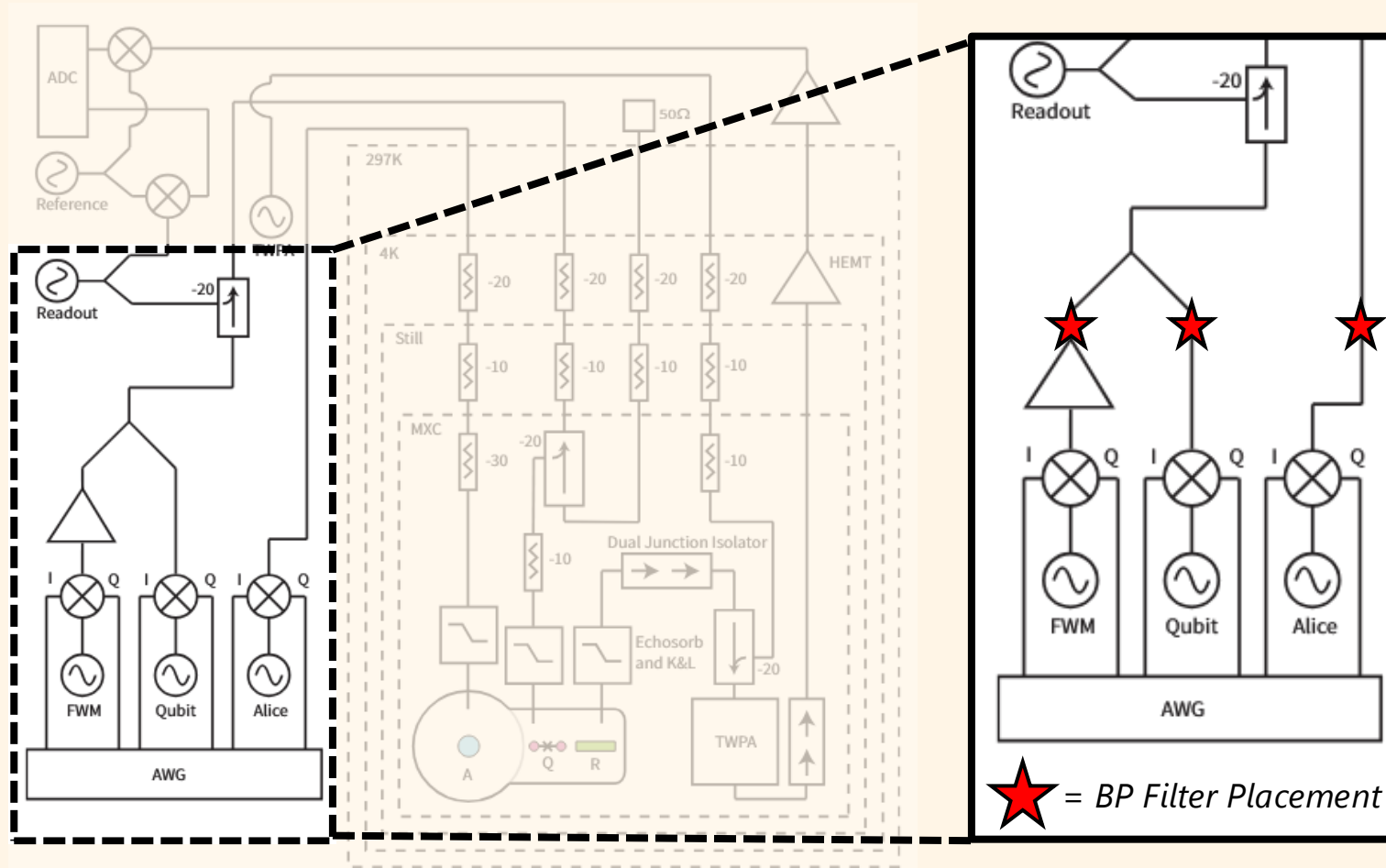


PReSPA Microwave Setup

Jeff Gertler et al. Protecting a Bosonic Qubit with Autonomous Quantum Error Correction. *Nature*. 2021.



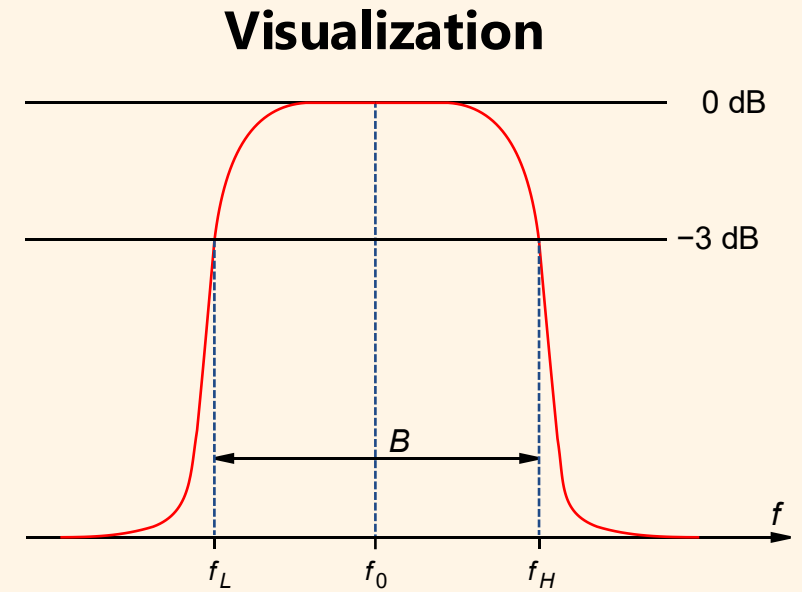
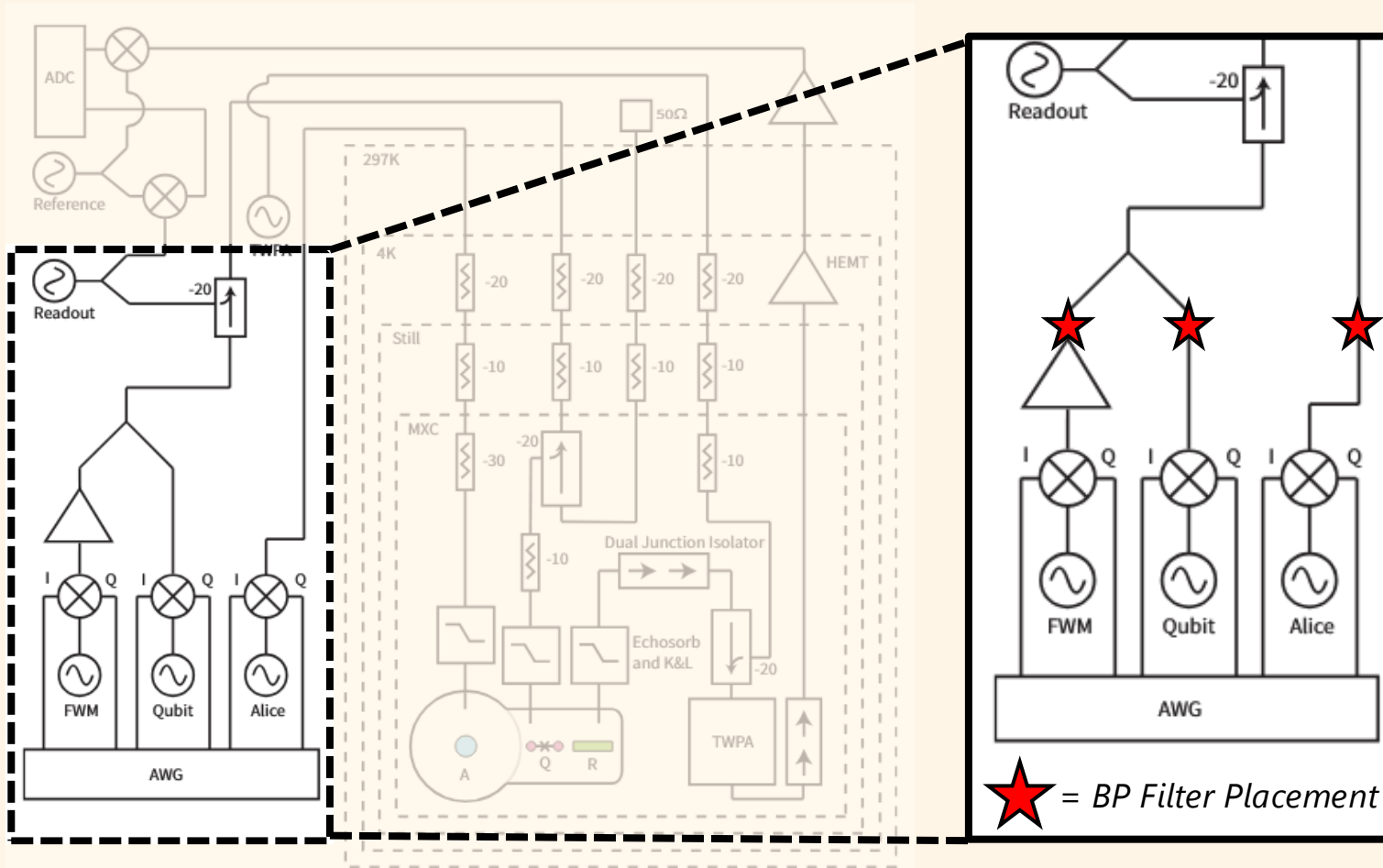
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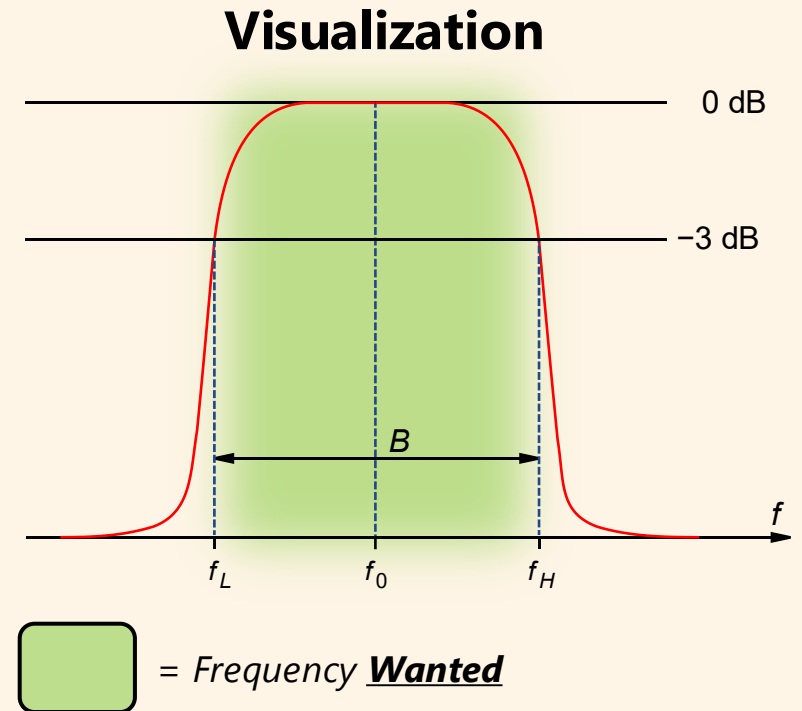
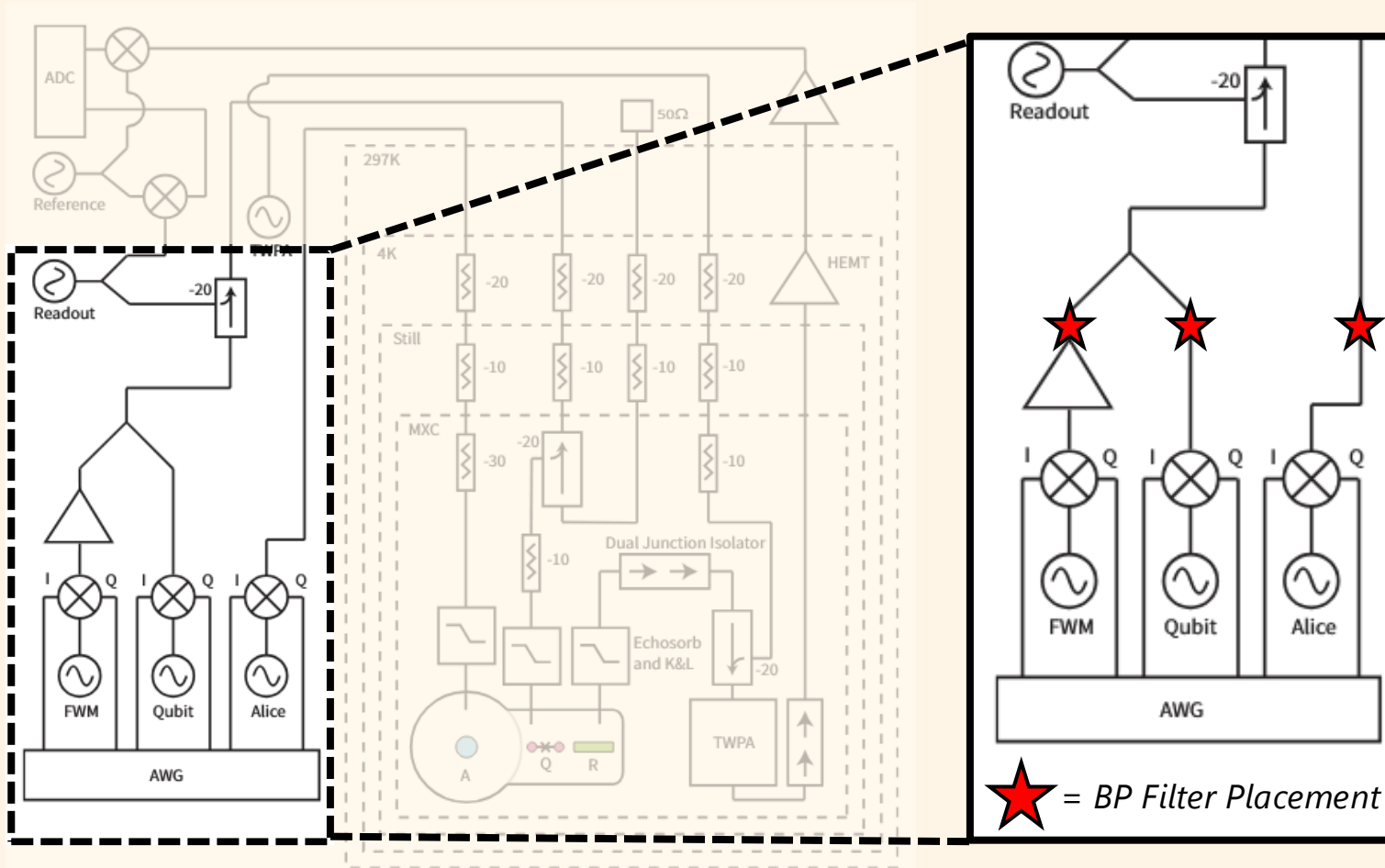


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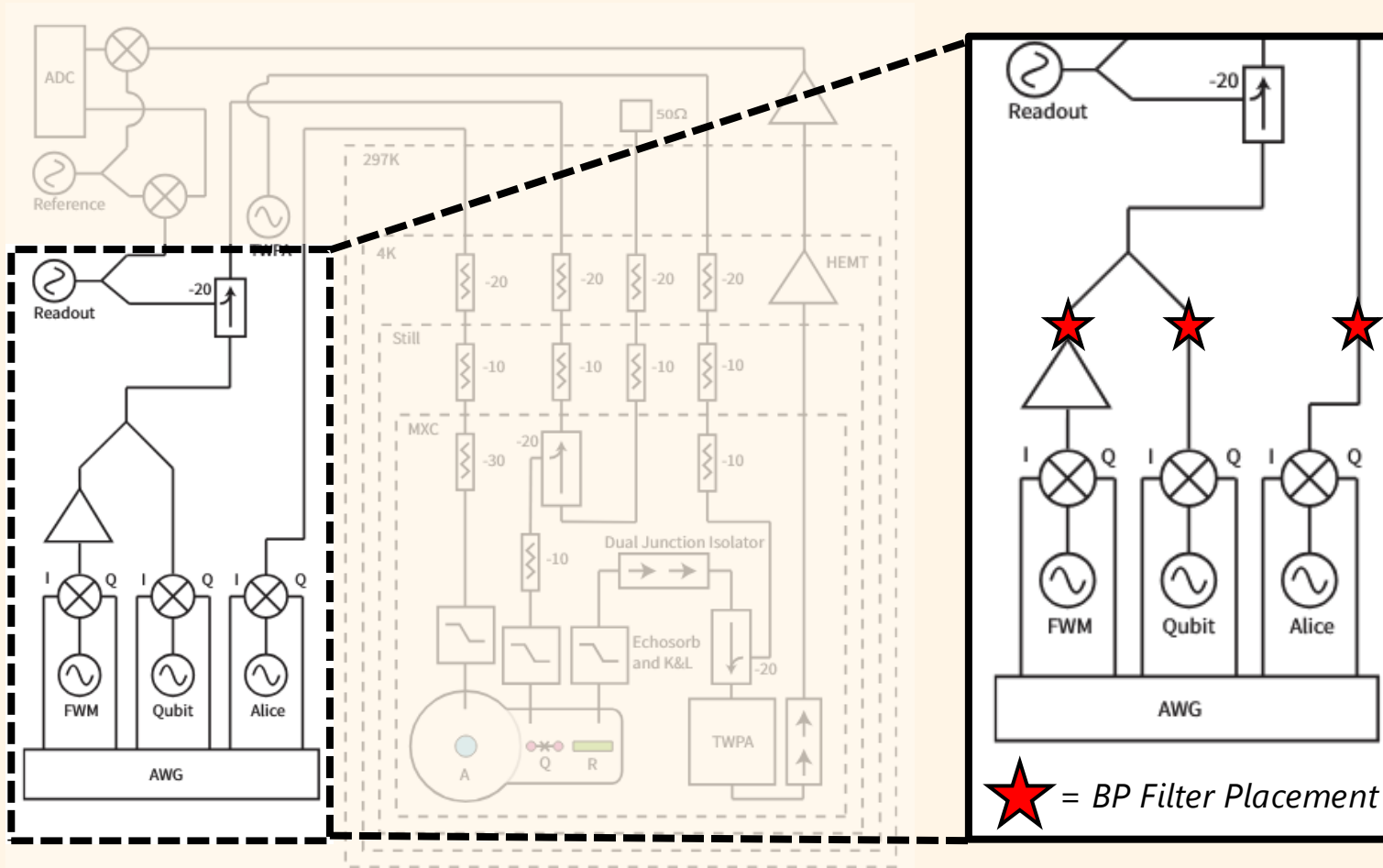


 = Frequency **Wanted**

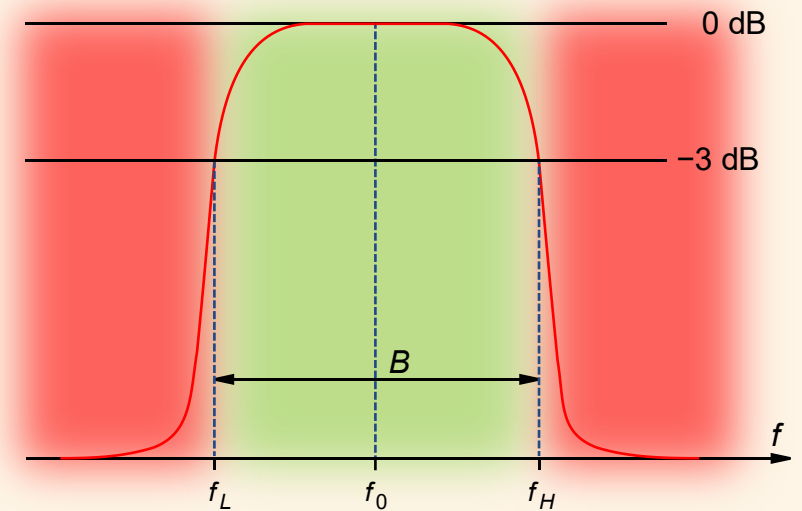
 = BP Filter Placement



Where Does a BP Filter Go?



Visualization



-  = Frequency **Wanted**
-  = Frequency **NOT** Wanted



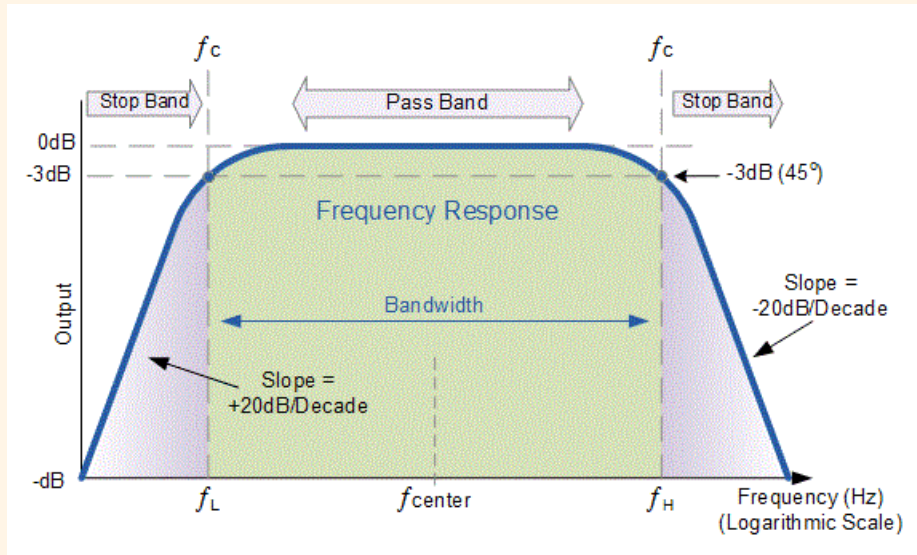
Key Metrics to Define a “Good” Filter

- Some Key Metrics are as Follows:



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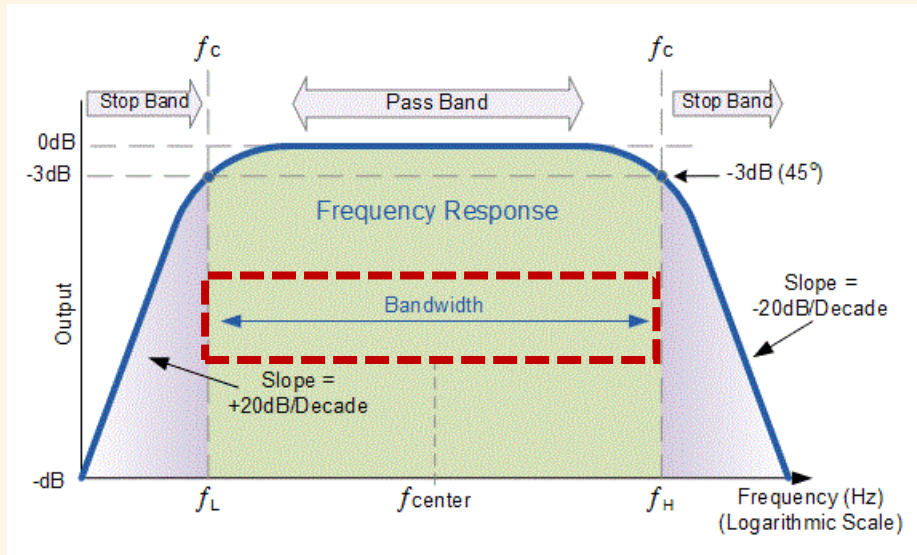
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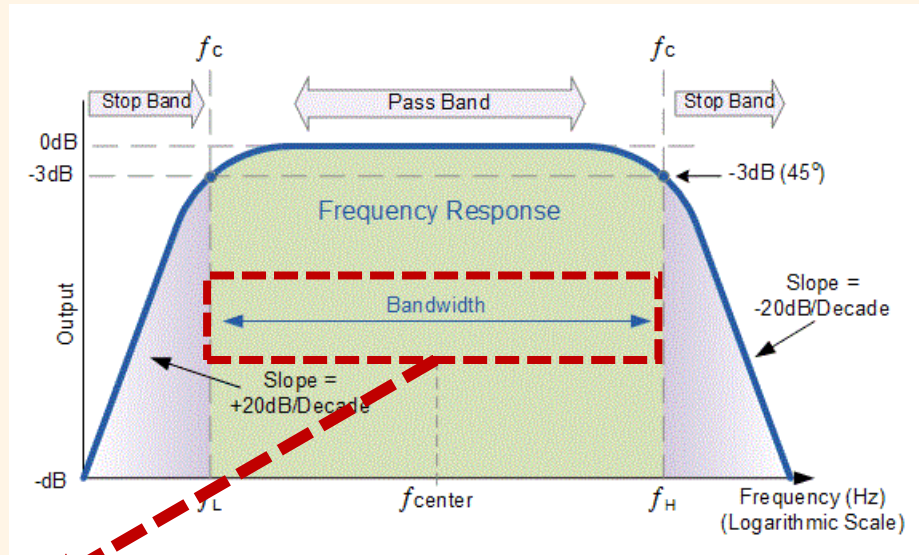
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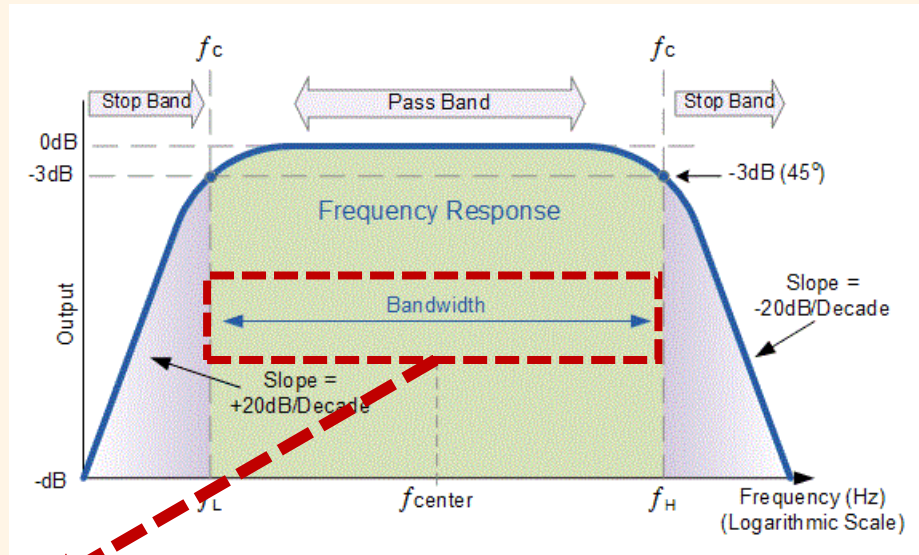


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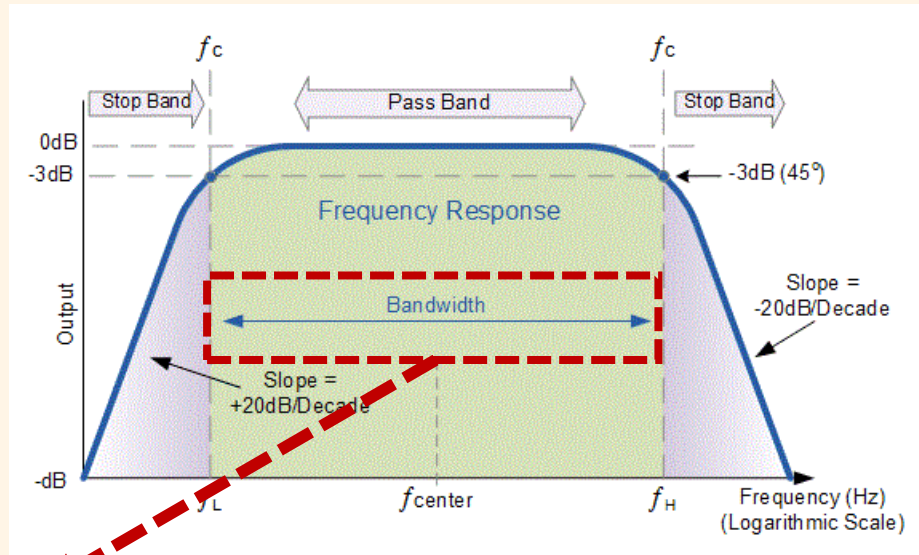
f_0 : Resonance frequency of the cavity mode

Q : Quality Factor



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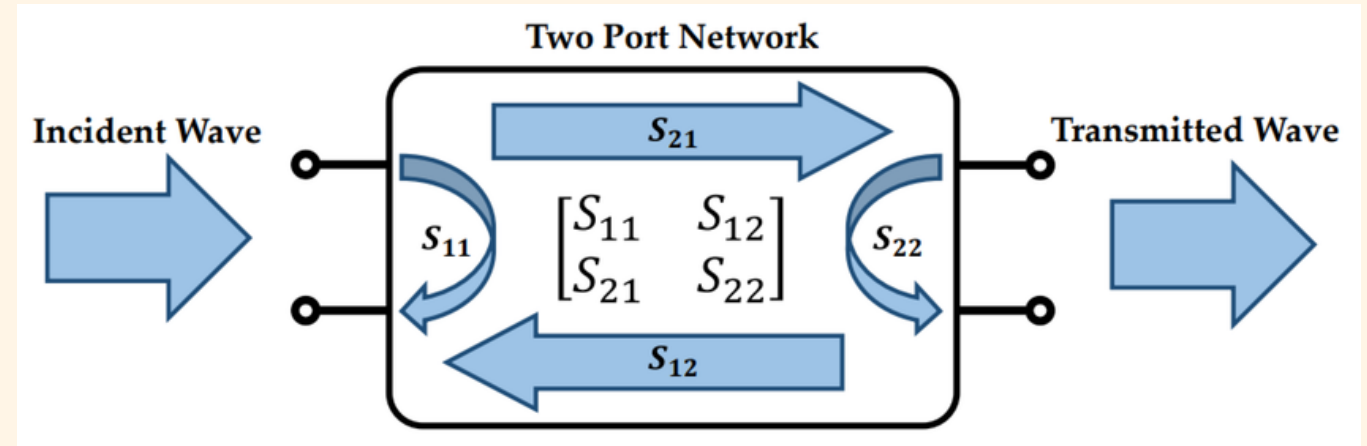
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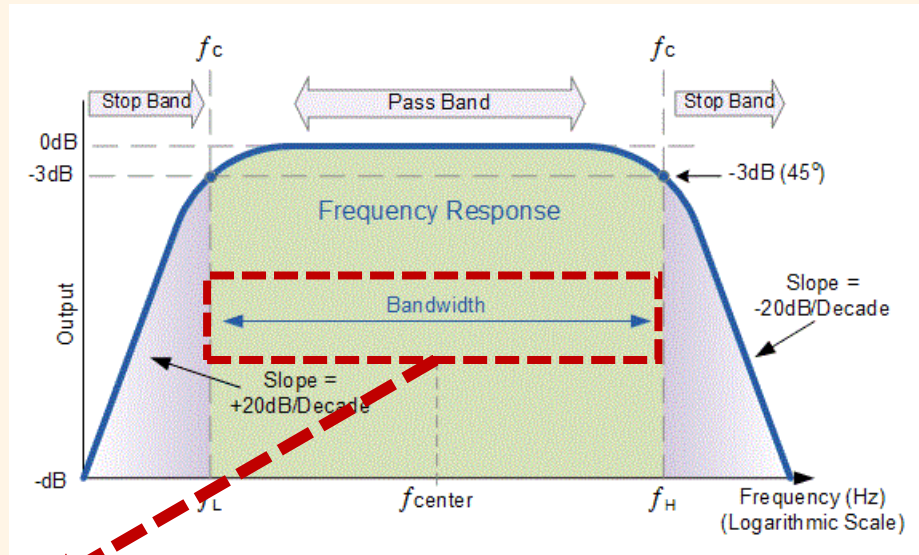
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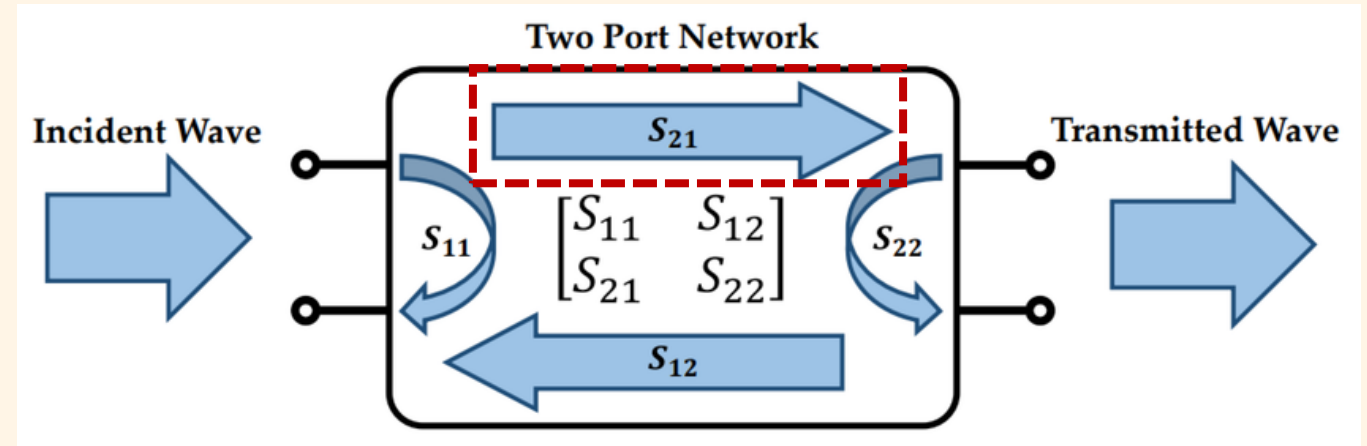
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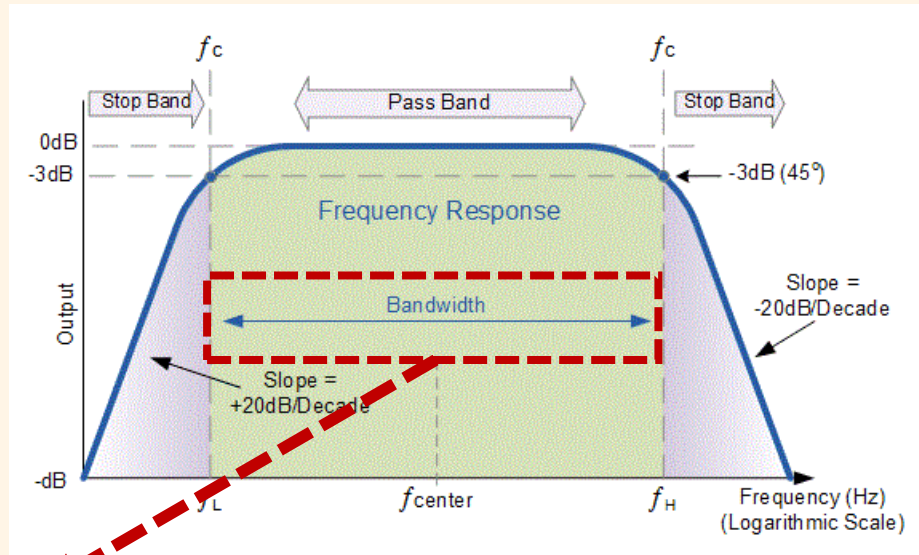
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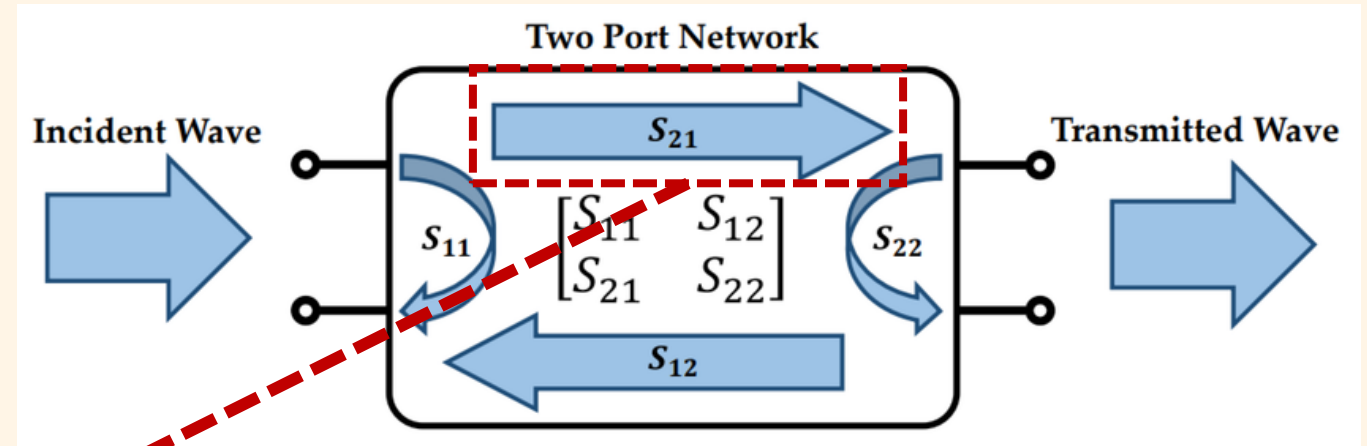
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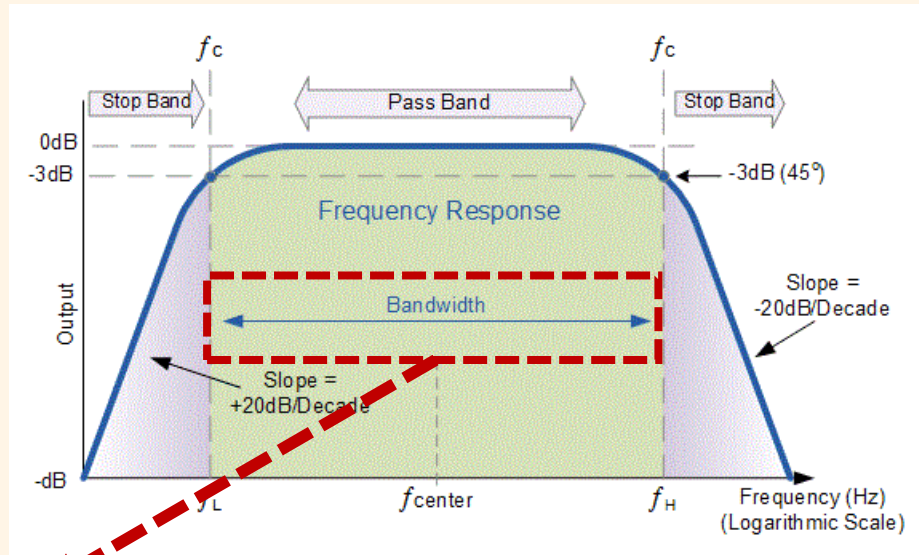


$$S_{21}(\omega) = \frac{\sqrt{\kappa_{\text{ext},1} \kappa_{\text{ext},2}}}{\frac{\kappa}{2} - i(\omega - \omega_0)}$$



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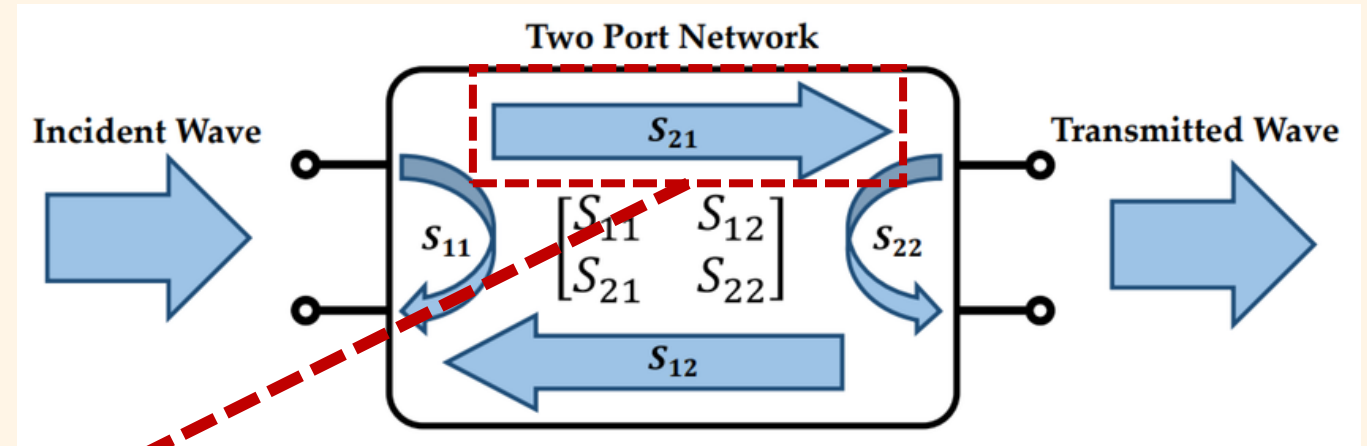
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Same For Us Considering Symmetric Ports

II. Literature Review & Objective



Understanding Tunability for BPFs

- N = Filter Order (number of resonators)



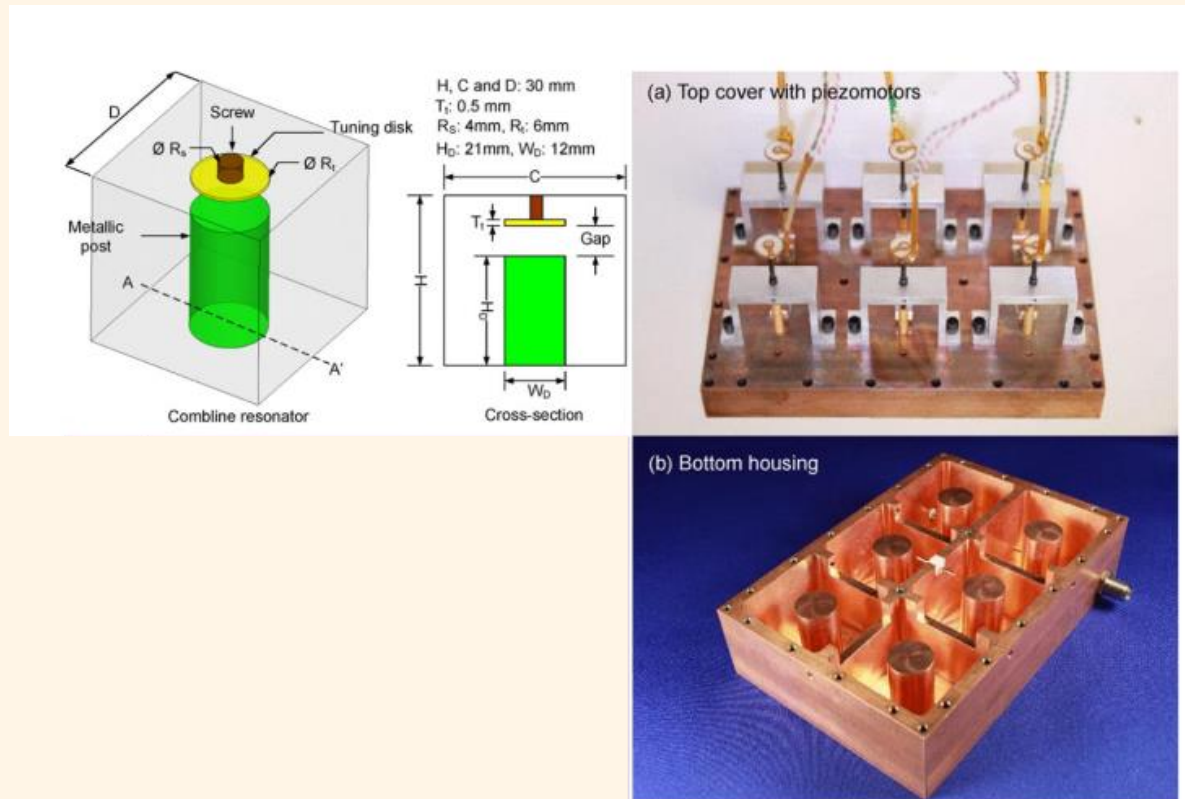
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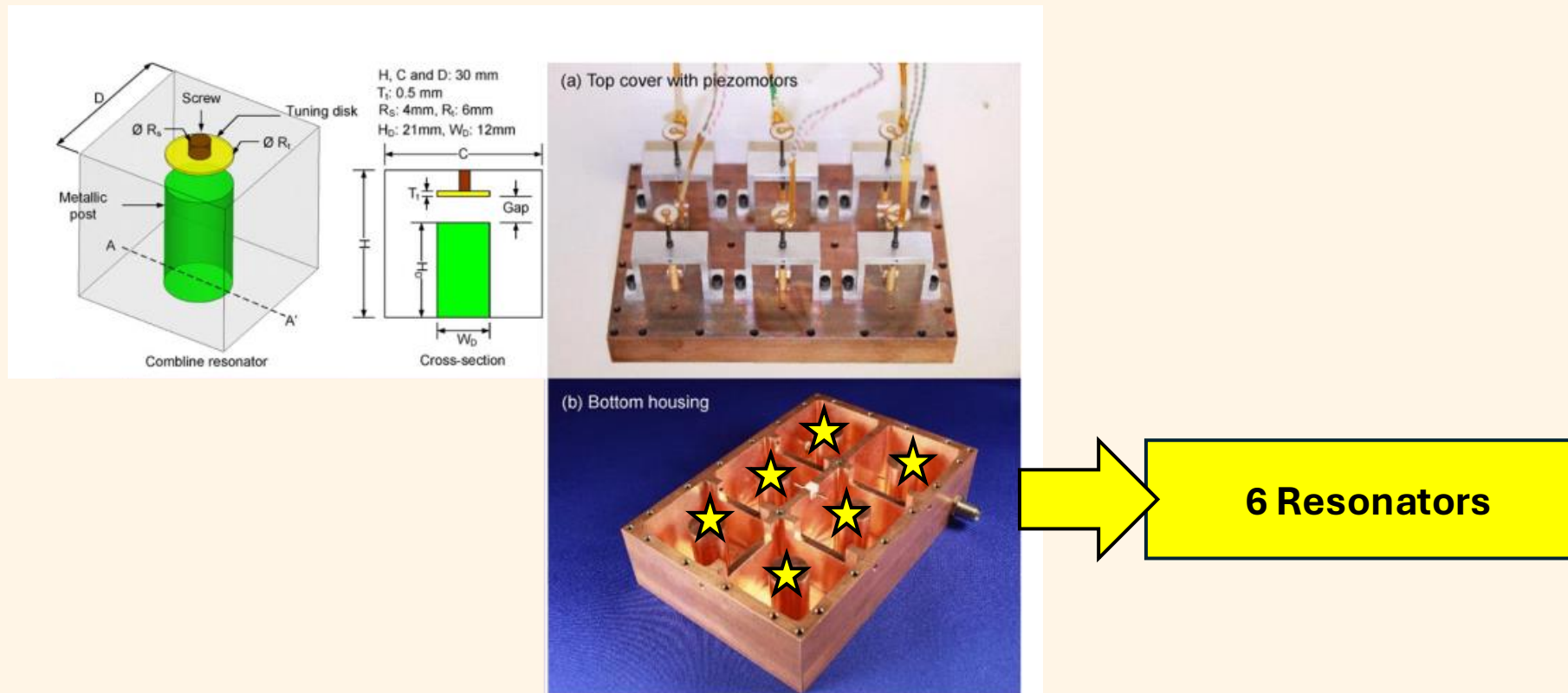


Gowrish Basavarajappa. Tunable Band Pass Filters for Communication Systems. 2021.



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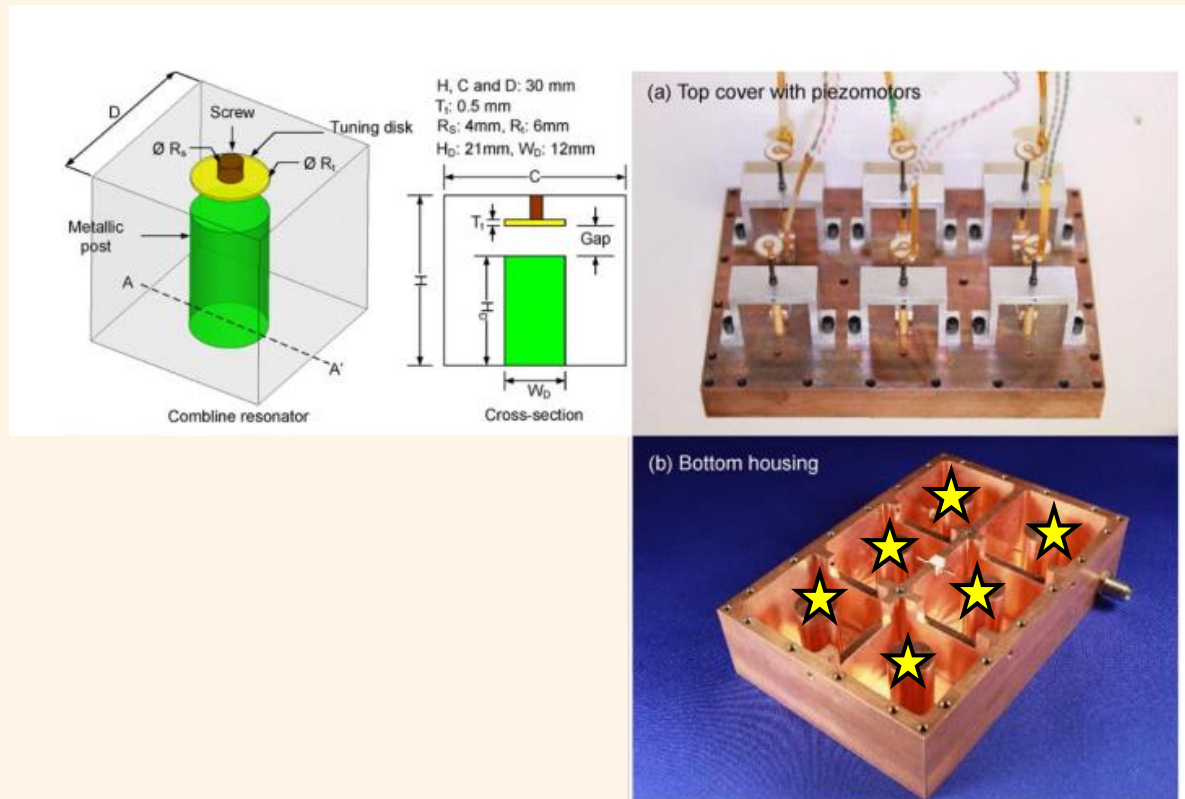
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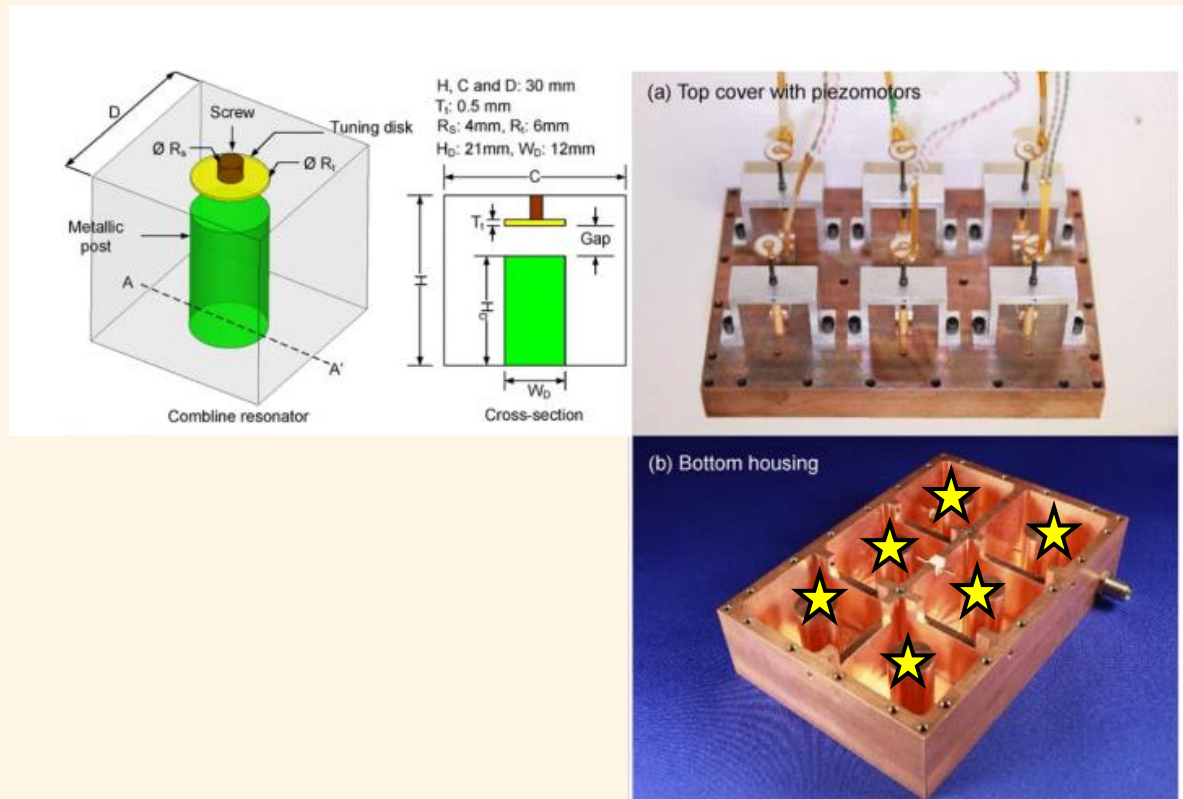


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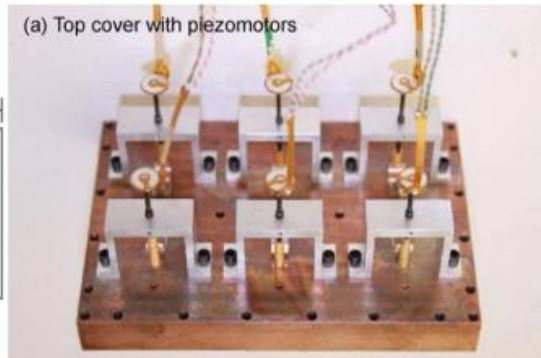
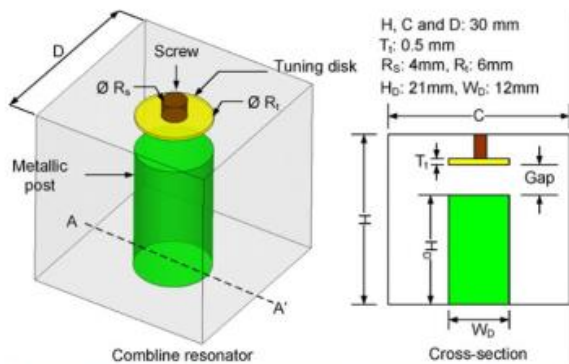


$$6 \text{ Resonators} = (6 * 2) + 1 \text{ Tunable Parts} \\ = 13 \text{ Tunable Parts}$$



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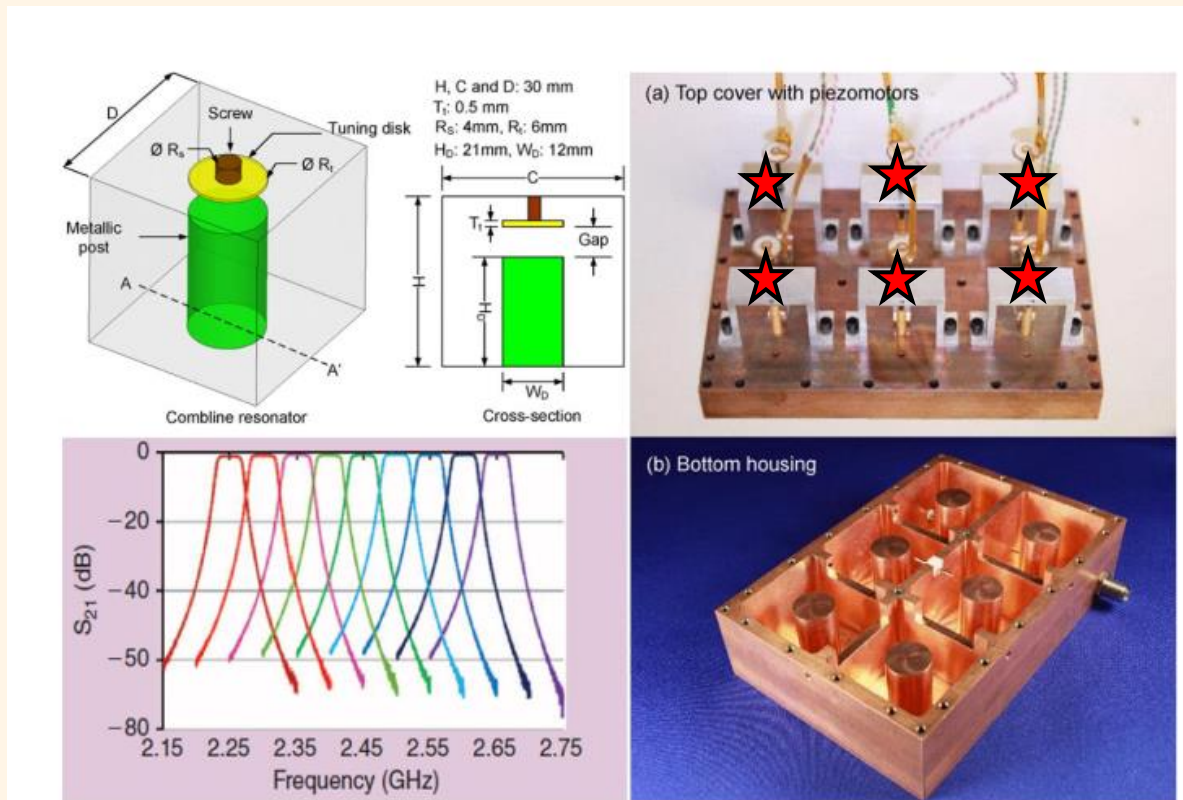
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**2013: Team From Waterloo
Worked to See If They
Could Reduce this
Complexity Problem**



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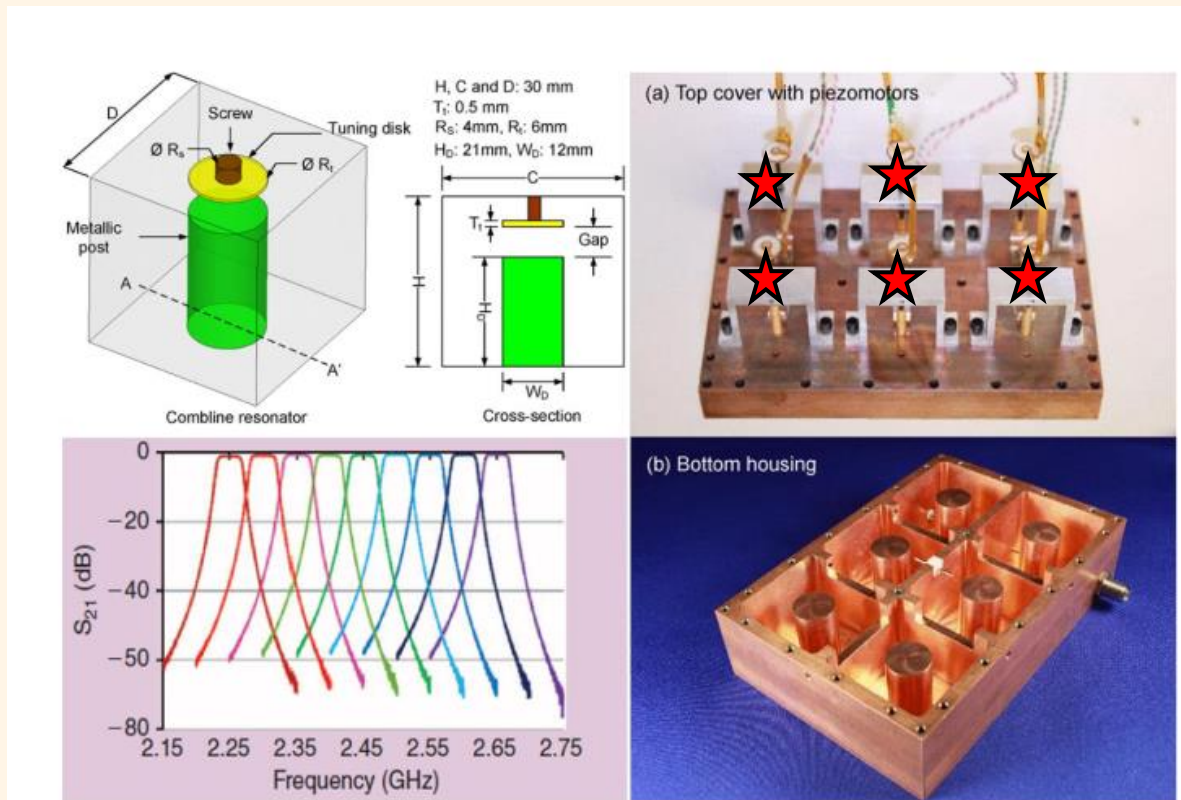


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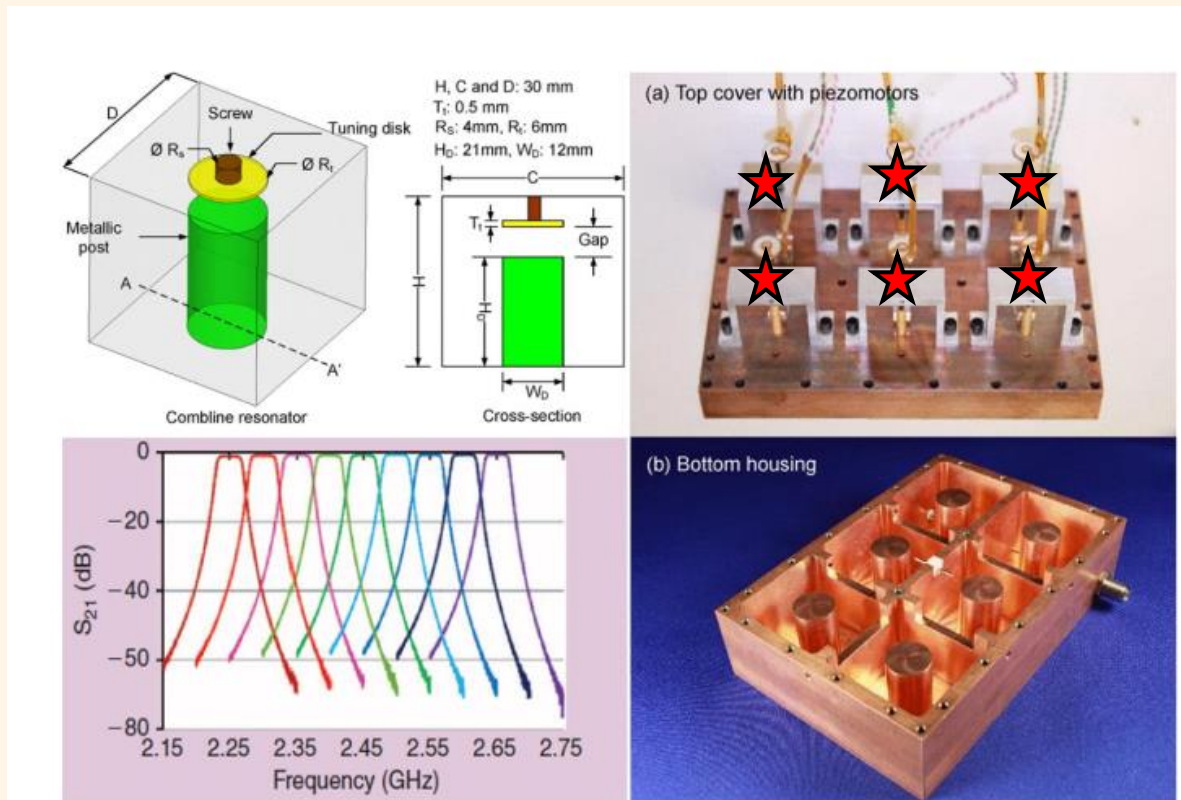


6 Resonators = 6 Tunable Parts

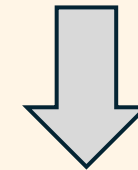


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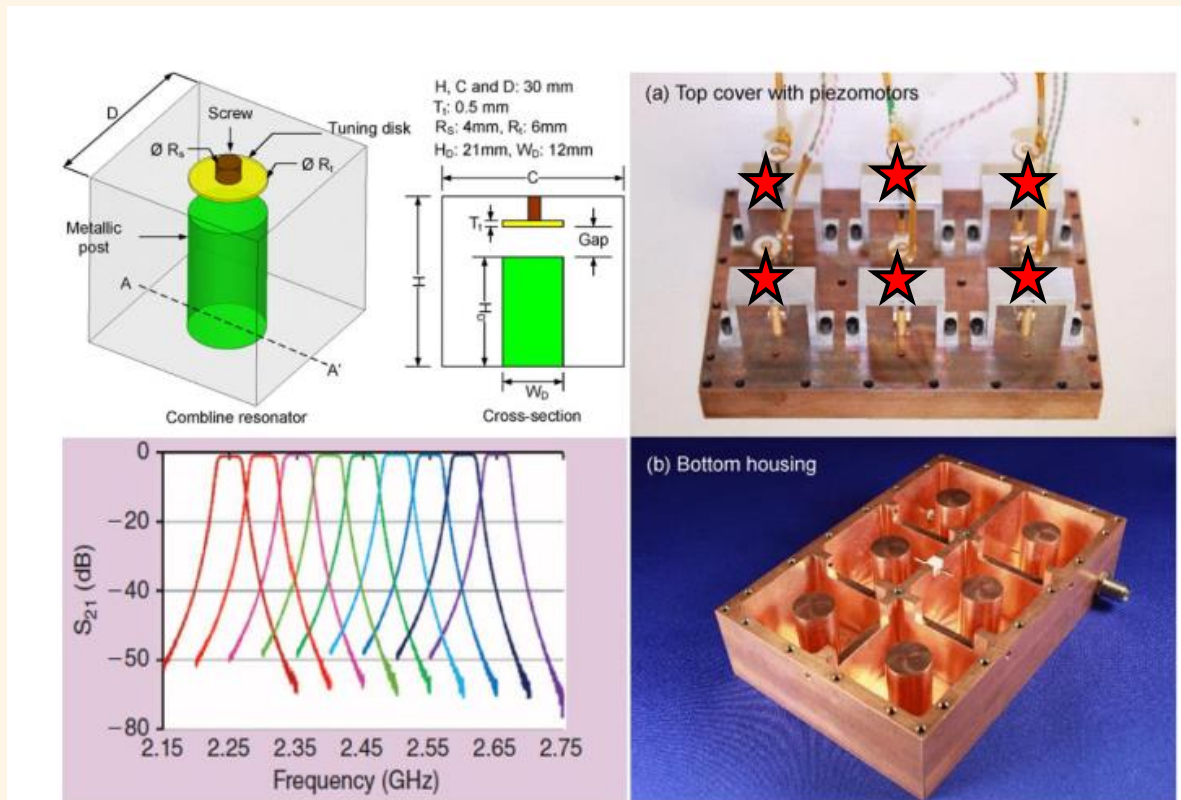


Reduced to N Independent Tuning Elements

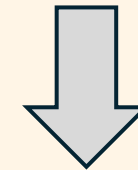


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Reduced to N Independent Tuning Elements

It is Possible!



Prior Work for Tunable Filters

Ref.	Freq.	Q	Tun. Range	BW and Var.	Tun. Elem. and Mech.
[12]	2.5 GHz	> 2250	400 MHz / (16%)	30 MHz / $<\pm 5\%$	N / robust
[13]	3.6 GHz	> 5650	430 MHz / (12%)	75 MHz / $<\pm 5\%$	N / robust
[14]	19.8 GHz	> 6700	200 MHz / (1%)	54 MHz / $<\pm 5\%$	2*N+1 / complex
[15]	19.5 GHz	> 4500	340 MHz / (2%)	180 MHz / $<\pm 5\%$	N / robust
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[18]	4.7 GHz	> 510	160 MHz / (4%)	21 MHz / $<\pm 5\%$	N / robust
[19]	4.8 GHz	> 460	2.8 GHz / (1.8:1)	not constant	N / complex
[20]	4.8 GHz	> 300	1.5 GHz / (1.4:1)	not constant	N / complex
[21]	21.5 GHz	> 1000	725 MHz / (3%)	350 MHz / $<\pm 5\%$	N / robust
[22]	12 GHz	> 9000	200 MHz / (2%)	60 MHz / $<\pm 10\%$	1 / robust
[23]	11 GHz	> 1400	2.2 GHz / (21%)	516 MHz / $<\pm 10\%$	1 / complex
[24]	12 GHz	> 850	700 MHz / (6%)	200 MHz / $<\pm 5\%$	1 / complex
[25]	2 GHz	> 2000	360 MHz / (18%)	25 MHz / $<\pm 5\%$	2*N-1 / complex

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TLDR; It is Incredibly Difficult To Maintain Consistent BW Across a Wide Frequency Range With A Simple Tuning Element

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III. Geometries Available



Types of Cavity Shapes: Rectangular

- ***Resonance Frequency of Bare Rectangular Cavity:***

$$f_{mnp} = \frac{c}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{d}\right)^2}$$



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a : x -width, b : y -height, d : z -length

m : x -mode index, n : y -mode index, p : z -mode index

c : speed of light in medium



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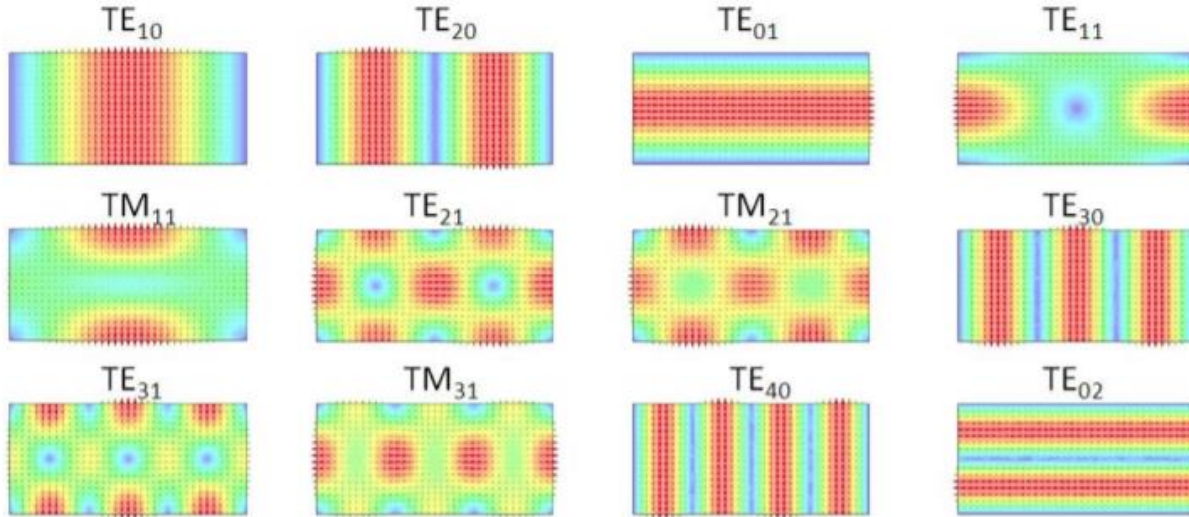


Fig. 5: The first 12 modes in a rectangular waveguide. The arrows of the electric fields are sketched



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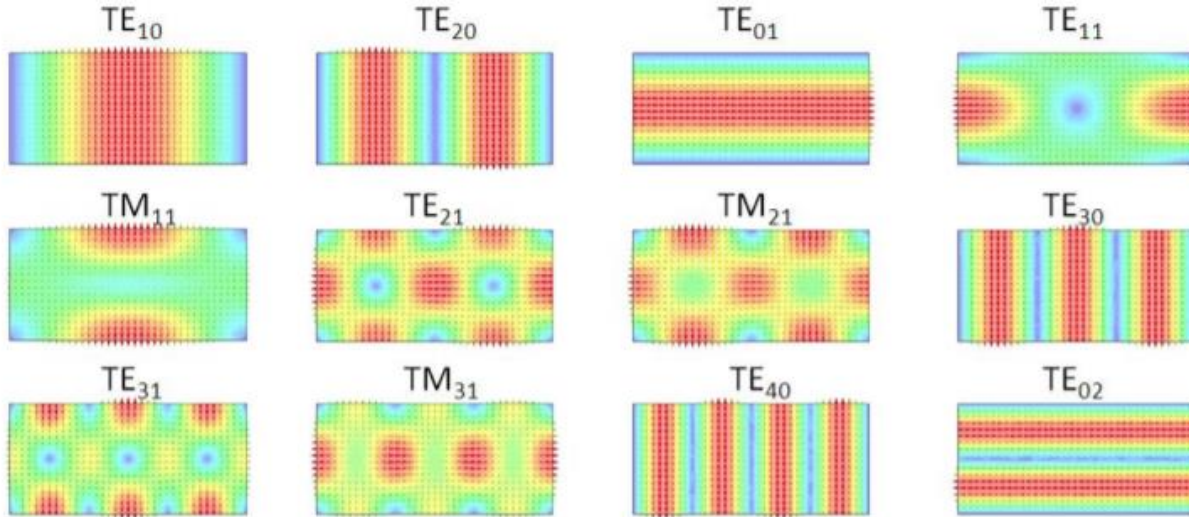
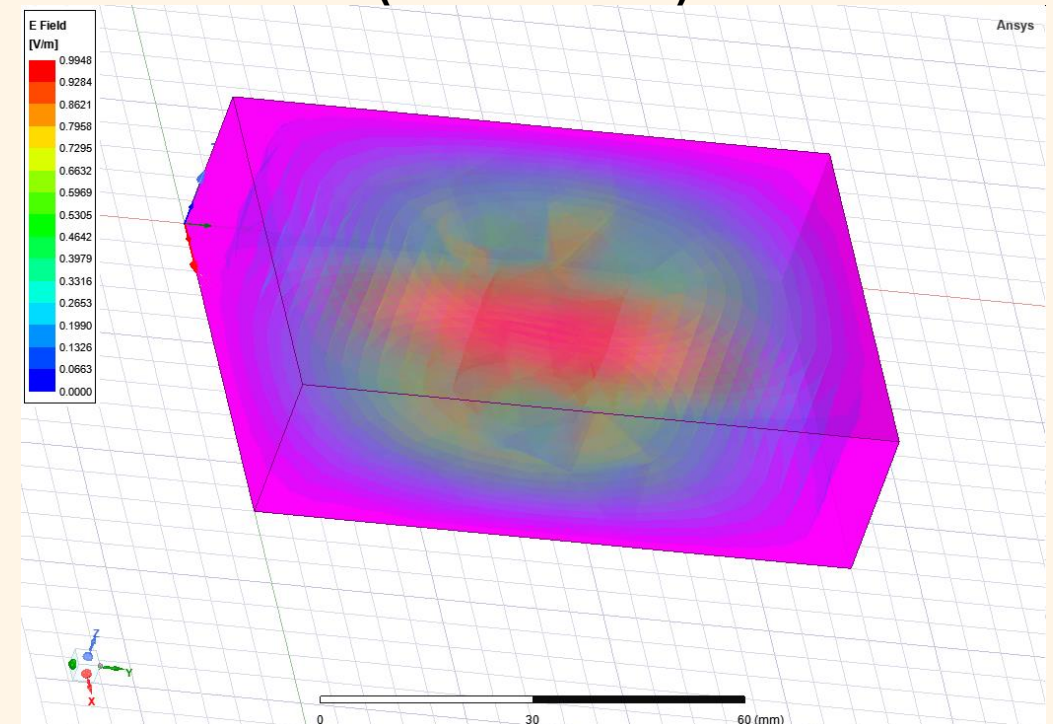


Fig. 5: The first 12 modes in a rectangular waveguide. The arrows of the electric fields are sketched

(Electric Field)





Types of Cavity Shapes: Cylindrical

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m : azimuthal index, n : radial index, p : axial index

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X'_{mn} : n th root of $J'_m = 0$ (TE modes)

X_{mn} : n th root of $J_m = 0$ (TM modes)



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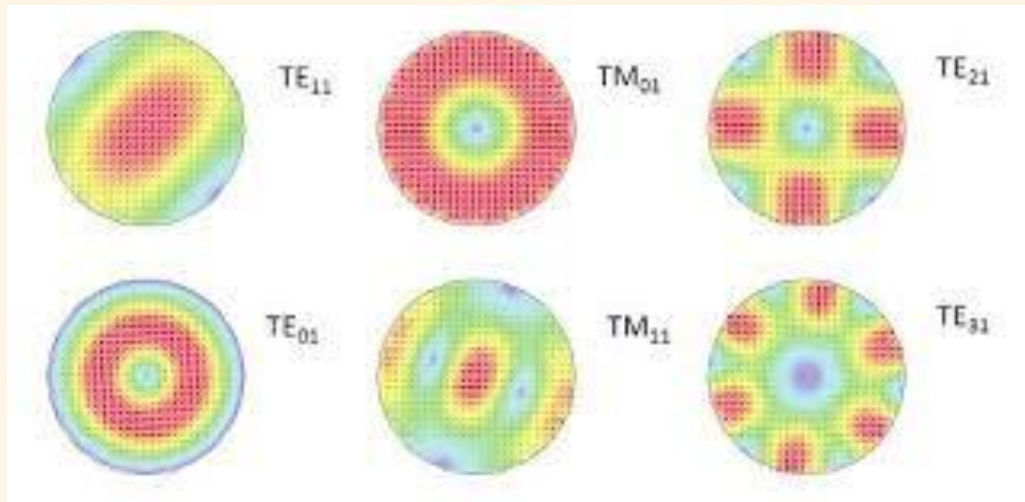
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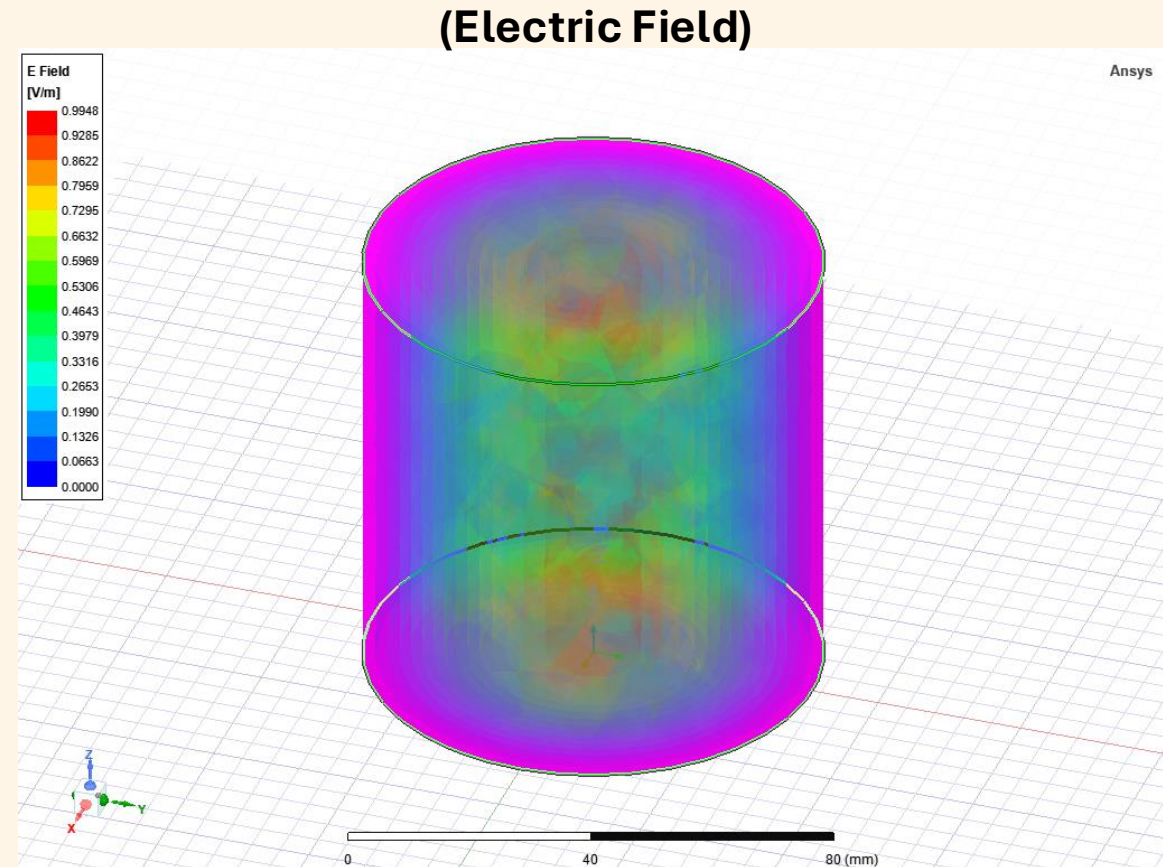
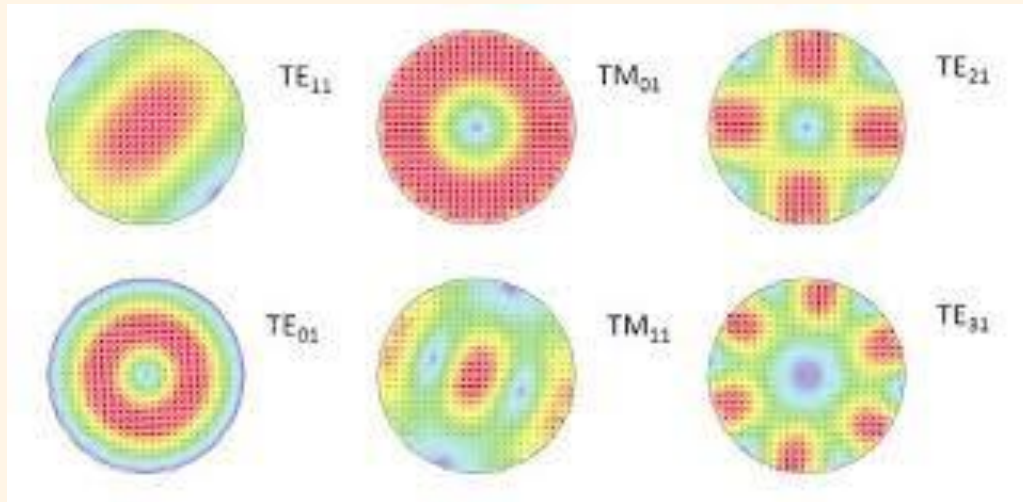
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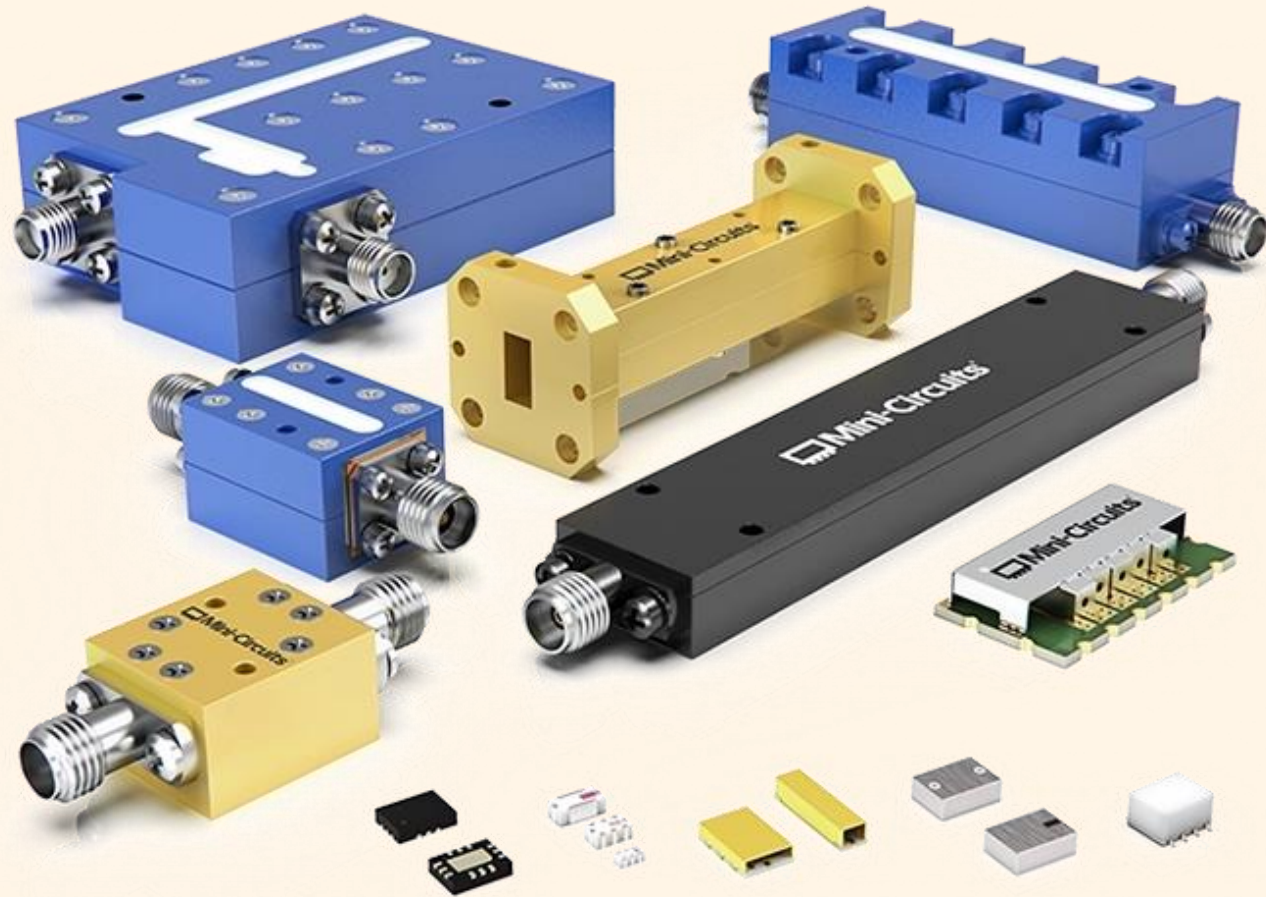
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IV. Simulating the Proposed Design



The Goal

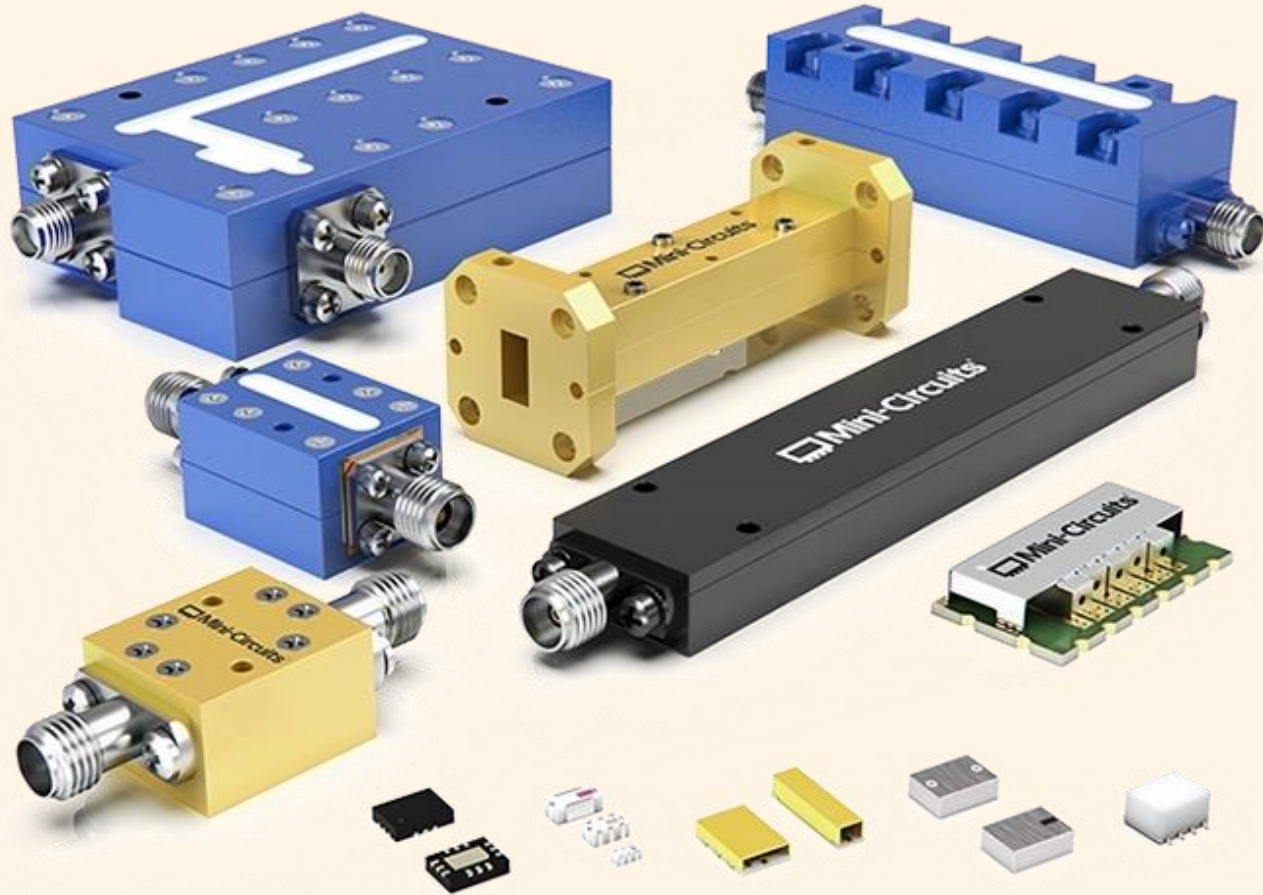


Large Bandwidth (Low Q)

$$\kappa \approx 100 \text{ MHz}$$



The Goal



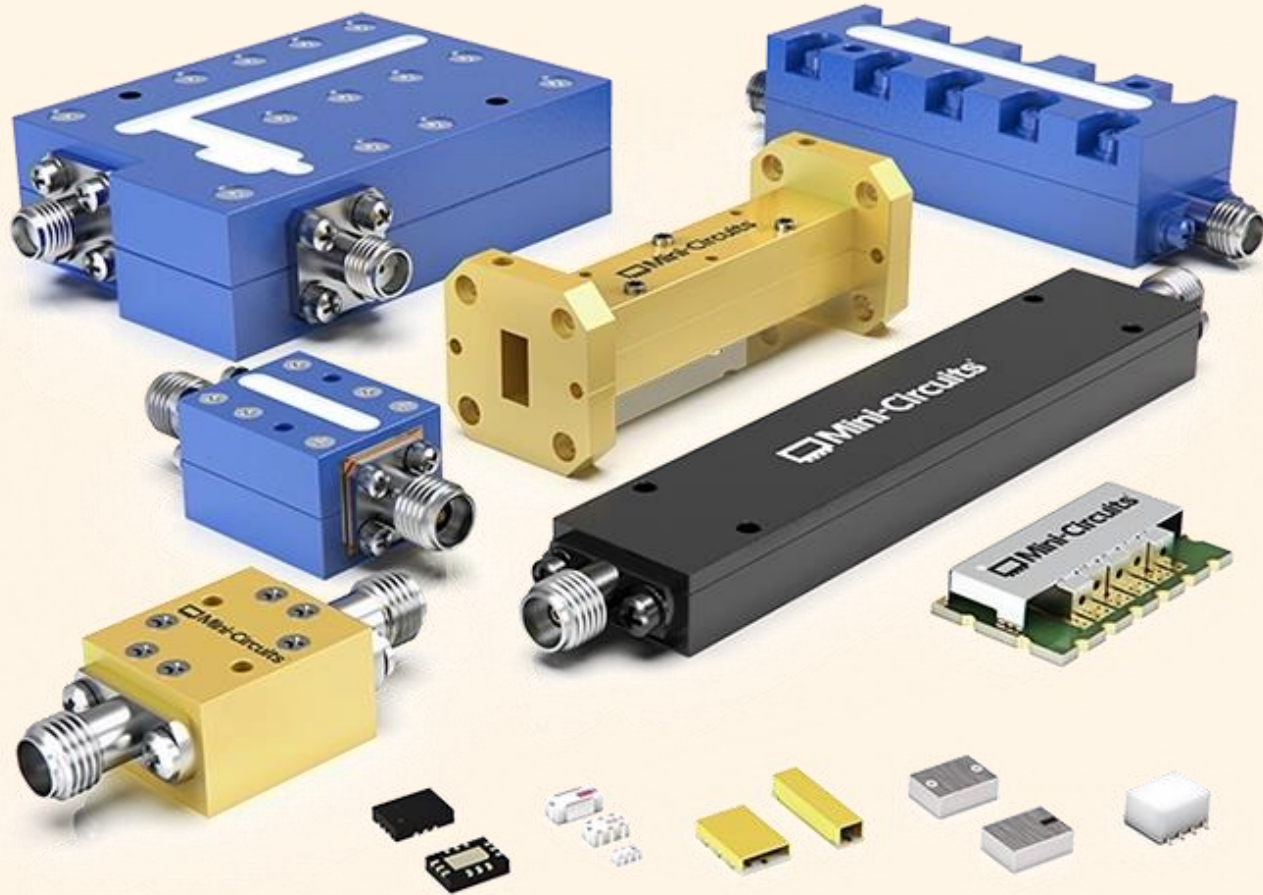
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$$\kappa \approx 100 \text{ MHz}$$

Tunable Across $\pm 200 \text{ MHz}$



The Goal



Large Bandwidth (Low Q)

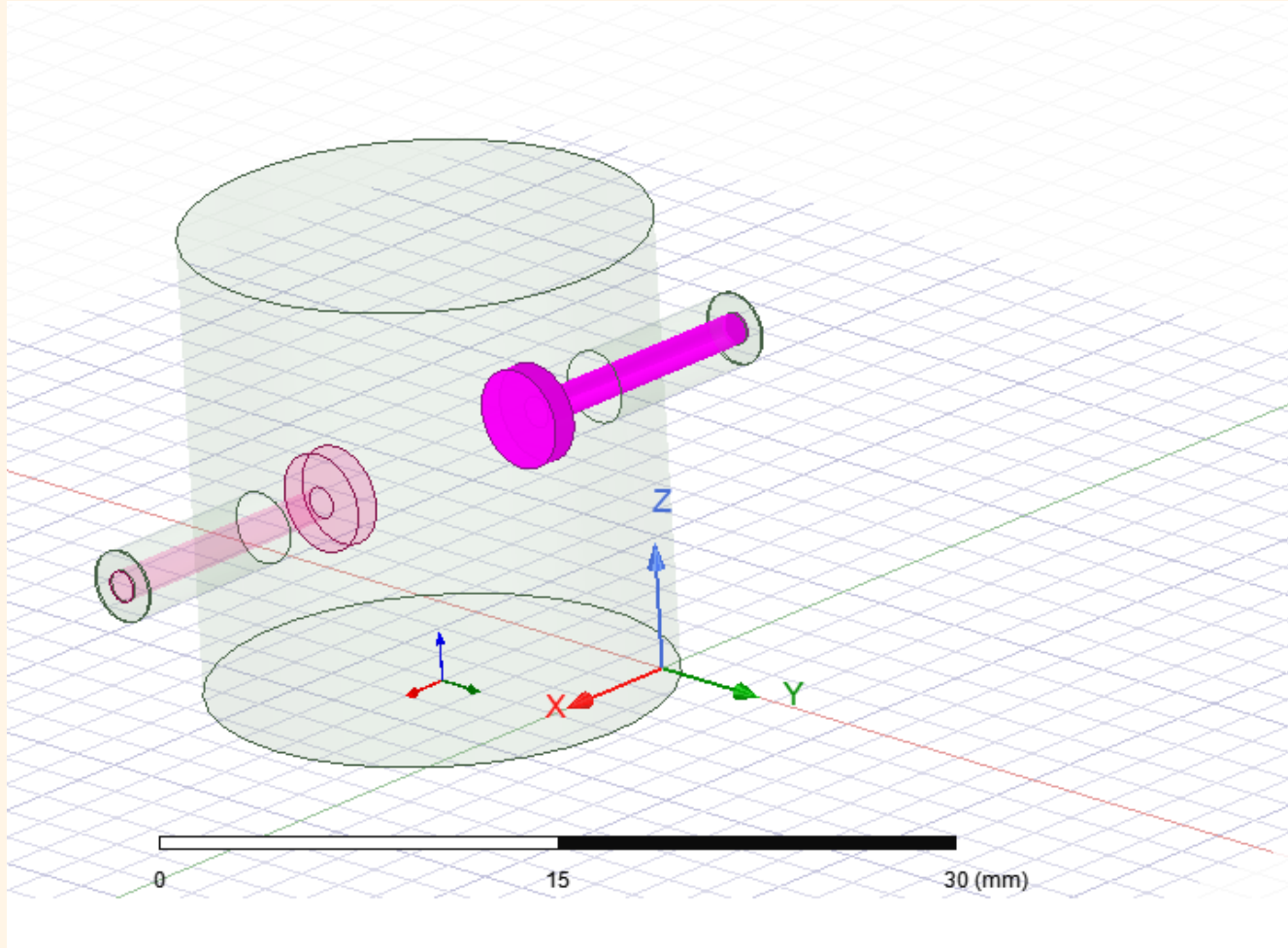
$$\kappa \approx 100 \text{ MHz}$$

Tunable Across $\pm 200 \text{ MHz}$

With Only One Tuning Element

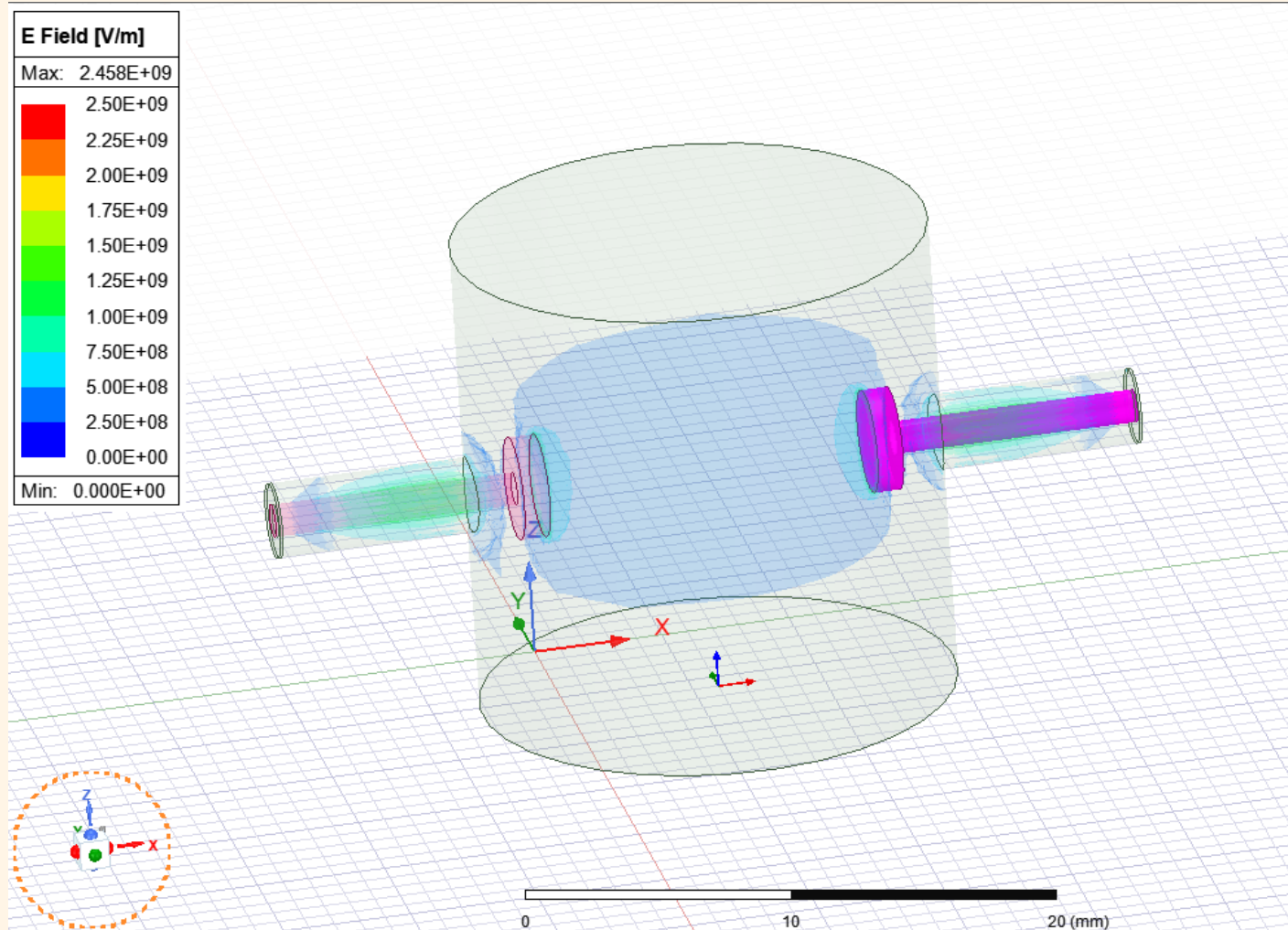


Step 1: Starting with Cylindrical



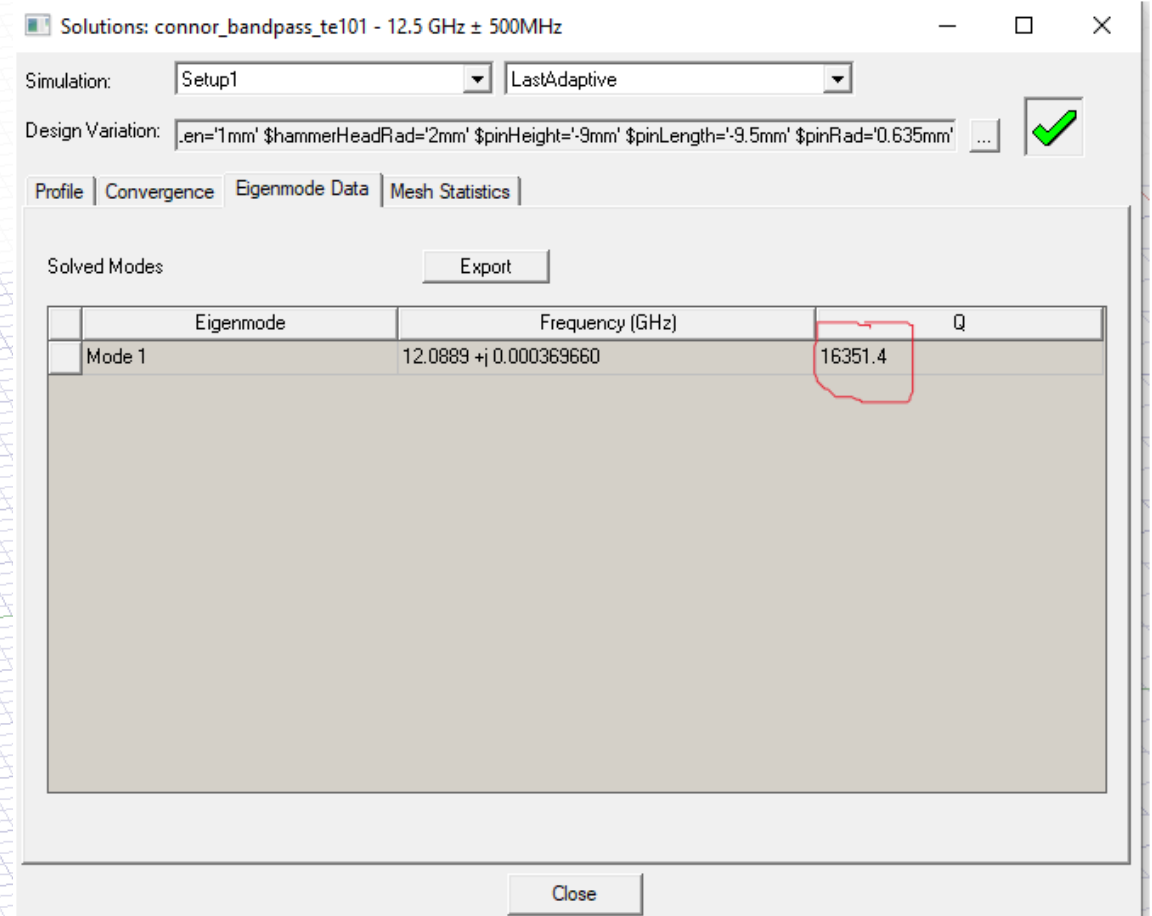
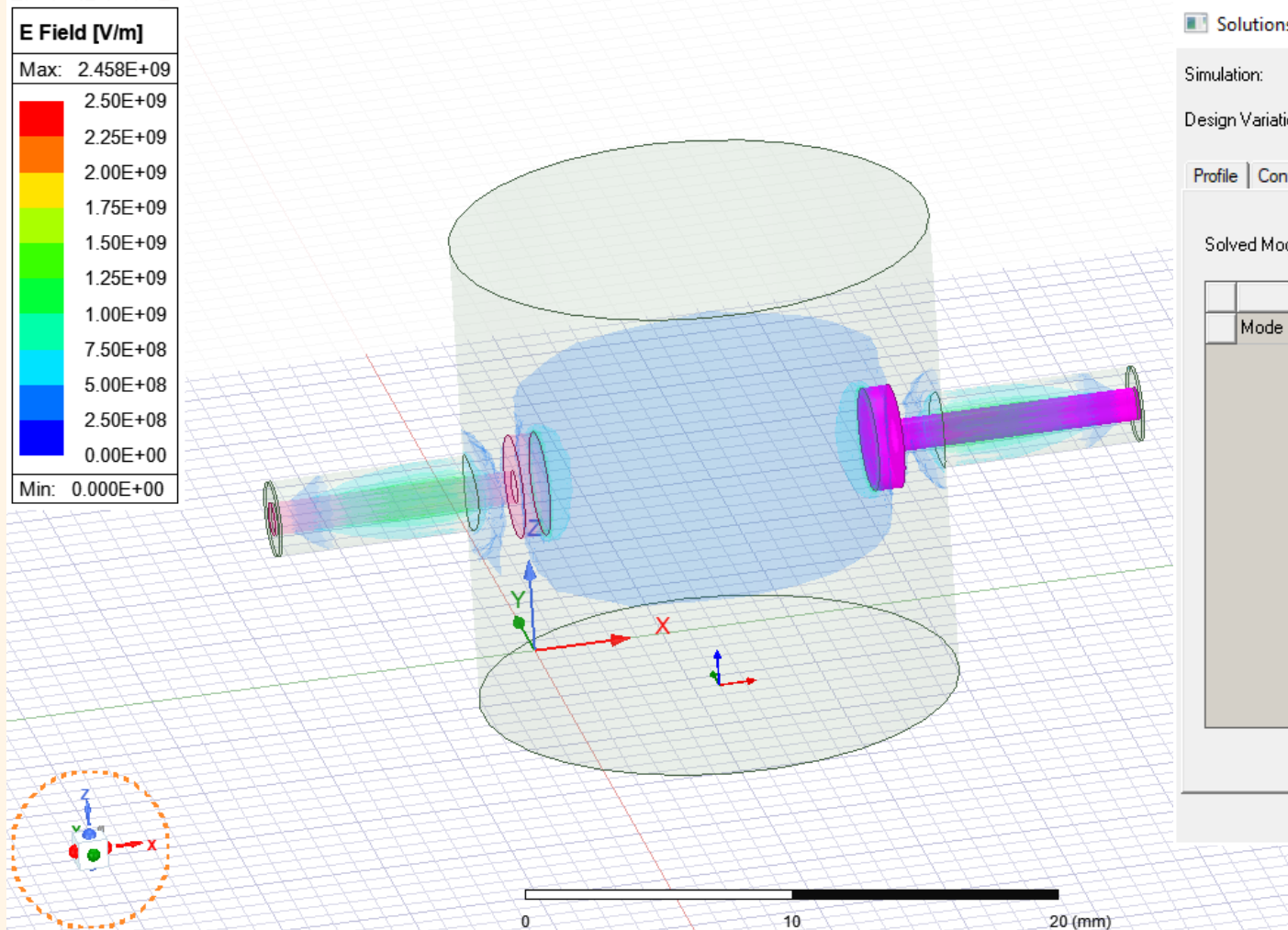


Step 2: Shortening Pin Depth In Cavity



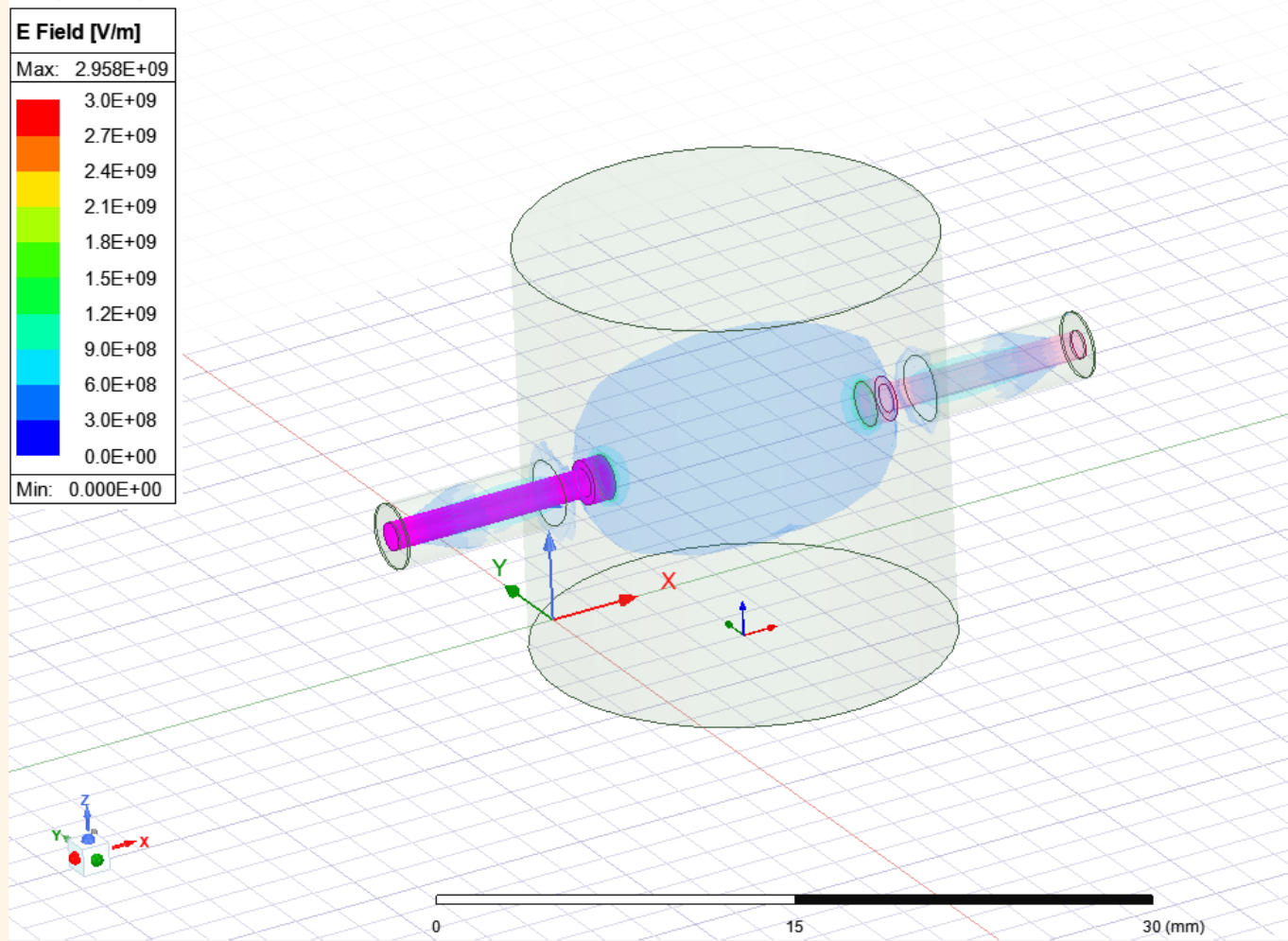


Step 2: Shortening Pin Depth In Cavity



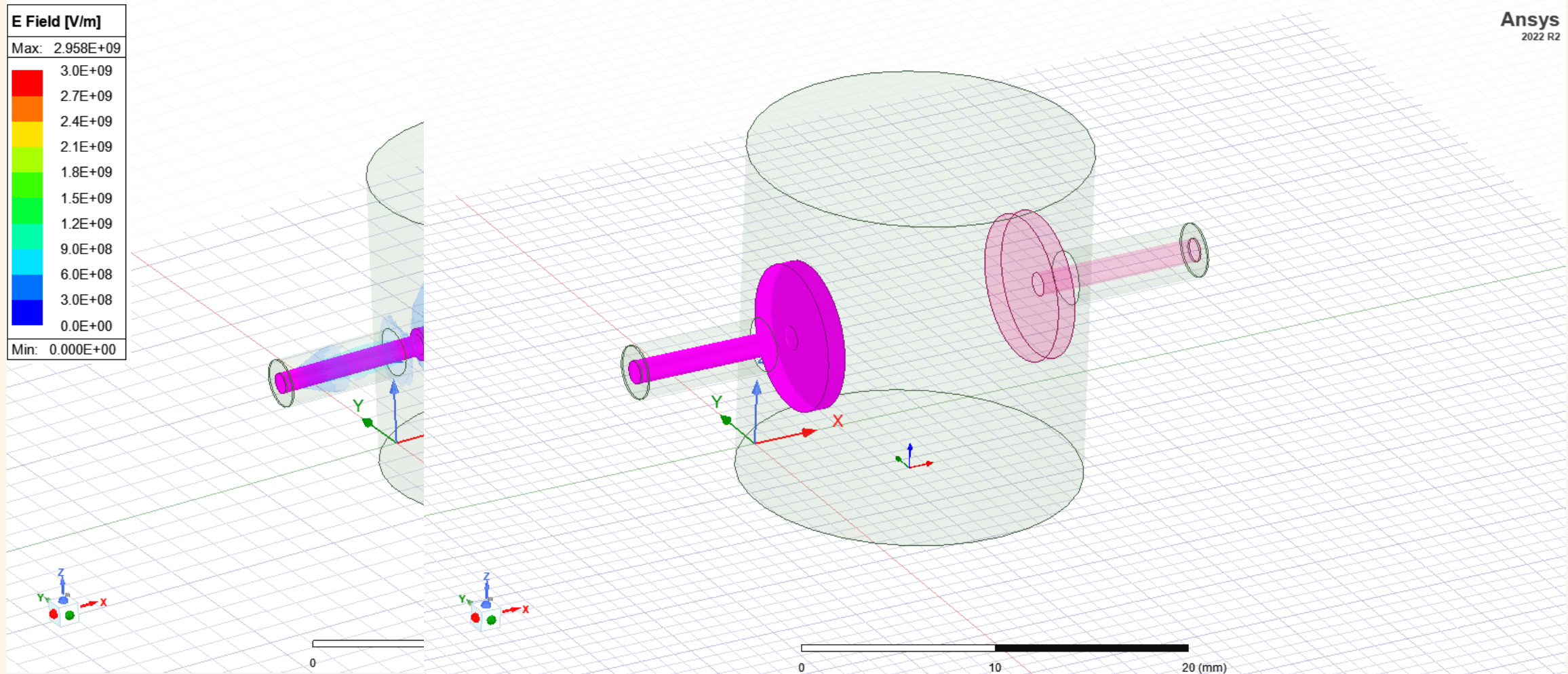


Step 3: Checking Different DoF



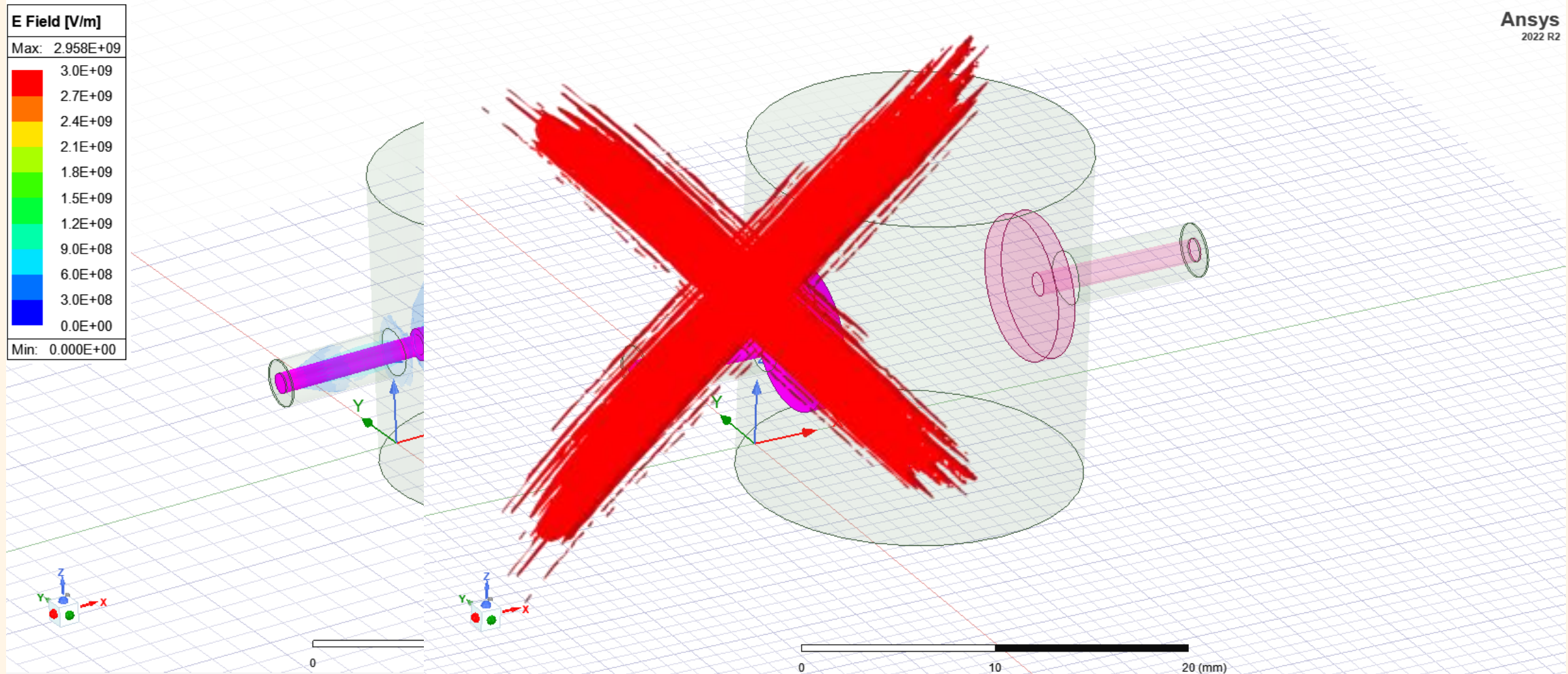


Step 3: Checking Different DoF





Step 3: Checking Different DoF



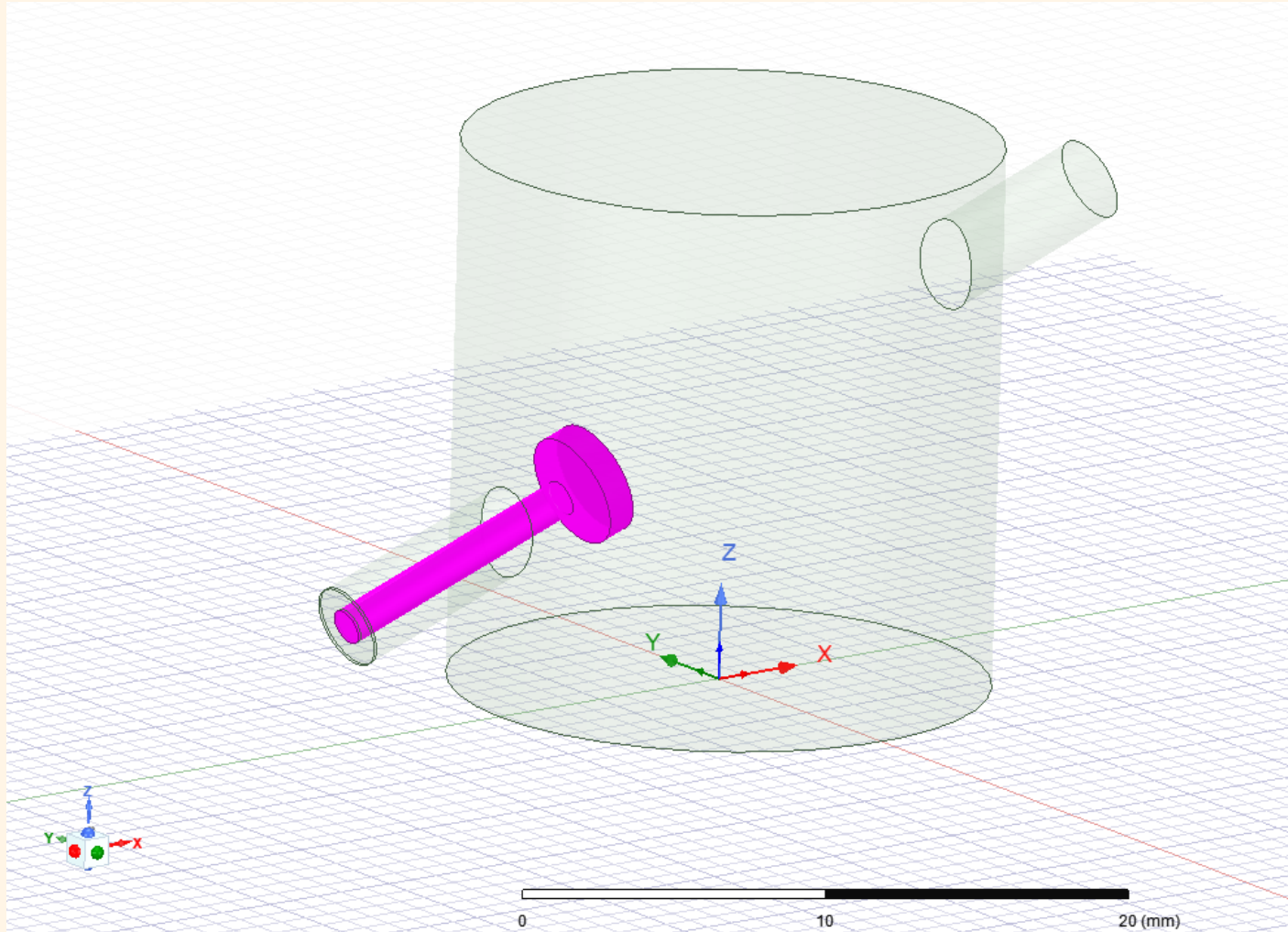


Step 4: Debugging Low Coupling

Because the modes are radial, side entry with symmetric placement leads to extremely low coupling — thanks, Sean!

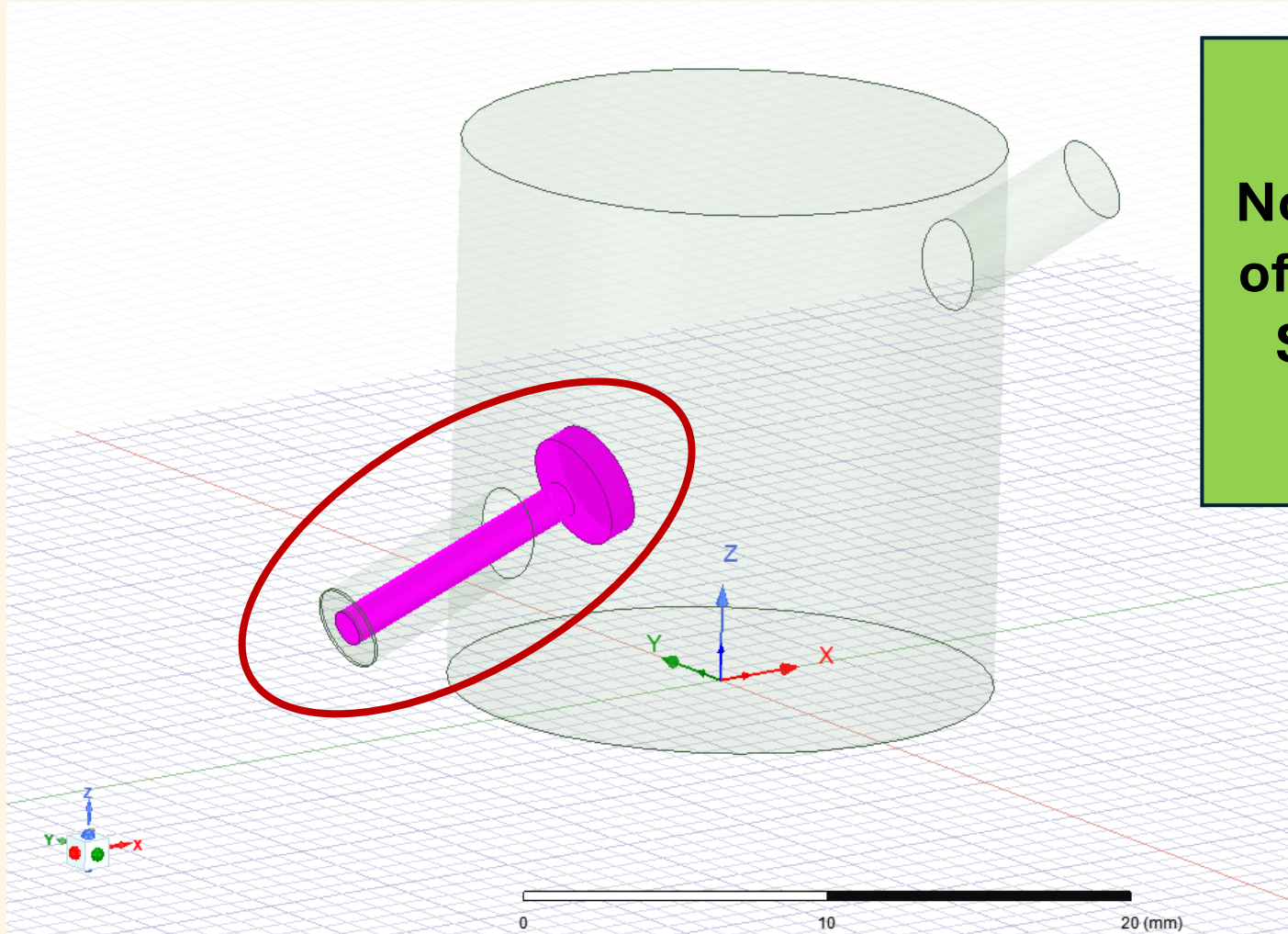


Step 5: Debugging Low Coupling





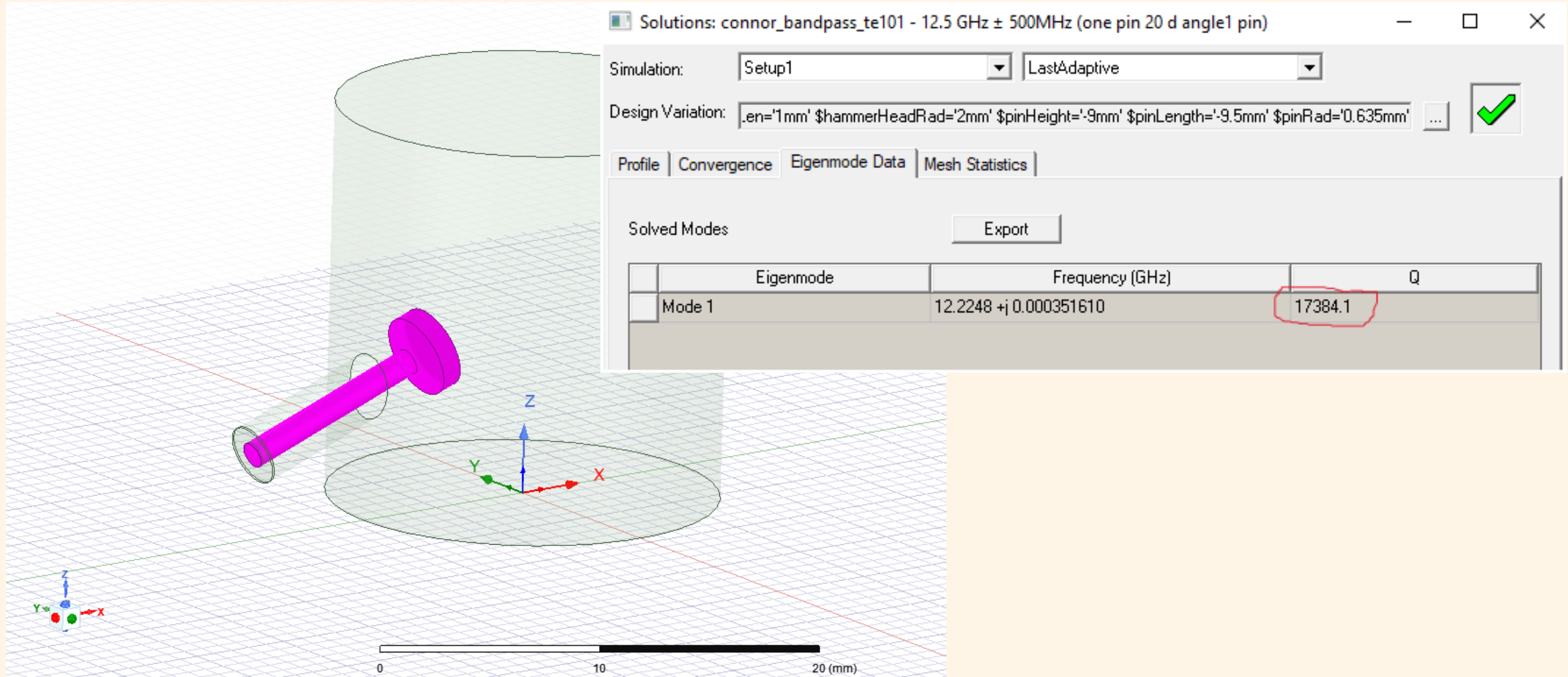
Step 5: Debugging Low Coupling



Note the One Pin for the Next Set of Simulations (Computationally Simpler but Q is $\approx 2\times$ Higher)

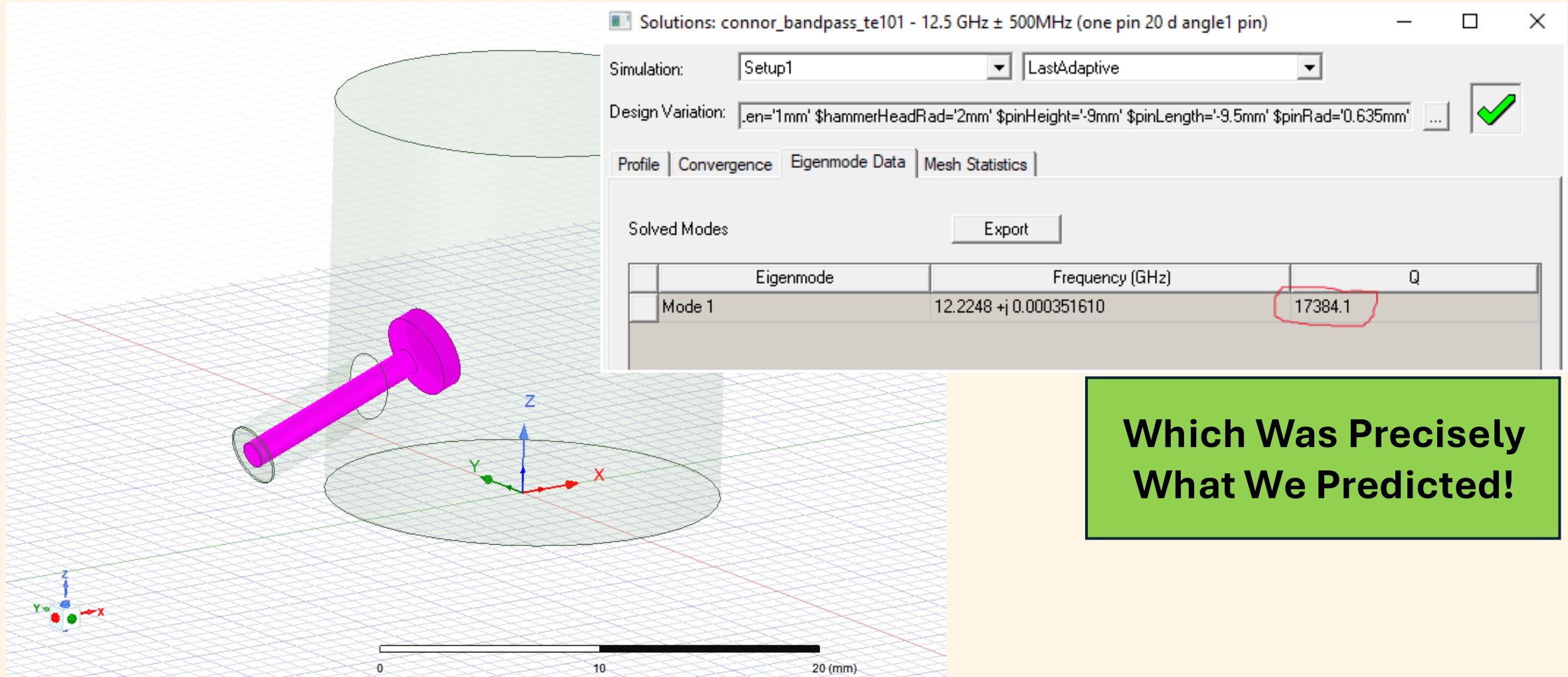


Step 5: Debugging Low Coupling





Step 5: Debugging Low Coupling





Step 6: Sweeping Angle DoF

Degree Angle	Q Factor
10	2.19×10^6
15	225
20	286
22.5	339
25	418
30	724
35	887
40	1229



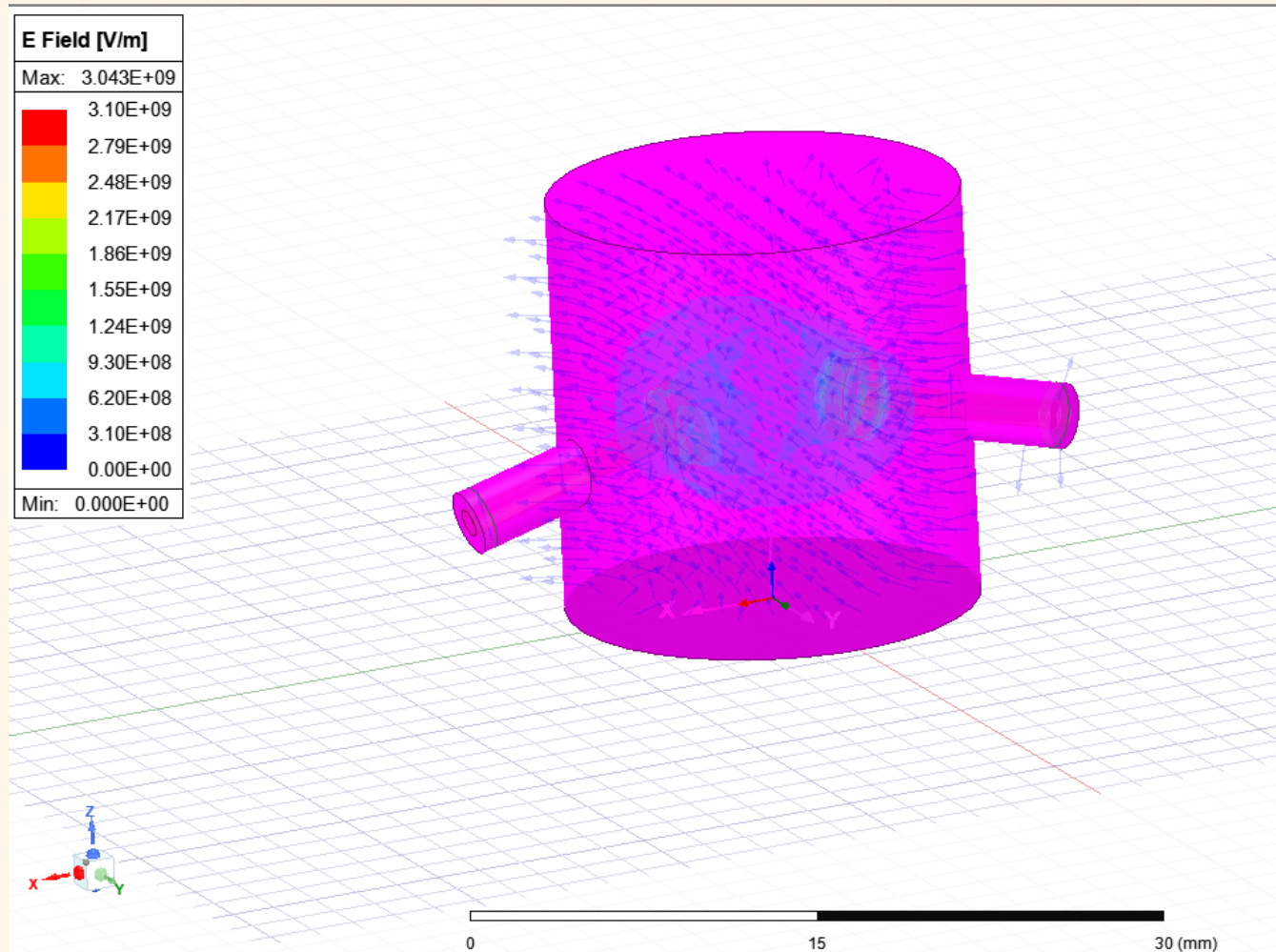
Step 6: Sweeping Angle DoF

Degree Angle	Q Factor
10	2.19×10^6
15	225
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40	1229

Sticking With This 15 Degree Angle...

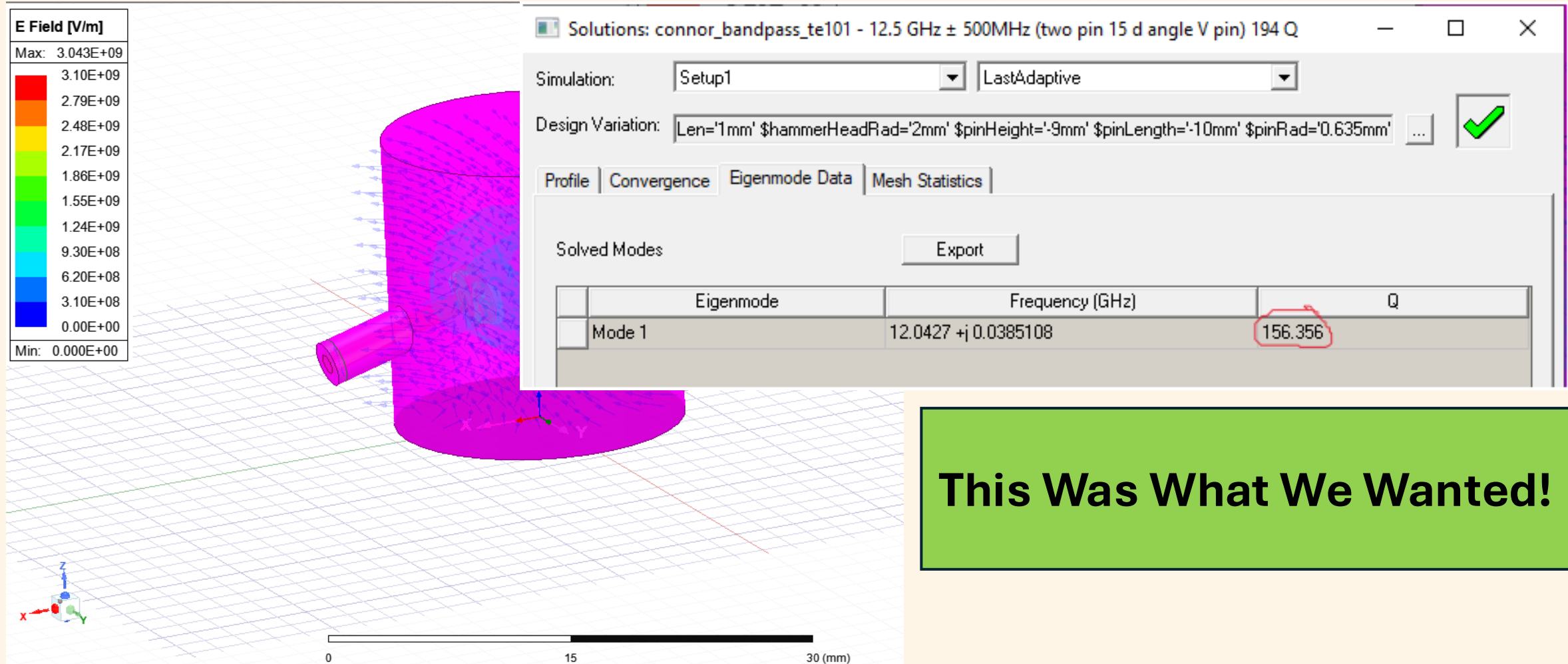


Step 7: Sticking with 15 Degree Angle





Step 7: Sticking with 15 Degree Angle



This Was What We Wanted!



Step 8: Now Checking Tunability

Sweep	Height (mm)	Frequency (GHz)	Q Factor	Implied κ (MHz)
Default	18.8	12.0732	178.50	67.64
Sweep 1	18.3	12.1546	171.16	71.01
Sweep 2	18.0	12.2085	167.64	72.82
Sweep 3	17.8	12.2424	166.21	73.75
Sweep 4	17.6	12.2787	173.24	70.88
Sweep 4.1	17.5	12.3041	192.90	63.78
Sweep 4.2	17.46	12.3117	205.21	60.00
Sweep 4.3	17.455	12.3215	209.47	58.82
Sweep 4.4	17.45	12.3200	592.01	20.81
Sweep 5	17.4	13.3169	740.31	17.99



Step 8: Now Checking Tunability

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Sweep 5	17.4	13.3169	740.31	17.99

Not Bad Overall For A Simple Design...



Step 9: Focusing on S21 Simulations

But the Celebration Was *Very* Short Lived...



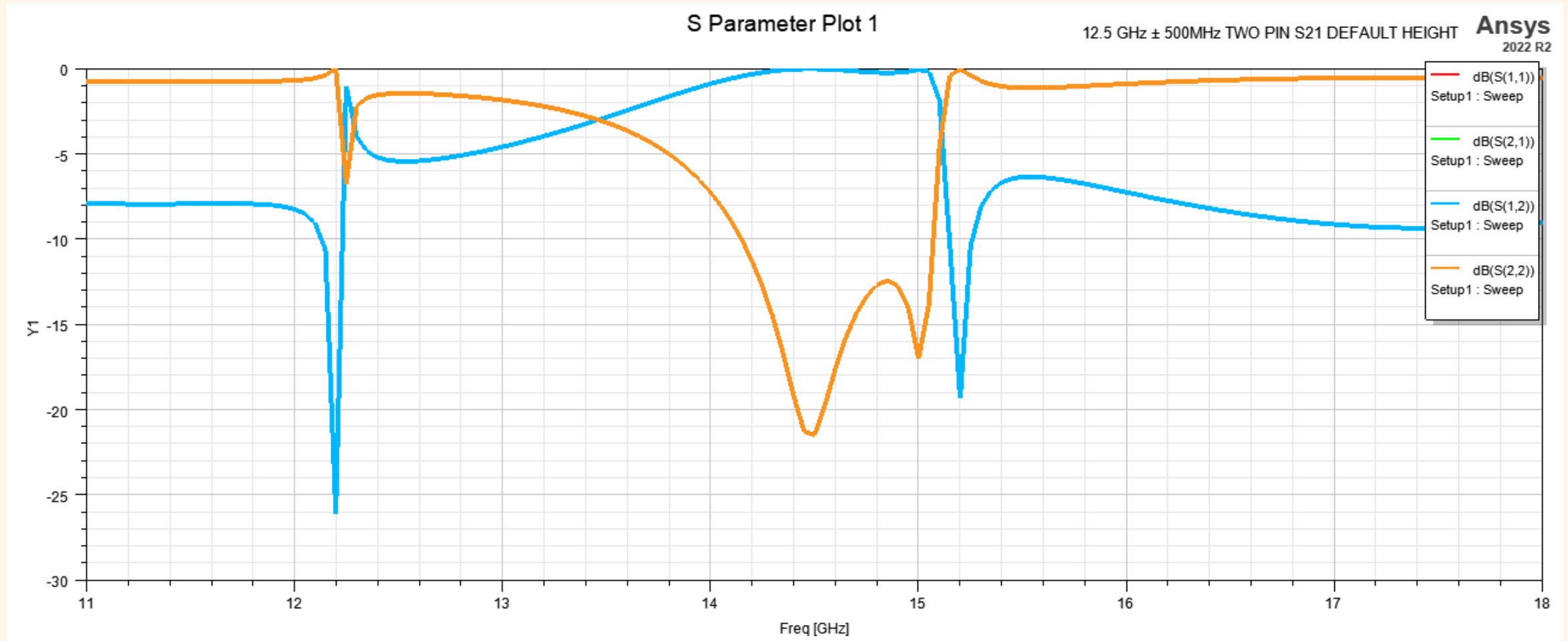
Step 9: Focusing on S21 Simulations

But the Celebration Was *Very* Short Lived...

Why You Ask?



Step 9: Focusing on S21 Simulations





Step 10: Debugging S21

First, I Checked the Modes



Step 10: Debugging S21

Mode	Type	Analytic (GHz)	Simulation (GHz)
1	TE 111	12.408	12.4497
2	TE 11	12.408	12.5190
3	TM 010	12.418	12.5542
4	TM 011	14.757	14.8955
5	TE 211	17.672	17.2945
6	TE 211	17.672	17.8843
7	TE 112	18.866	18.6018
8	TE 112	18.866	18.6532
9	TM 111	19.760	19.9606
10	TM 110	19.760	20.0605



Step 10: Debugging S21

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10	TM 110	19.760	20.0605

Which All Made Sense...



Step 10: Debugging S21

Next, I Ensured Pins Were Symmetric



Step 10: Debugging S21

Solutions: connor_bandpass_te101 - 12.5 GHz $\pm$ 500MHz (one pin 15 d angle1 pin v2) 221 Q

Simulation: Setup1 LastAdaptive

Design Variation: Len='1mm' \$hammerHeadRad='2mm' \$pinHeight='9mm' \$pinLength='10mm' \$pinRad='0.635mm' ...

Profile | Convergence | Eigenmode Data | Mesh Statistics

Solved Modes Export

	Eigenmode	Frequency (GHz)	Q
	Mode 1	12.1694 +j 0.0274008	222.063



Step 10: Debugging S21

Solutions: connor_bandpass_te101 - 12.5 GHz $\pm$ 500MHz (one pin 15 d angle1 pin v2) 221 Q

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Solved Modes Export

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Profile | Convergence | Eigenmode Data | Mesh Statistics

Solved Modes Export

	Eigenmode	Frequency (GHz)	Q
Mode 1		12.1693 +j 0.0273962	222.111

Which Also Made Sense...



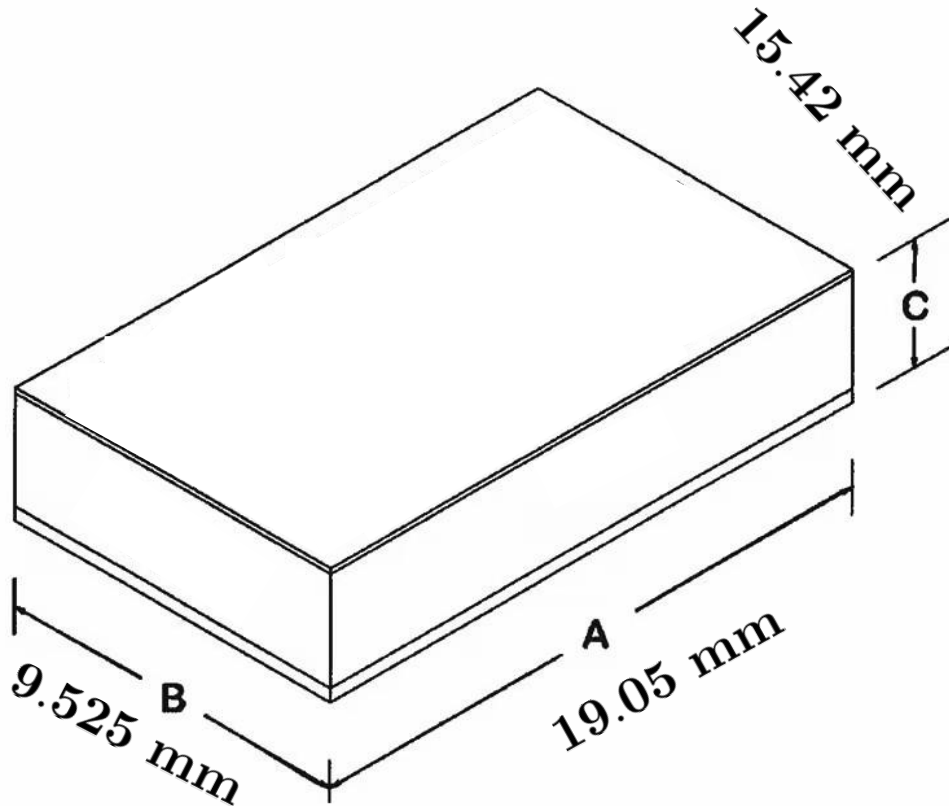
Step 11: Switching to Rectangular

To Sanity Check, I Switched to a Rectangular Geometry



Step 11: Switching to Rectangular

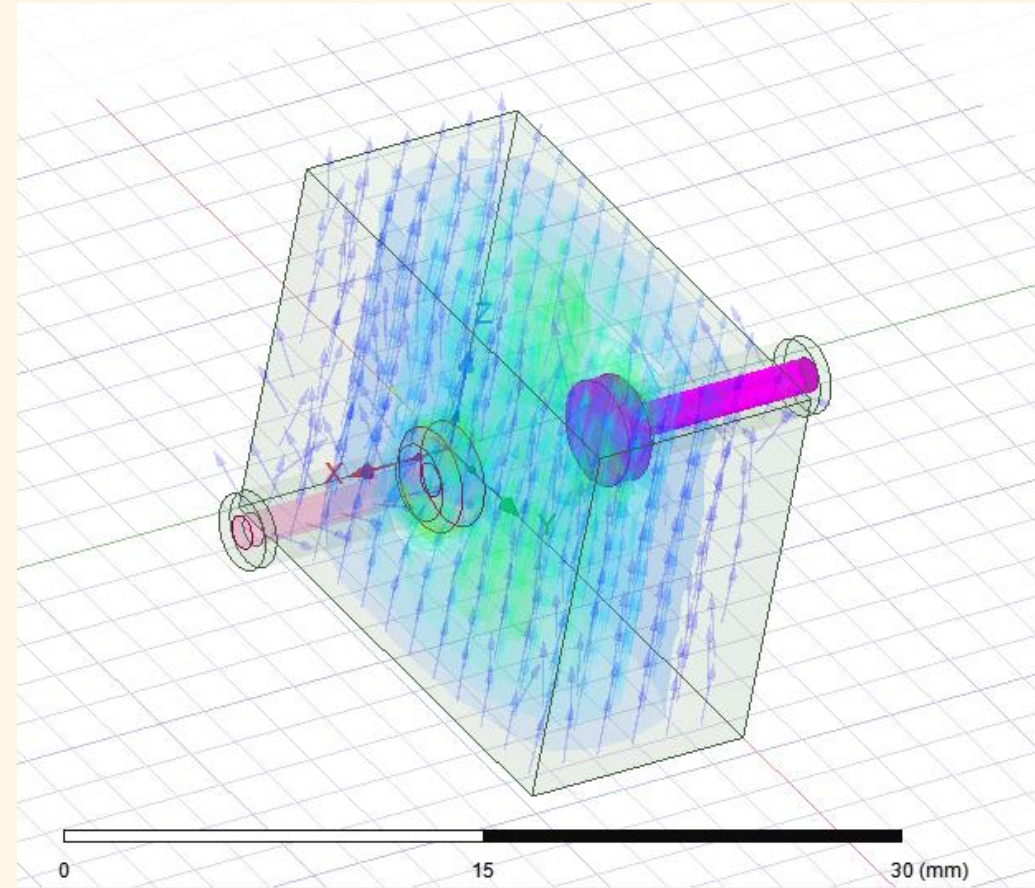
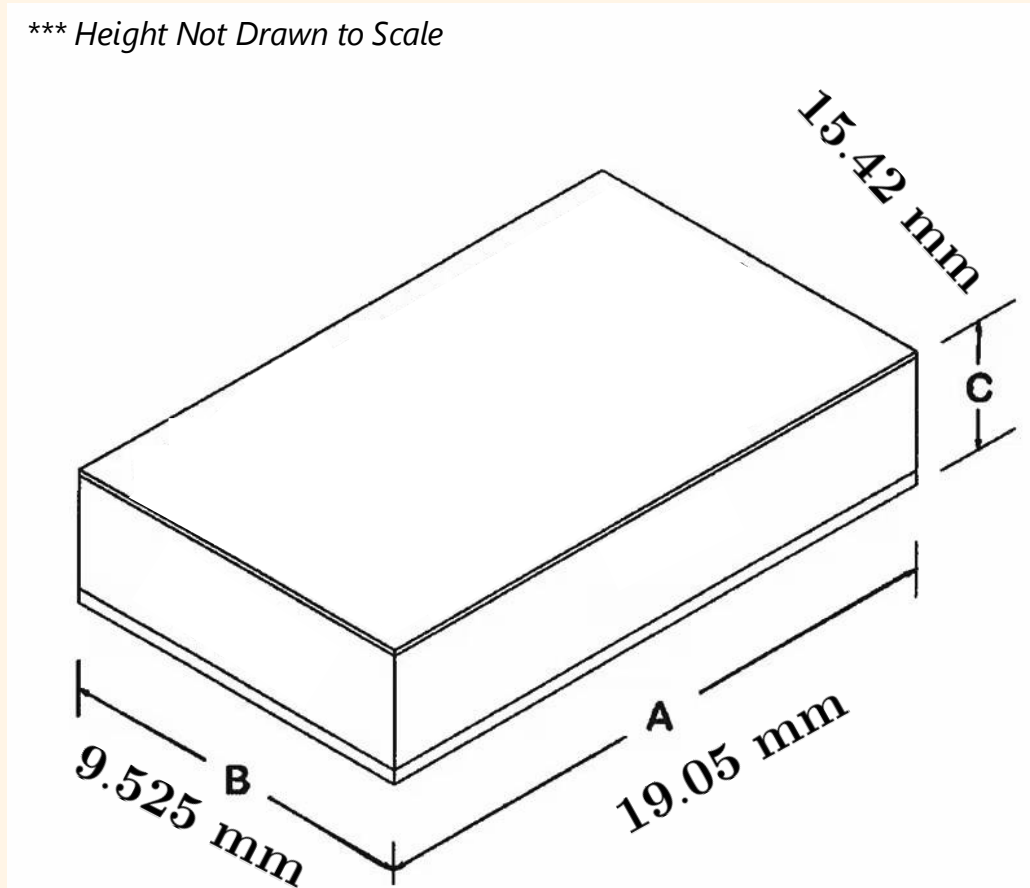
*** Height Not Drawn to Scale





Step 11: Switching to Rectangular

*** Height Not Drawn to Scale





Step 12: Rectangular Modes

$$f_{mnp} = \frac{c}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{d}\right)^2}$$



Step 12: Rectangular Modes

Solutions: connor_bandpass_te101 - RECTANGULAR CAVITY 12.5 GHz

Simulation: Setup1 LastAdaptive

Design Variation: Len='1mm' \$hammerHeadRad='2mm' \$pinHeight='9mm' \$pinLength='10mm' \$pinRad='0.635mm' ...

Profile Convergence Eigenmode Data Mesh Statistics

Solved Modes Export

	Eigenmode	Frequency (GHz)
	Mode 1	12.5065
	Mode 2	17.5951
	Mode 3	18.4978
	Mode 4	18.4982
	Mode 5	20.1024
	Mode 6	20.1026
	Mode 7	20.9748
	Mode 8	22.2569
	Mode 9	24.2892
	Mode 10	24.2894

$$f_{mnp} = \frac{c}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{d}\right)^2}$$



Step 12: Rectangular Modes

Solutions: connor_bandpass_te101 - RECTANGULAR CAVITY 12.5 GHz

Simulation: Setup1 LastAdaptive

Design Variation: Len='1mm' \$hammerHeadRad='2mm' \$pinHeight='9mm' \$pinLength='10mm' \$pinRad='0.635mm' ...

Profile Convergence Eigenmode Data Mesh Statistics

Solved Modes Export

	Eigenmode	Frequency (GHz)
	Mode 1	12.5065
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	Mode 9	24.2892
	Mode 10	24.2894

$$f_{mnp} = \frac{c}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{d}\right)^2}$$

**The Math
Matches, as
Expected!**



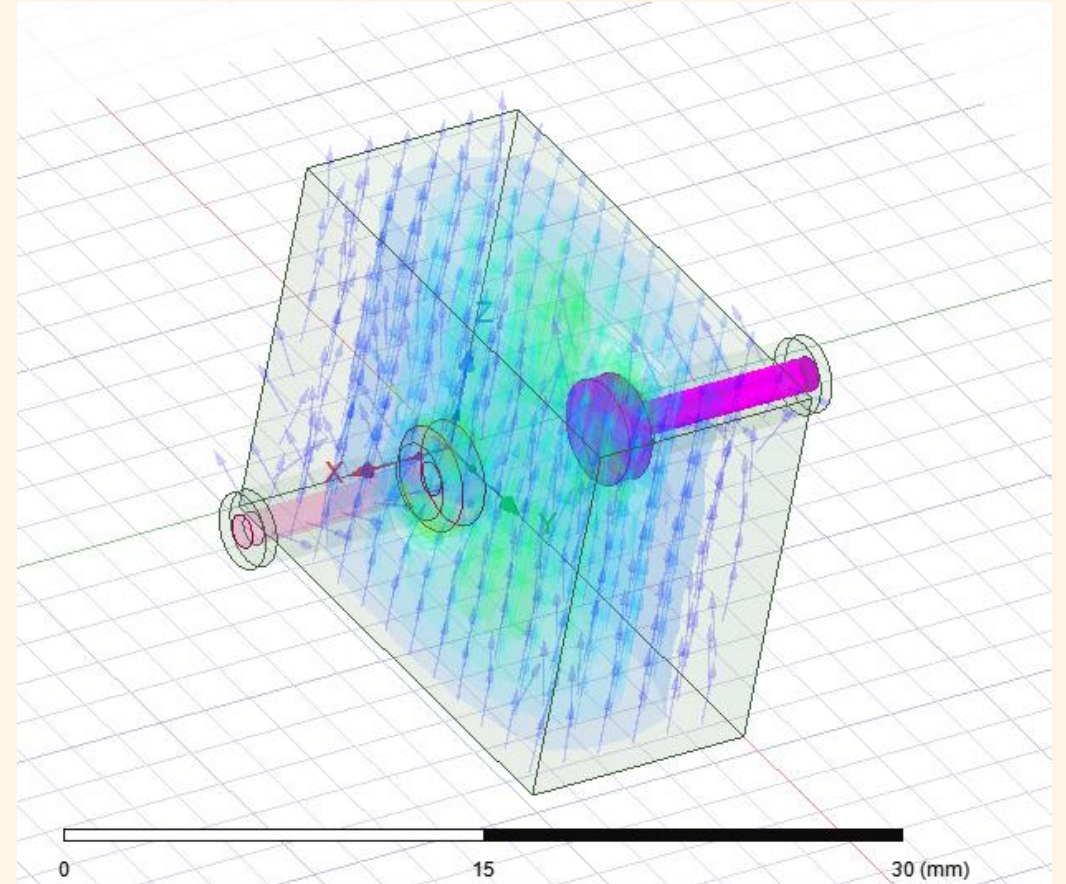
Step 12: Tuning Pin DoF

Position Fixes So Tuning Length...



Step 12: Tuning Pin DoF

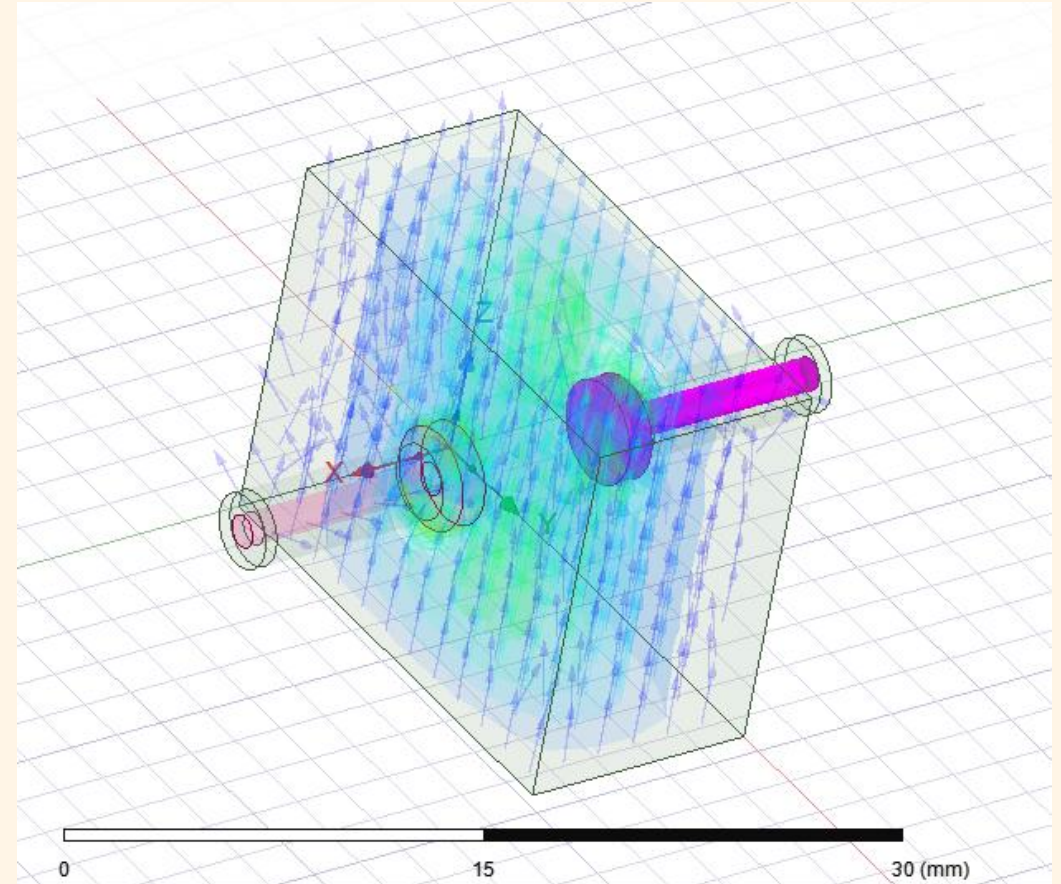
Position Fixes So Tuning Length...





Step 12: Tuning Pin DoF

Length Inside Cavity (mm)	Q Value	Frequency (GHz)
1.0	4424	12.43
1.5	2298	12.33
2.0	1324	12.18
3.5	172	16.70
4.0	174	16.94
4.5	172.58	16.72



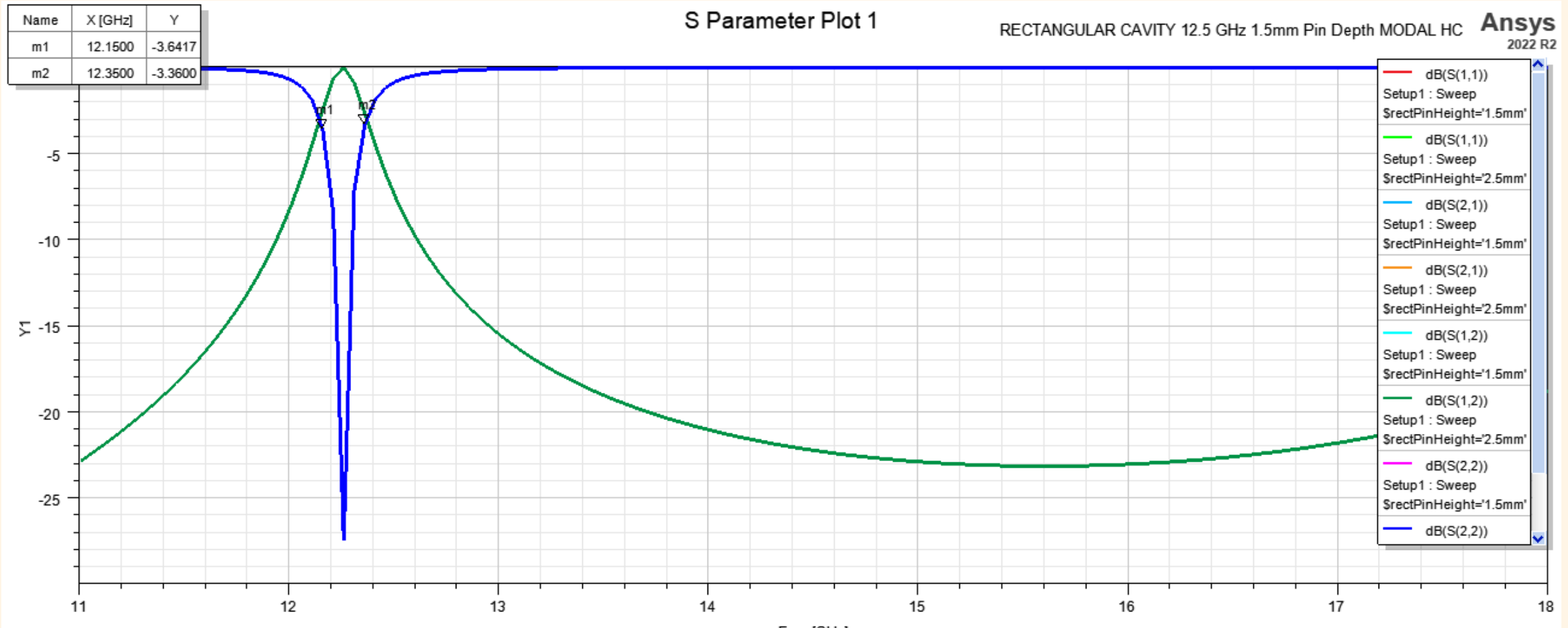


Step 13: Checking S21

This Time We Are Laser Focused on S21...

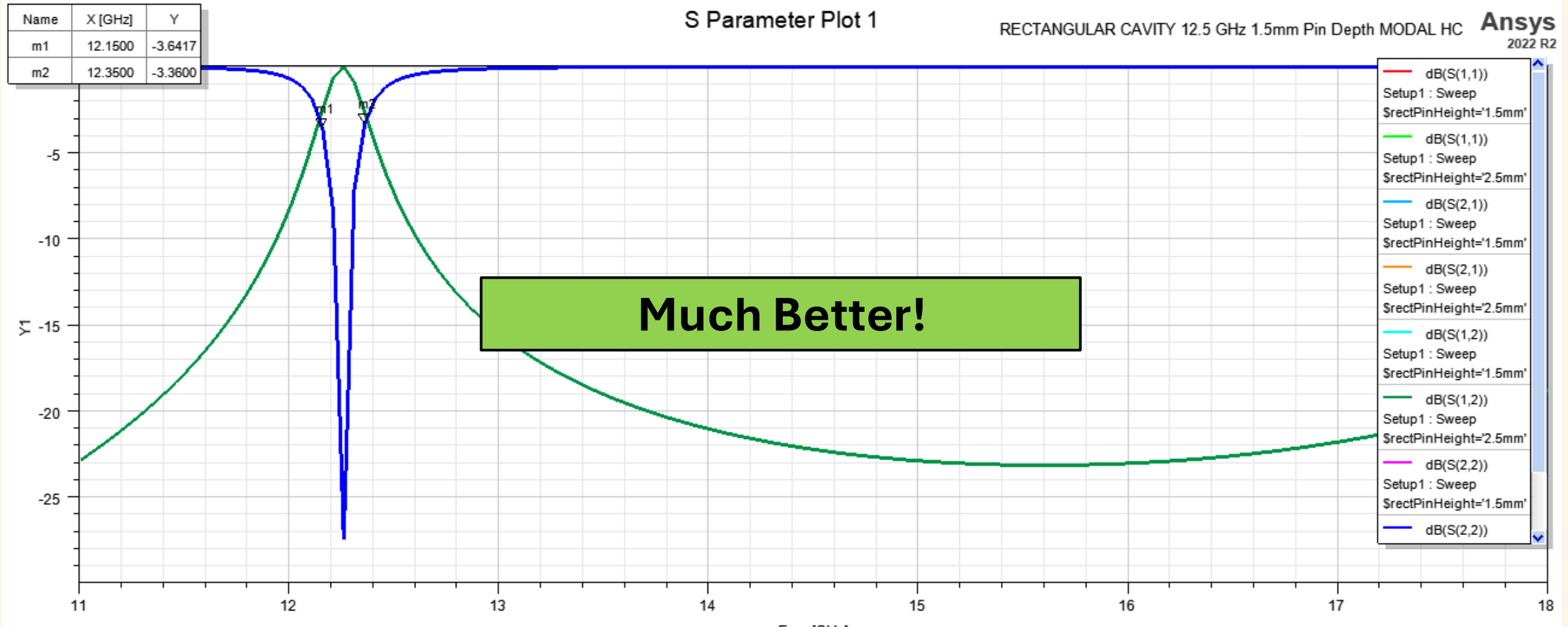


Step 13: Checking S21





Step 13: Checking S21



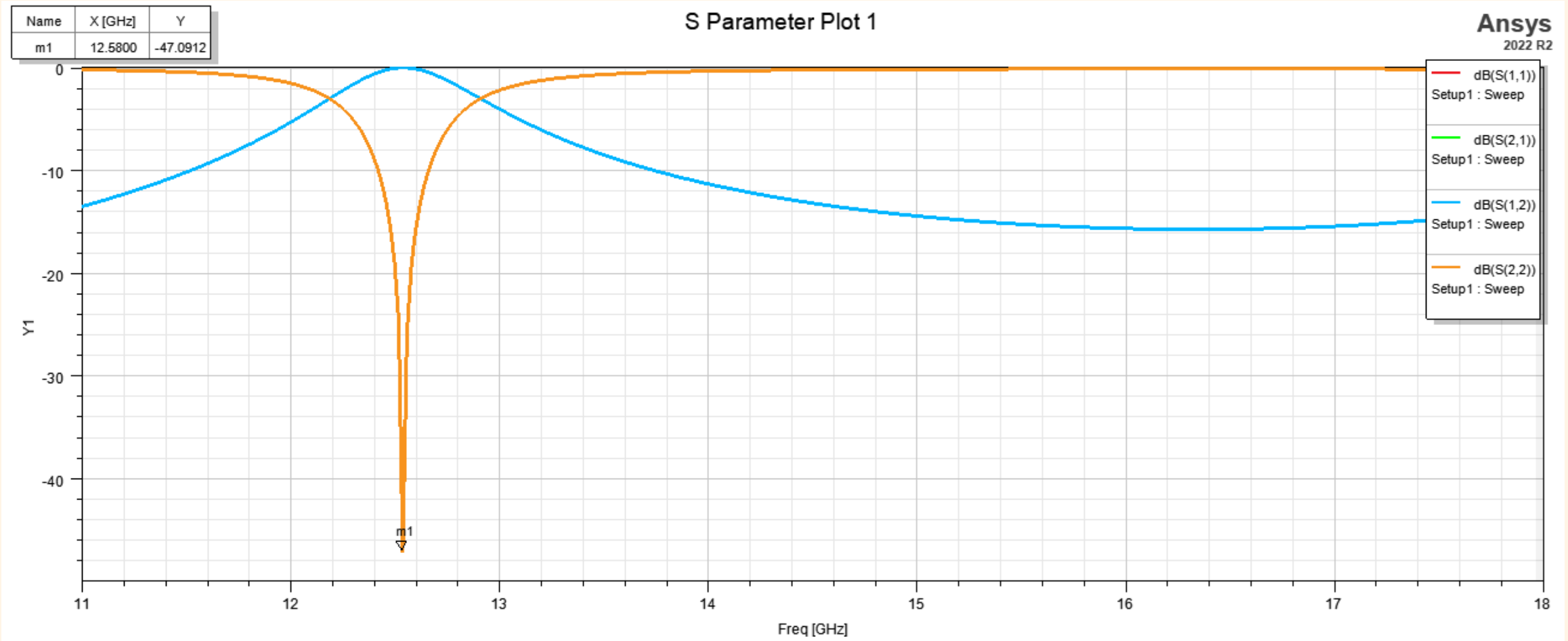


Step 13: Shifting Resonance (12.58 Ghz)

**PReSPA Needs a 12.58Ghz Filter, so we
Tweaked Some Parameters...**



Step 13: Shifting Resonance (12.58 GHz)





Step 14: Testing Tunability Mechanism

Freq (GHz)	Height (mm)	Δ Height (mm)	Q Value	κ (MHz)
16.28	10.2	-4.0	324	50.25
14.96	11.2	-3.0	539	27.76
13.98	12.2	-2.0	741	18.87
13.56	12.7	-1.5	867	15.64
13.20	13.2	-1.0	931	14.18
12.58	14.2	0.0	1119	11.24
12.32	14.7	0.5	1211	10.17
12.16	15.2	1.0	1295	9.39
6.00	15.4	1.2	722905	0.01



Step 14: Testing Tunability Mechanism

Freq (GHz)	Height (mm)	Δ Height (mm)	Q Value	κ (MHz)
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12.16	15.2	1.0	1295	9.39
6.00	15.4	1.2	722905	0.01

**This is Good Just Narrower Than We
Originally Set Out to Get**

V. Design Challenges for Lower Frequencies



Using a Cylindrical Design: TE₁₀₁

Deriving Minimum Radius for 2.0 GHz

$$f_{mnp} = \frac{c}{2\pi} \sqrt{\left(\frac{x'_{mn}}{r}\right)^2 + \left(\frac{p\pi}{h}\right)^2}$$

$$\lim_{h \rightarrow \infty} f_{\min} = \frac{c}{2\pi} \frac{x'_{mn}}{r}$$

$$r_{\min} = \frac{c x'_{11}}{2\pi f}$$

$$= \frac{3 \times 10^8 \times 1.8412}{2\pi \times (2 \times 10^9)} \approx 0.0245 \text{ m} \quad (24.5 \text{ mm})$$

Minimum radius for 2.0 GHz



Using a Cylindrical Design: TE₁₀₁

2.0 GHz @ 50 mm radius

$$f = \frac{c}{2\pi} \sqrt{\left(\frac{x'_{11}}{r}\right)^2 + \left(\frac{p\pi}{h}\right)^2}$$

$$\left(\frac{2\pi f}{c}\right)^2 = \left(\frac{x'_{11}}{r}\right)^2 + \left(\frac{p\pi}{h}\right)^2$$

$$\left(\frac{2\pi f}{c}\right)^2 - \left(\frac{x'_{11}}{r}\right)^2 = \left(\frac{p\pi}{h}\right)^2$$

$$h = \frac{p\pi}{\sqrt{\left(\frac{2\pi f}{c}\right)^2 - \left(\frac{x'_{11}}{r}\right)^2}}$$



Using a Cylindrical Design: TE₁₀₁

Substitute $f = 2.0$ GHz, $r = 0.050$ m, $x'_{11} = 1.8410$, $p = 1$, $c = 3 \times 10^8$ m/s:

$$h = \frac{\pi}{\sqrt{\left(\frac{2\pi(2 \times 10^9)}{3 \times 10^8}\right)^2 - \left(\frac{1.8410}{0.050}\right)^2}}$$

$$h \approx 350 \text{ mm}$$

Required for 2.0 GHz.



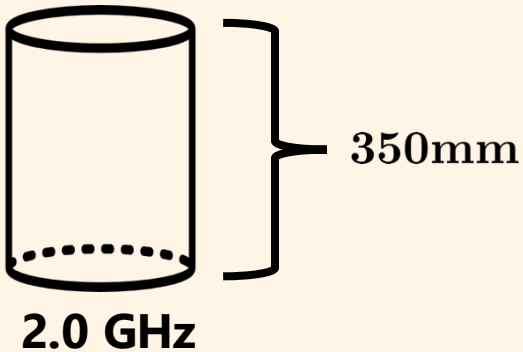
Using a Cylindrical Design: TE₁₀₁

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Required for 2.0 GHz.





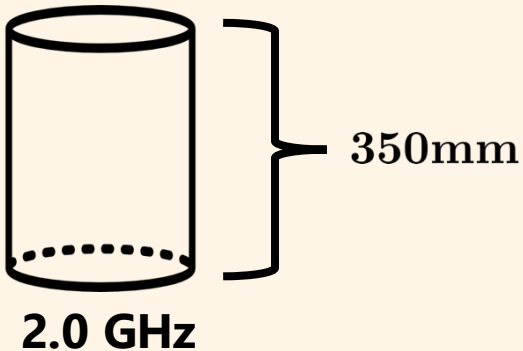
Using a Cylindrical Design: TE101

Substitute $f = 2.0$ GHz, $r = 0.050$ m, $x'_{11} = 1.8410$, $p = 1$, $c = 3 \times 10^8$ m/s:

$$h = \frac{\pi}{\sqrt{\left(\frac{2\pi(2 \times 10^9)}{3 \times 10^8}\right)^2 - \left(\frac{1.8410}{0.050}\right)^2}}$$

$h \approx 350$ mm

Required for 2.0 GHz.



Same Math Yields the Following



Using a Cylindrical Design: TE101

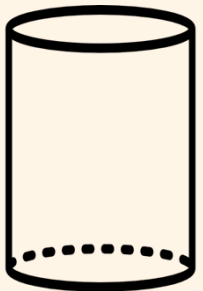
Substitute $f = 2.0$ GHz, $r = 0.050$ m, $x'_{11} = 1.8410$, $p = 1$, $c = 3 \times 10^8$ m/s:

$$h = \frac{\pi}{\sqrt{\left(\frac{2\pi(2 \times 10^9)}{3 \times 10^8}\right)^2 - \left(\frac{1.8410}{0.050}\right)^2}}$$

$$h \approx 350 \text{ mm}$$

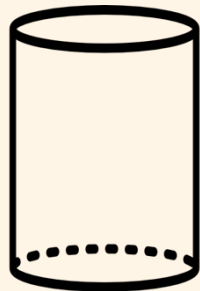
Required for 2.0 GHz.

Same Issue as Rectangular as 3.4 GHz will be Similar Story!



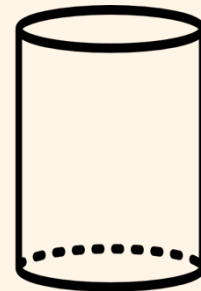
2.0 GHz

350mm



7.8 GHz

20mm



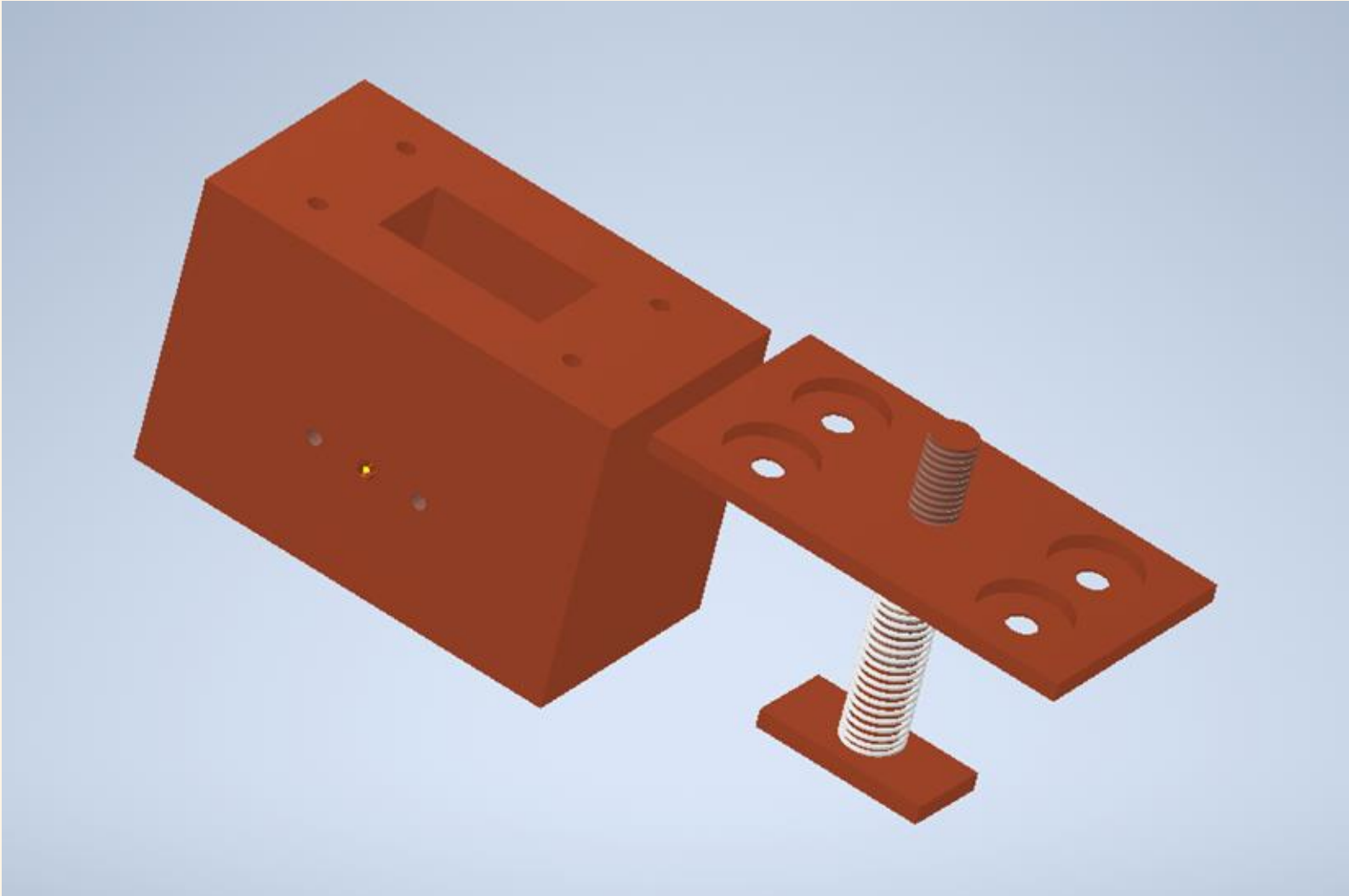
12.5 GHz

12mm

VI. The Actual Design

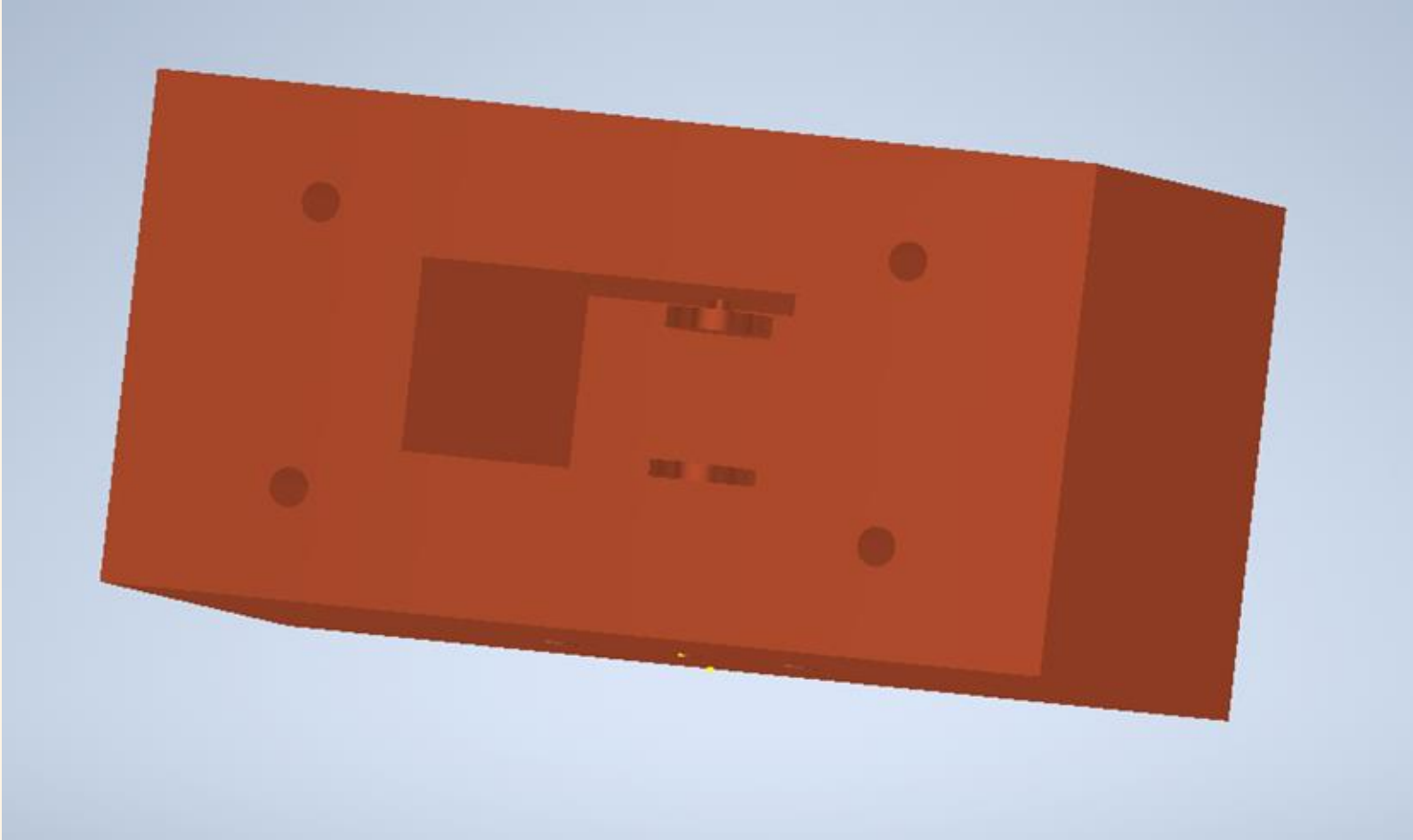


CAD Design of the Filter



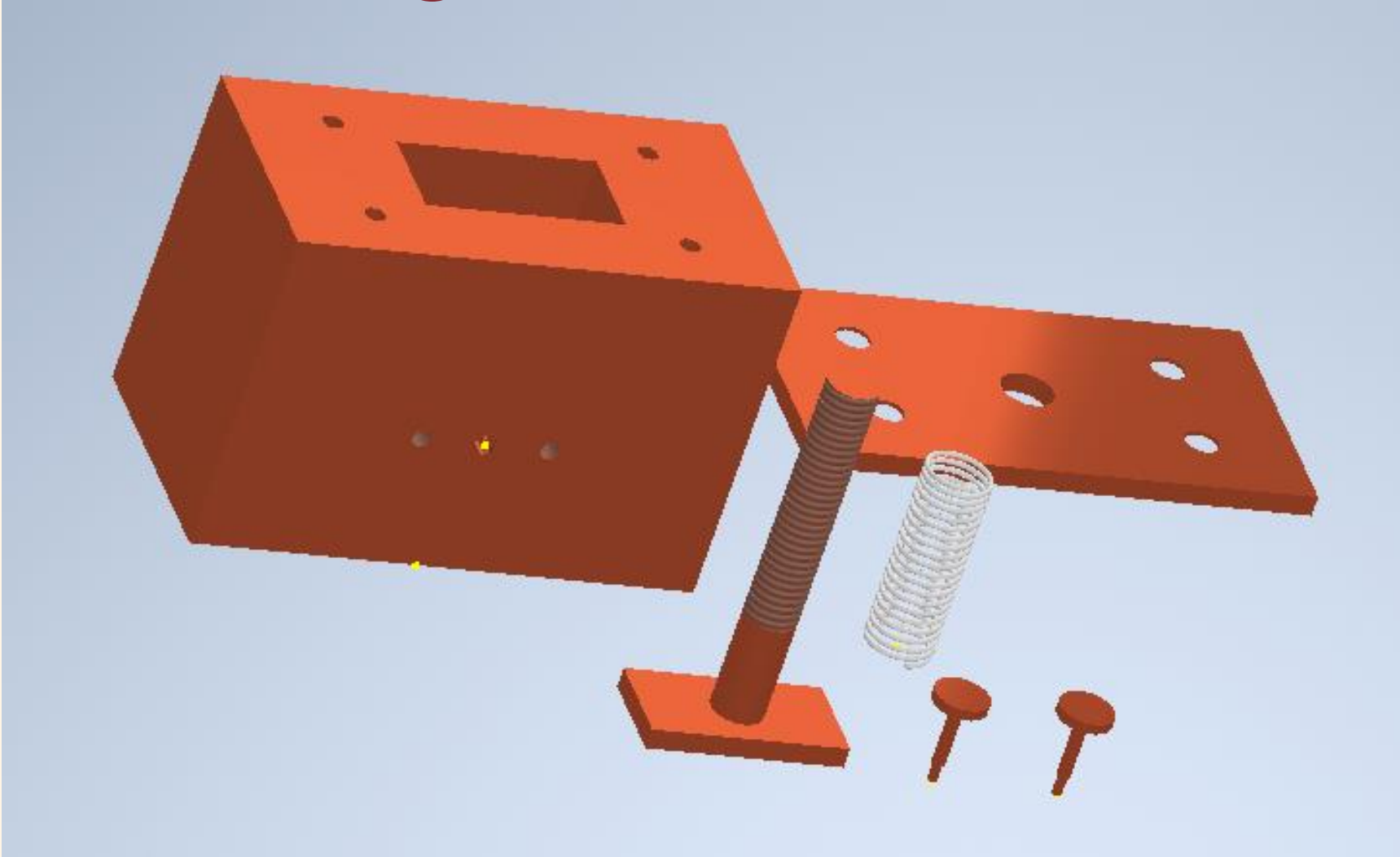


CAD Design of the Filter



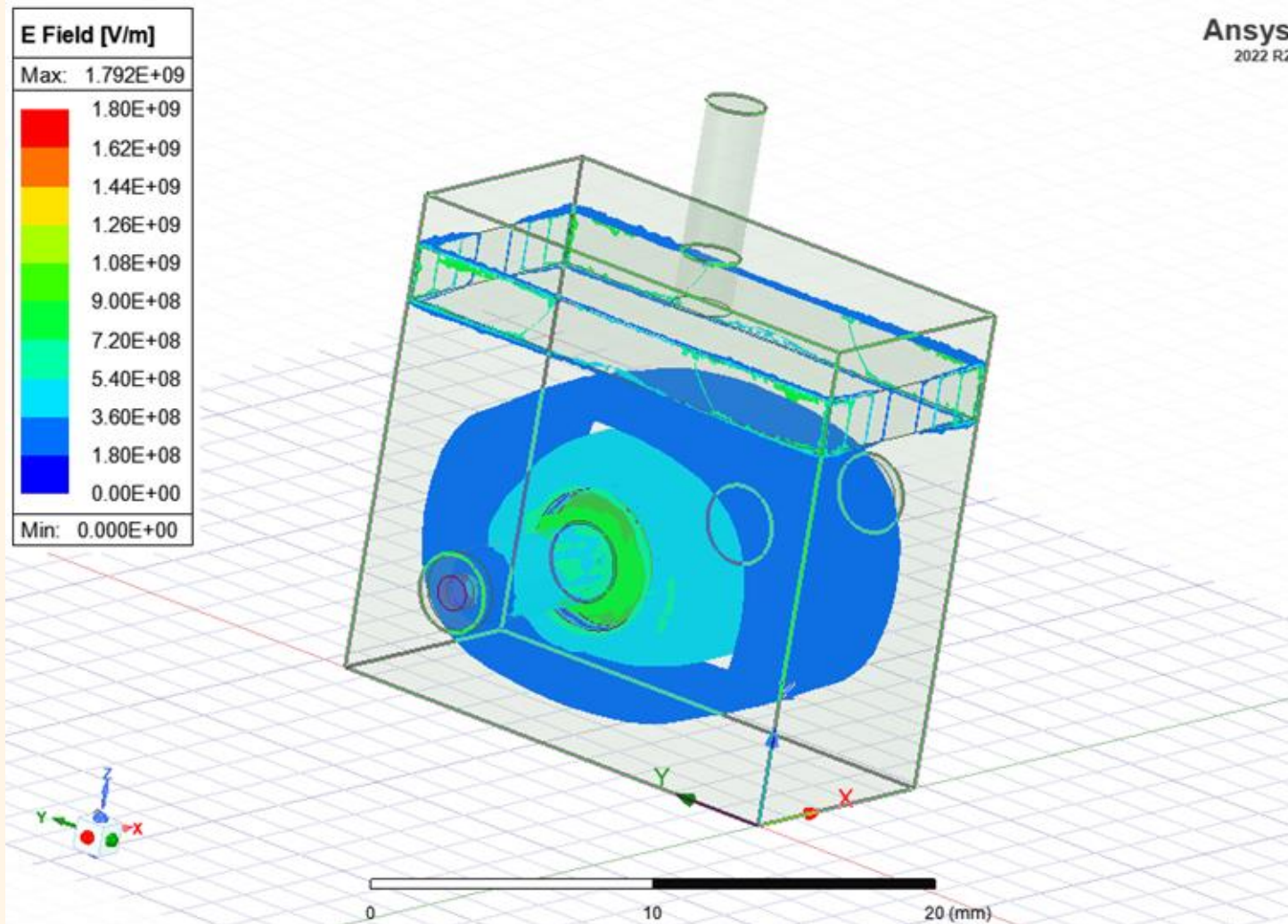


CAD Design of the Filter



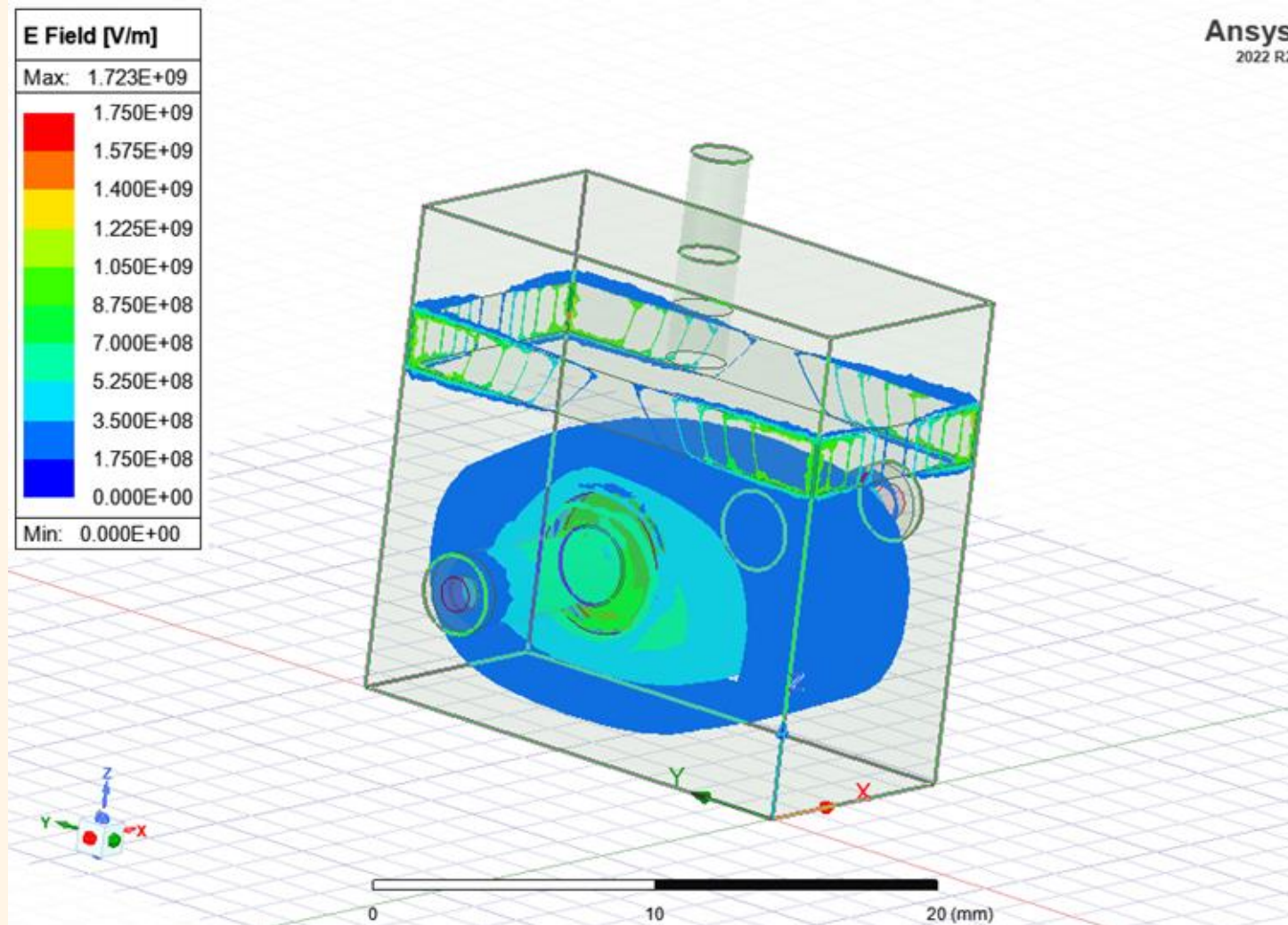


Expected Results



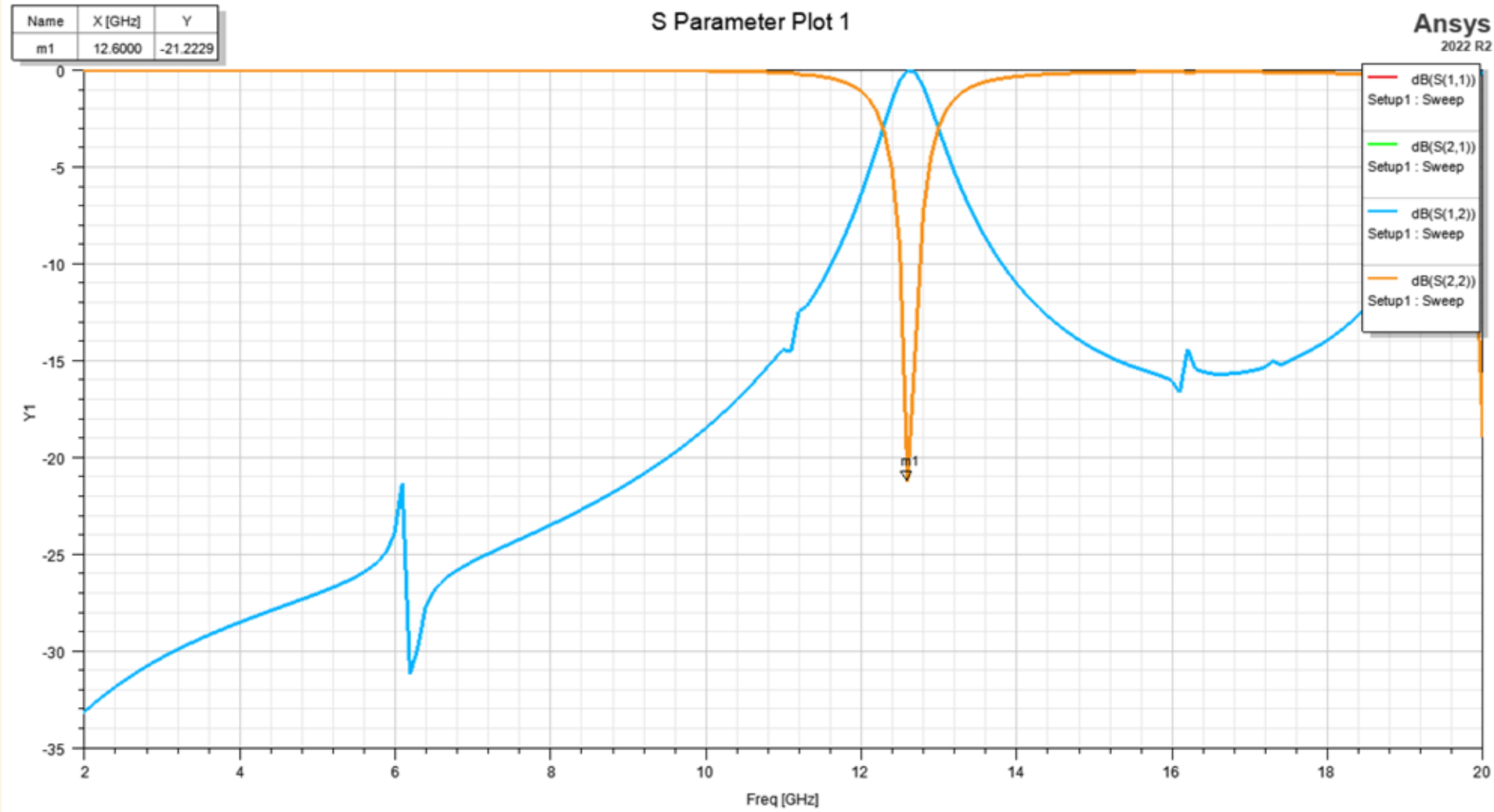


Expected Results



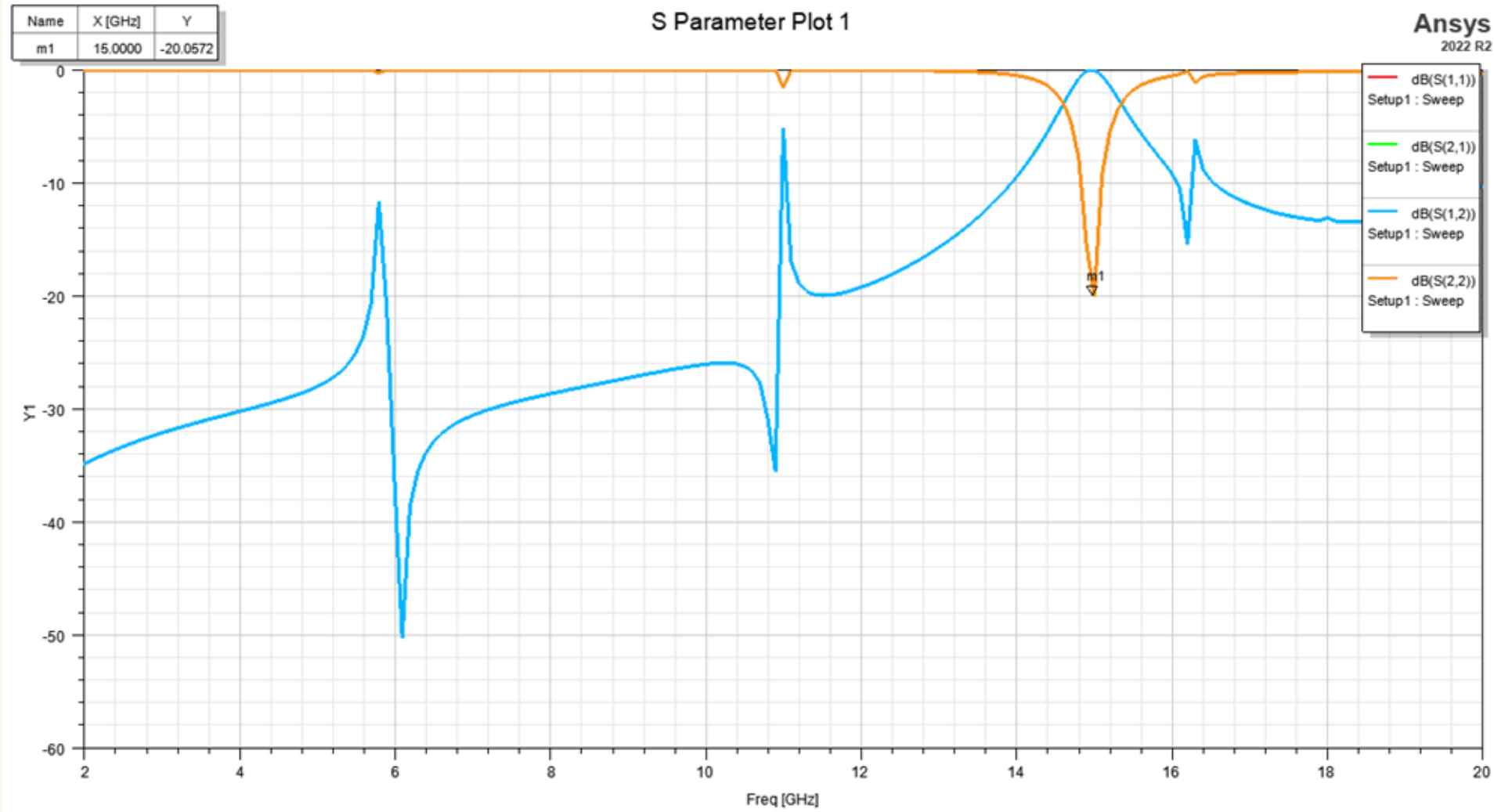


Expected Results



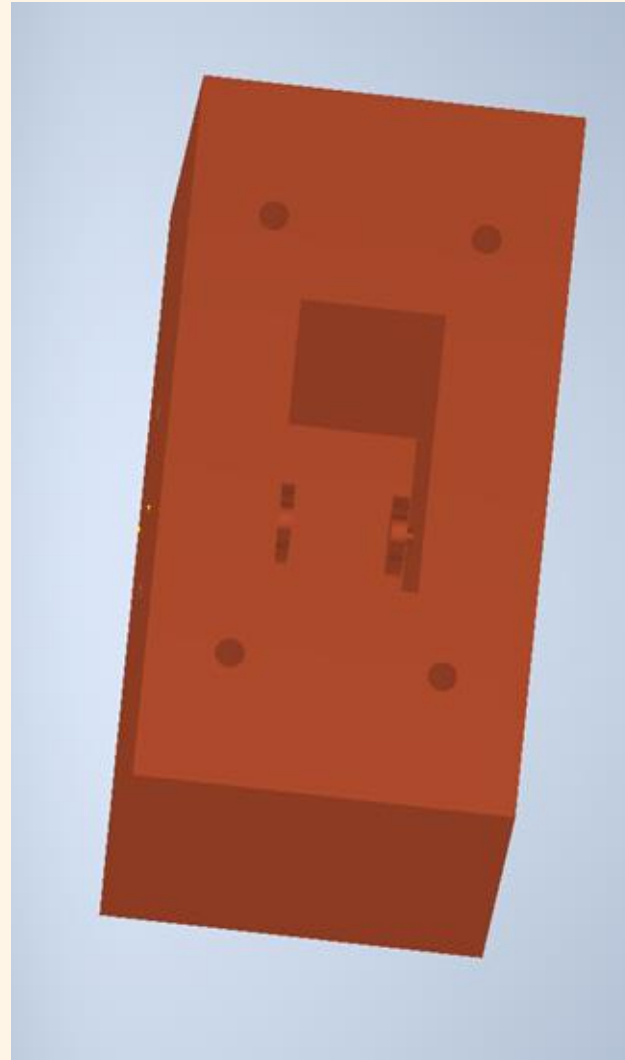
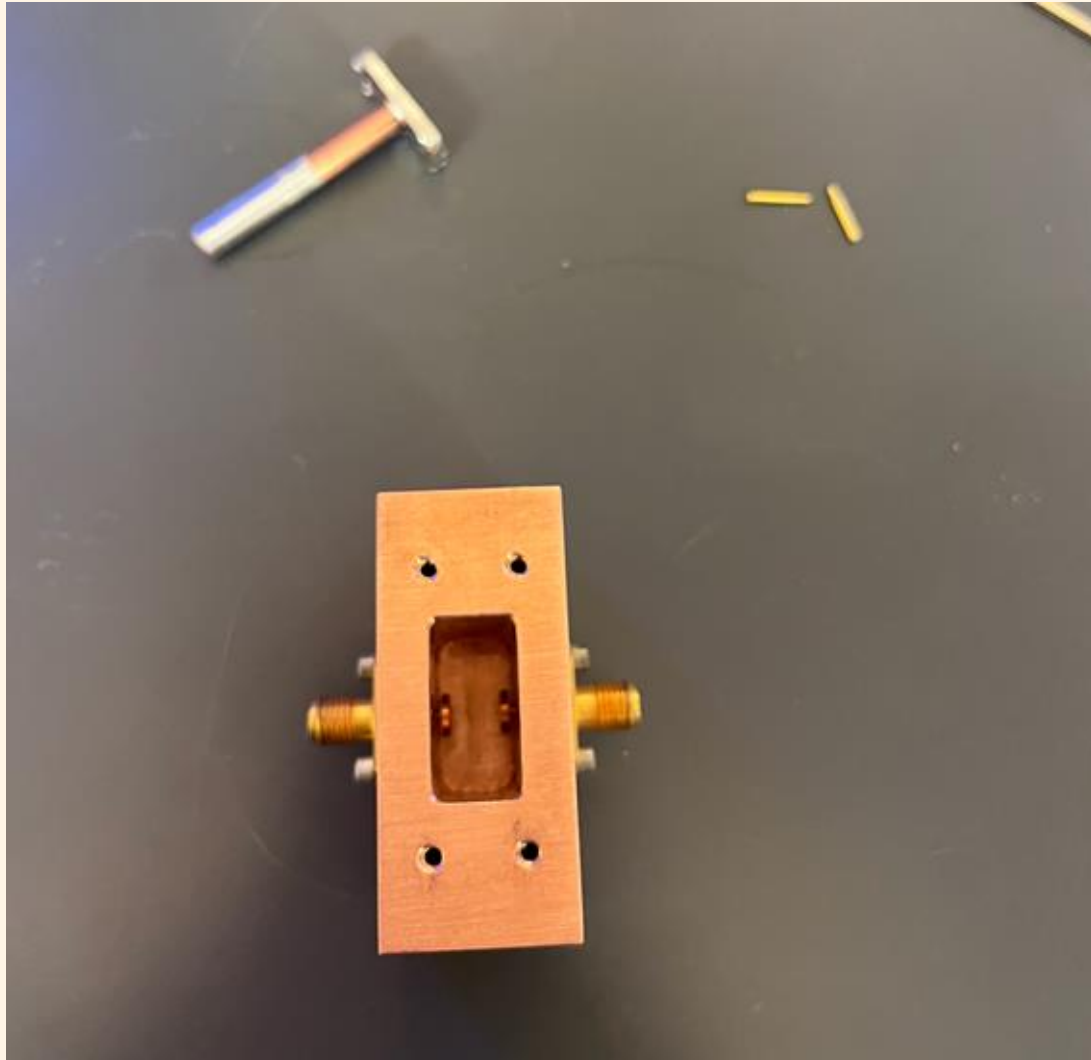


Expected Results



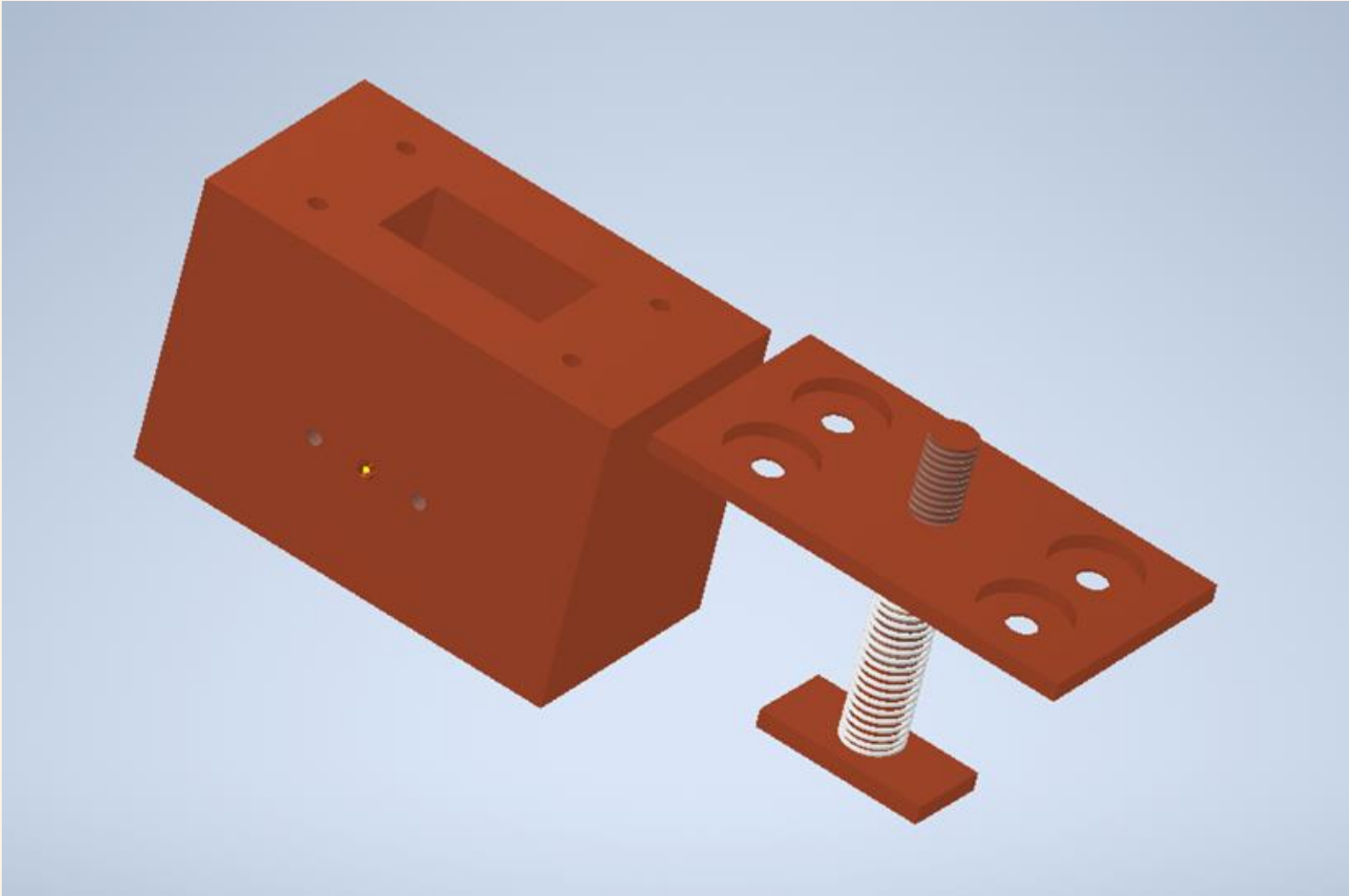


Fabricated Device



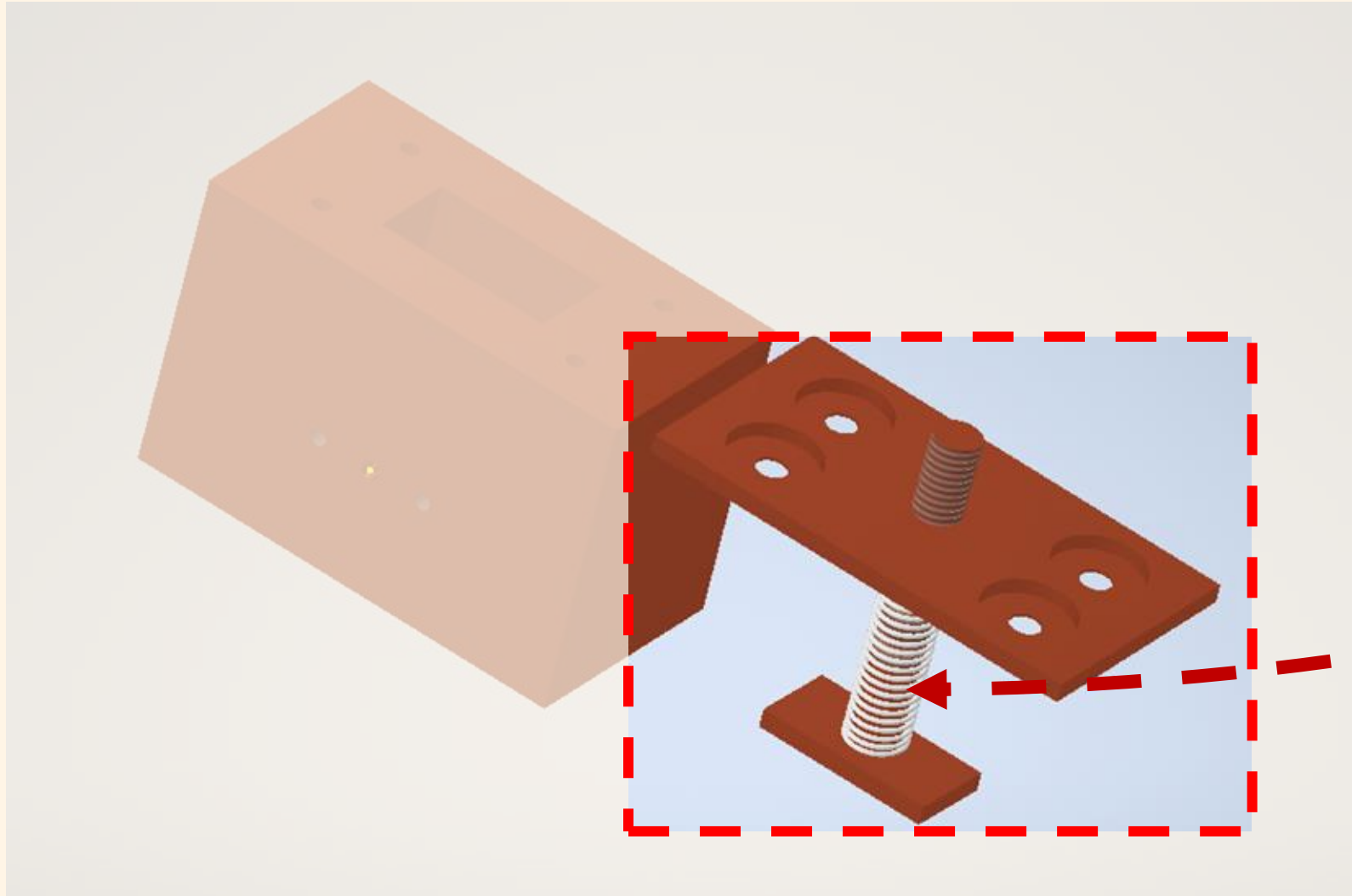


First Roadblock





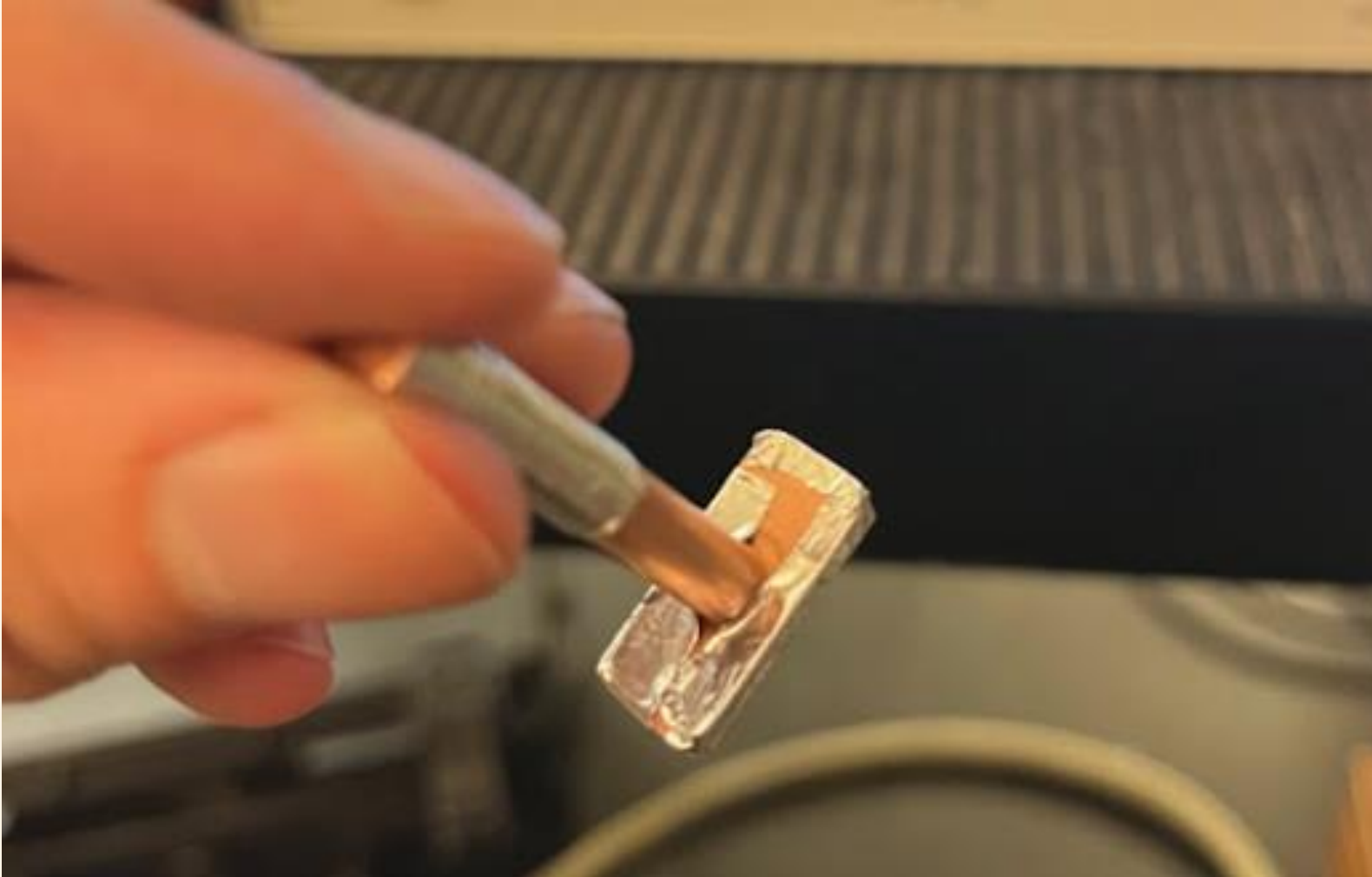
First Roadblock



Spring Did Not Function As Expected...

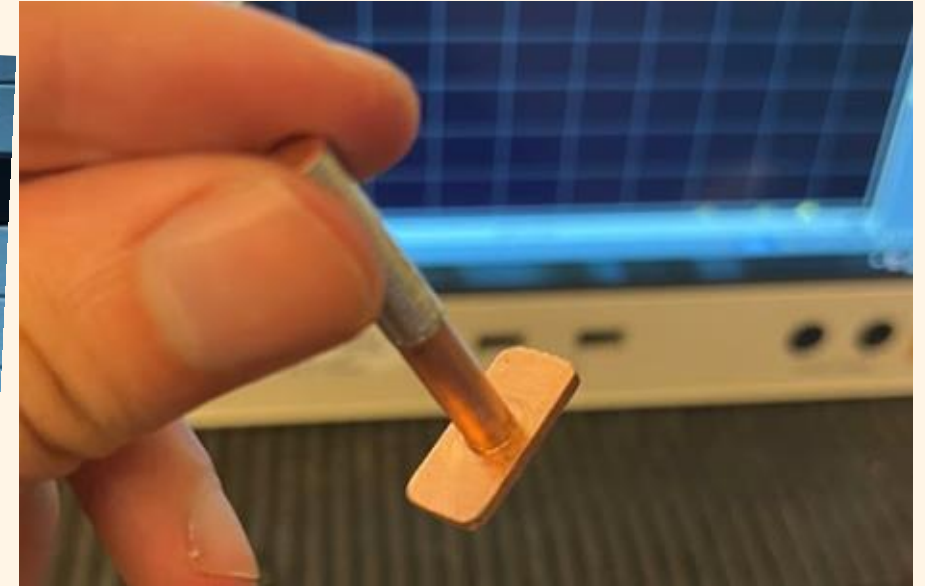
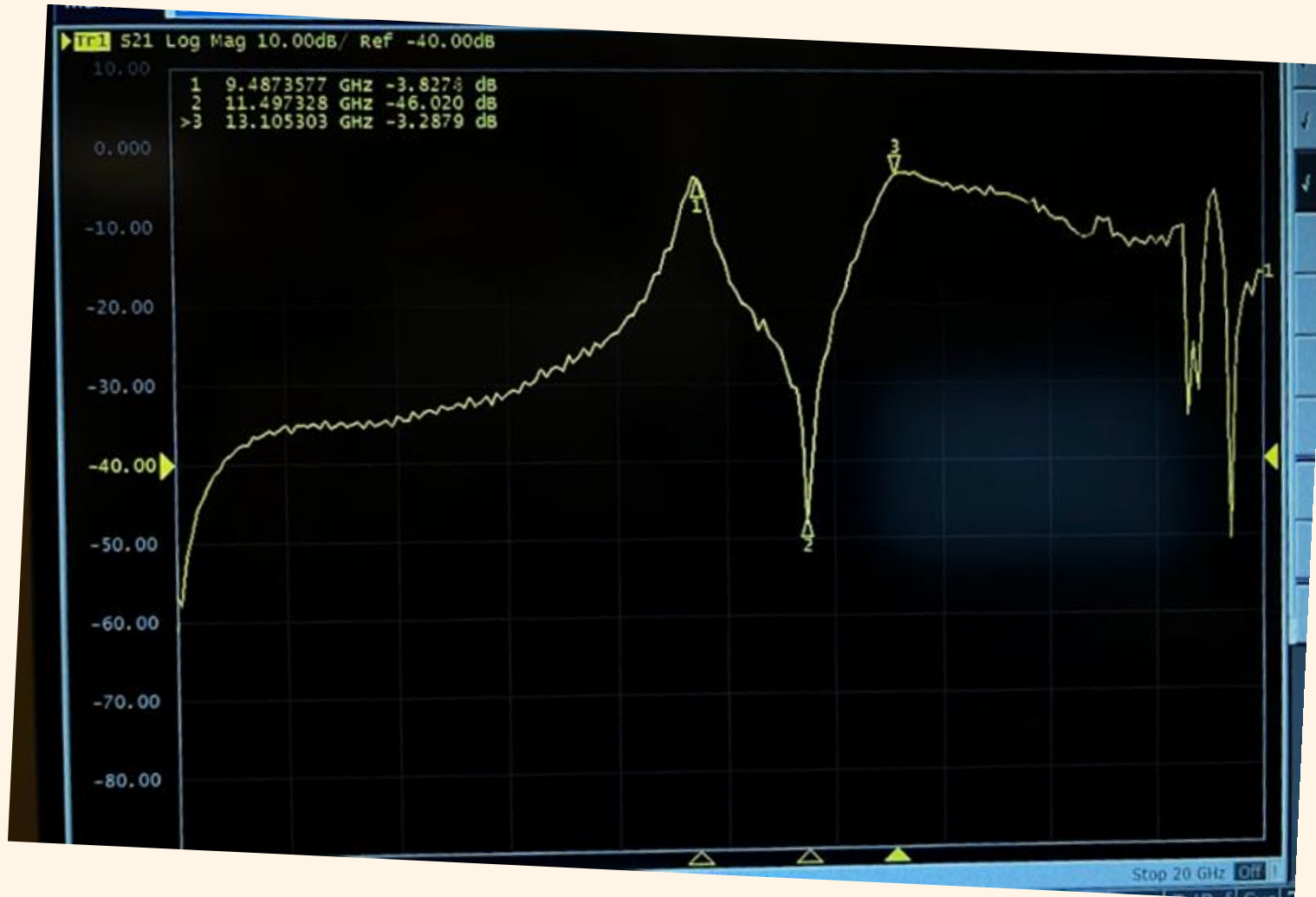


The Fix: Our Friend Friction



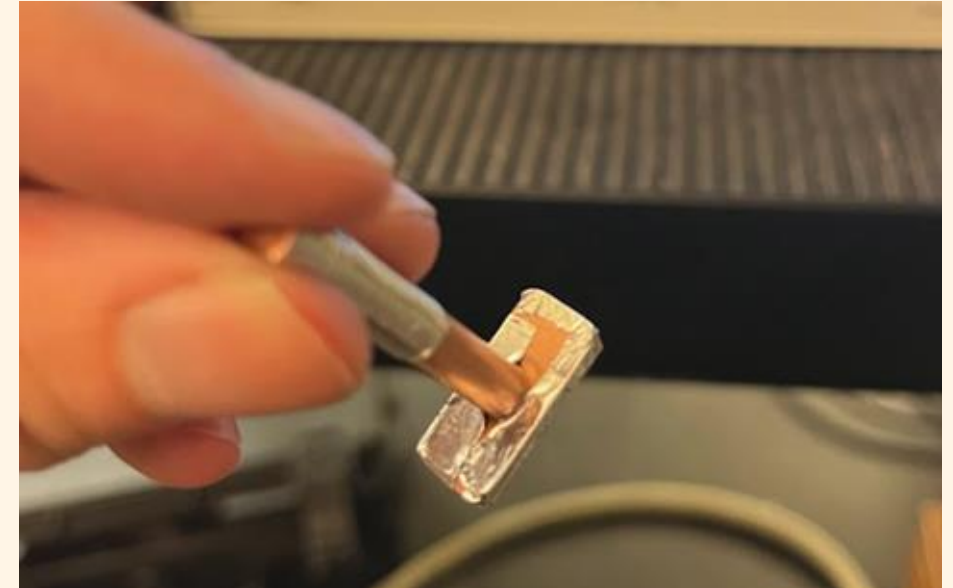
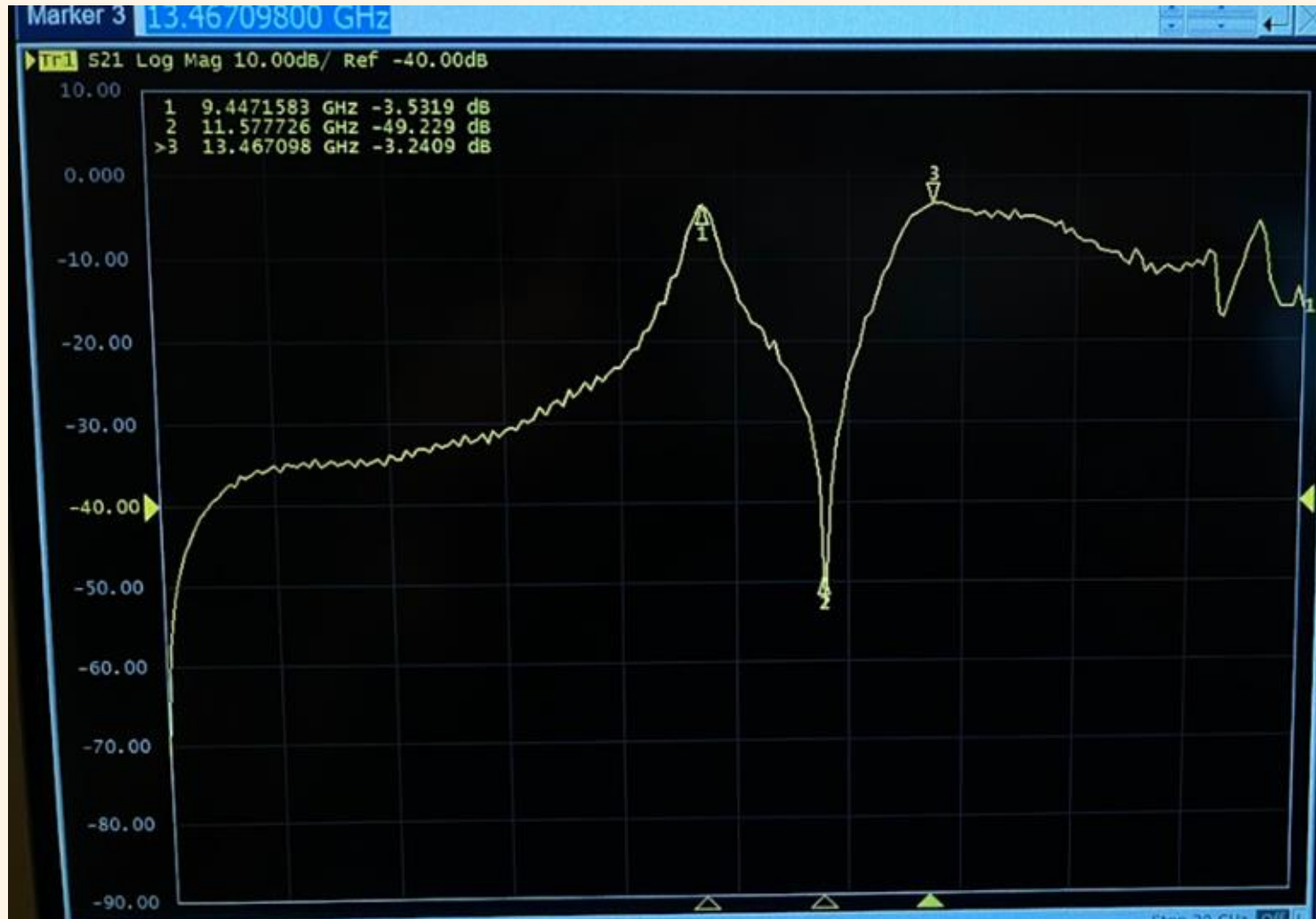


Do We Even Need a Cover?



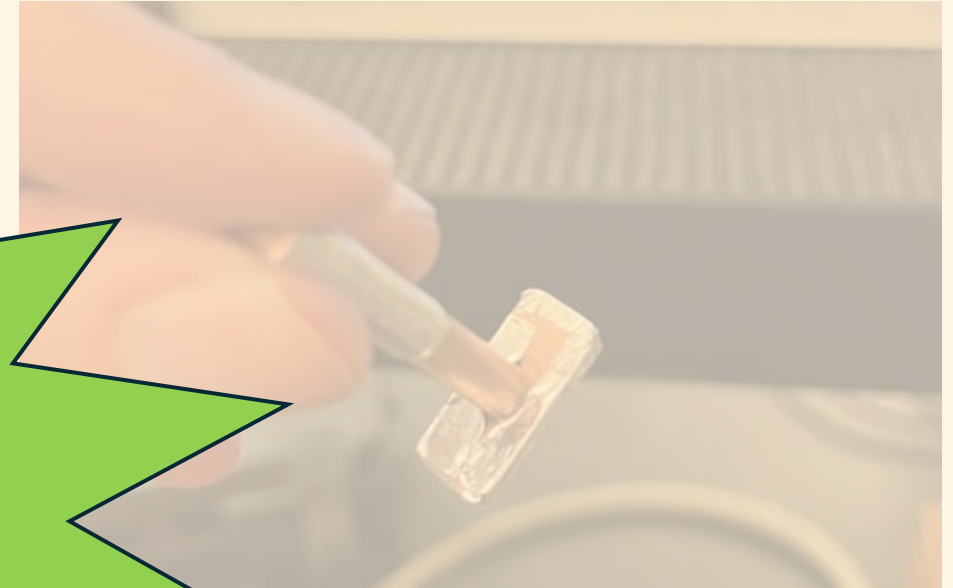


Do We Even Need a Cover?



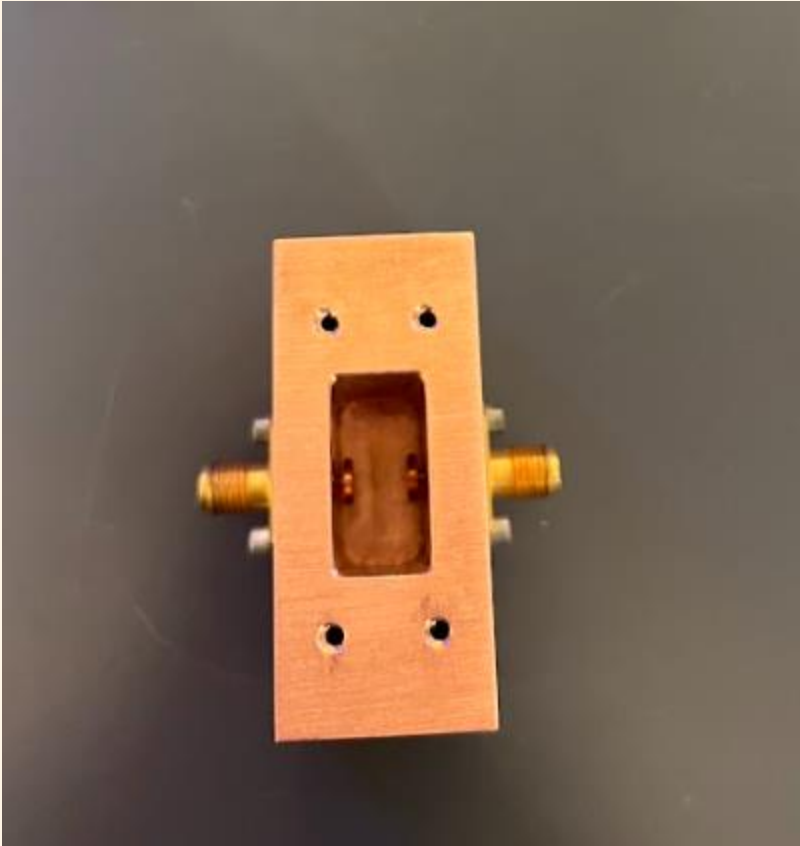


Do We Even Need a Cover?

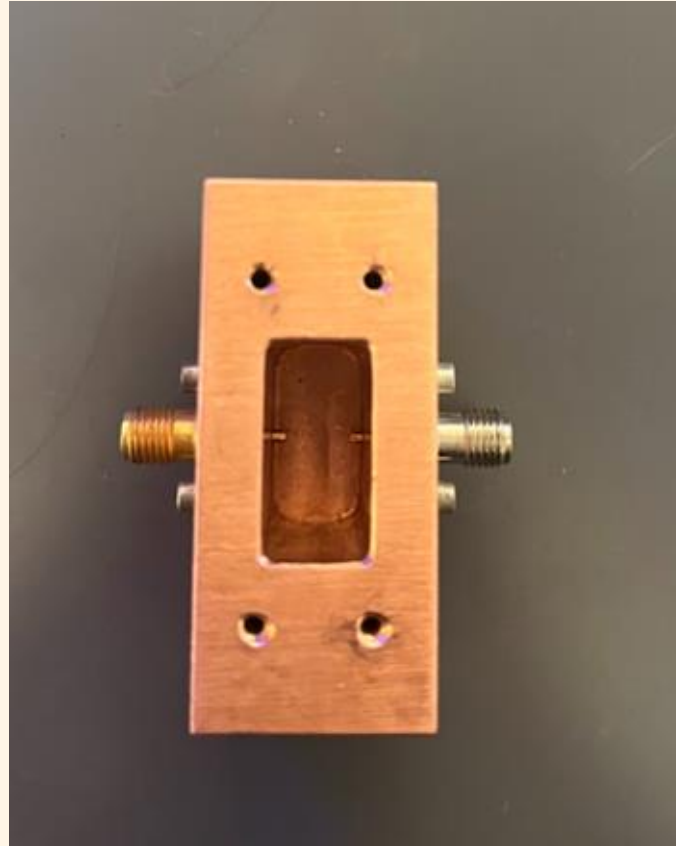




Let's Look at Different Types of Pins



The Hammerhead



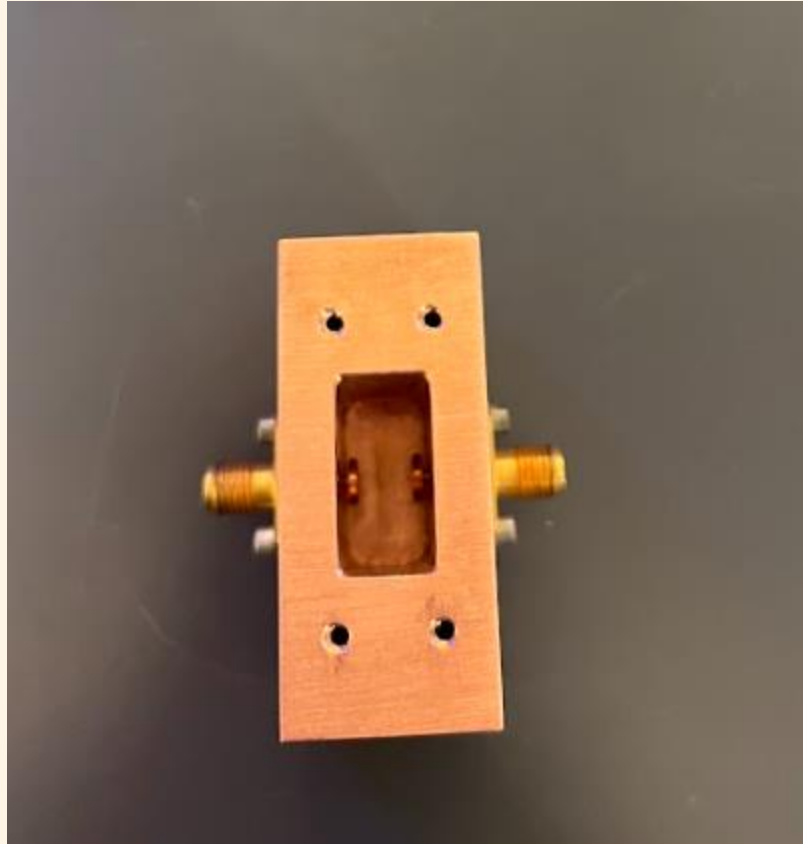
**Short SMA
(9.75mm)**



**Long SMA
(12.56mm)**



Let's Look at Different Types of Pins

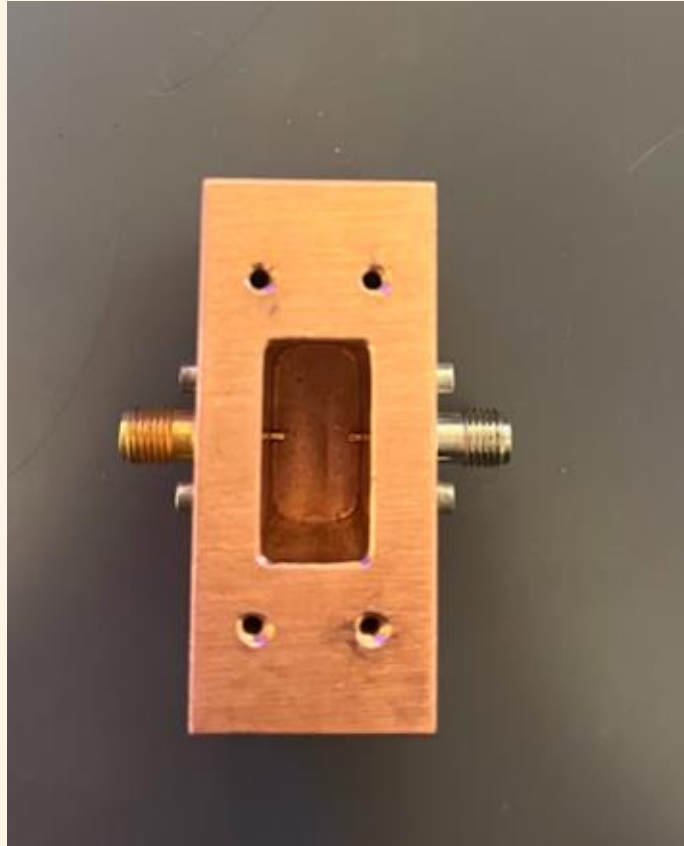


The Hammerhead

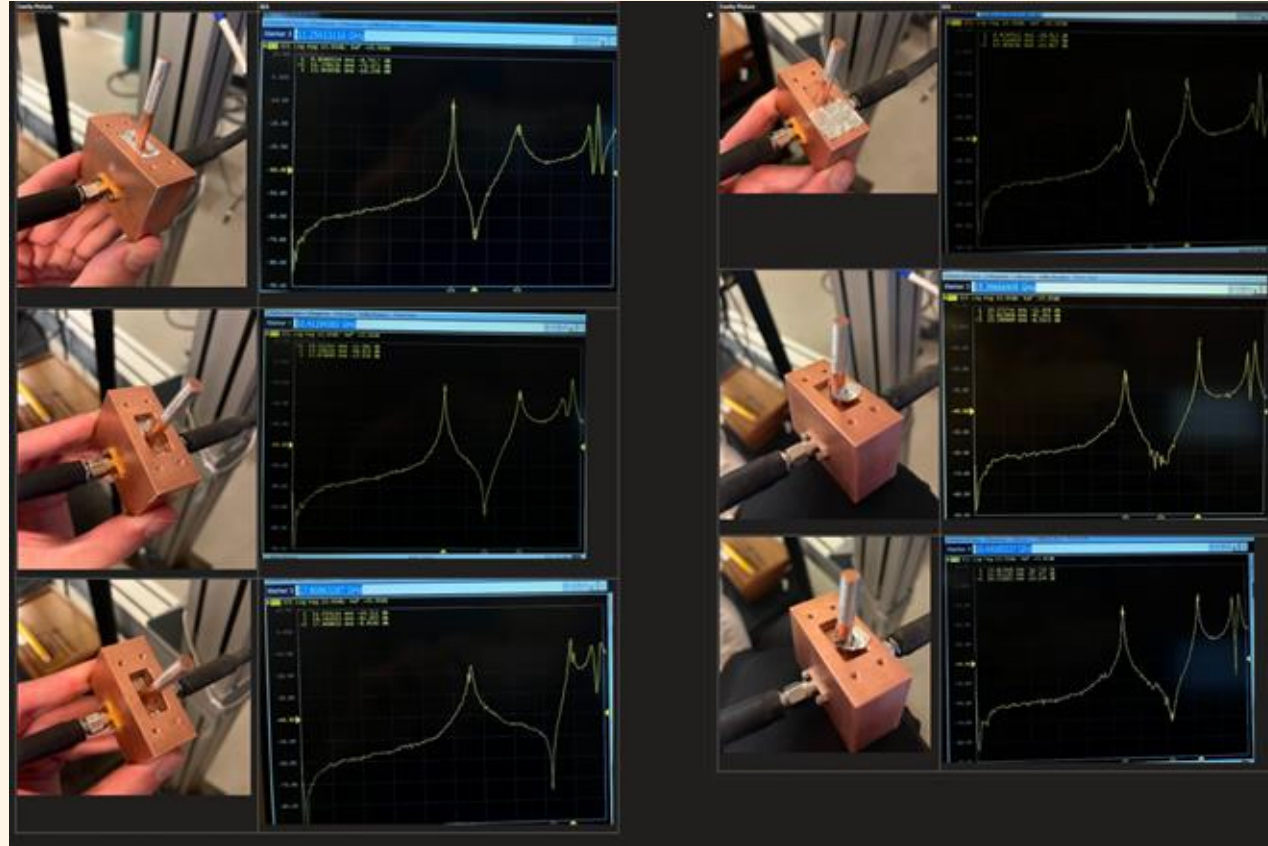




Let's Look at Different Types of Pins

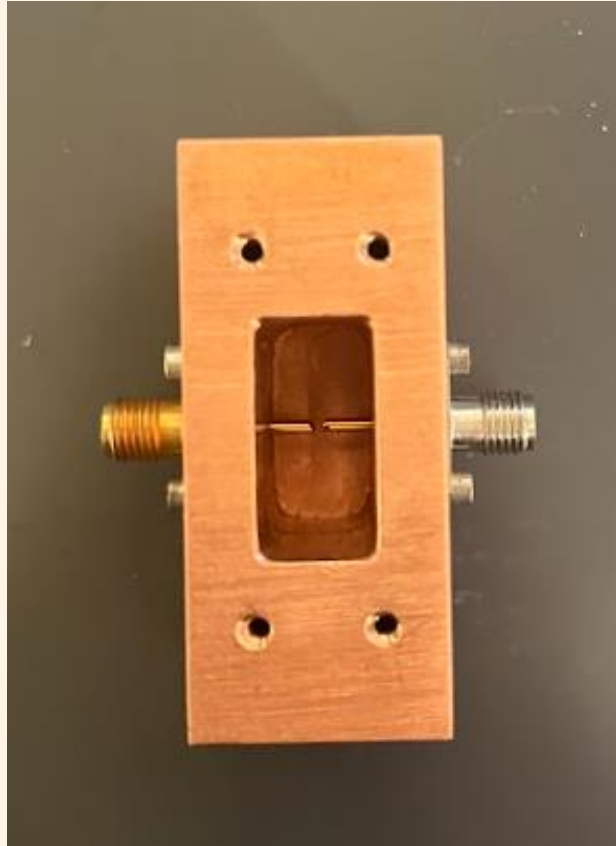


**Short SMA
(9.75mm)**

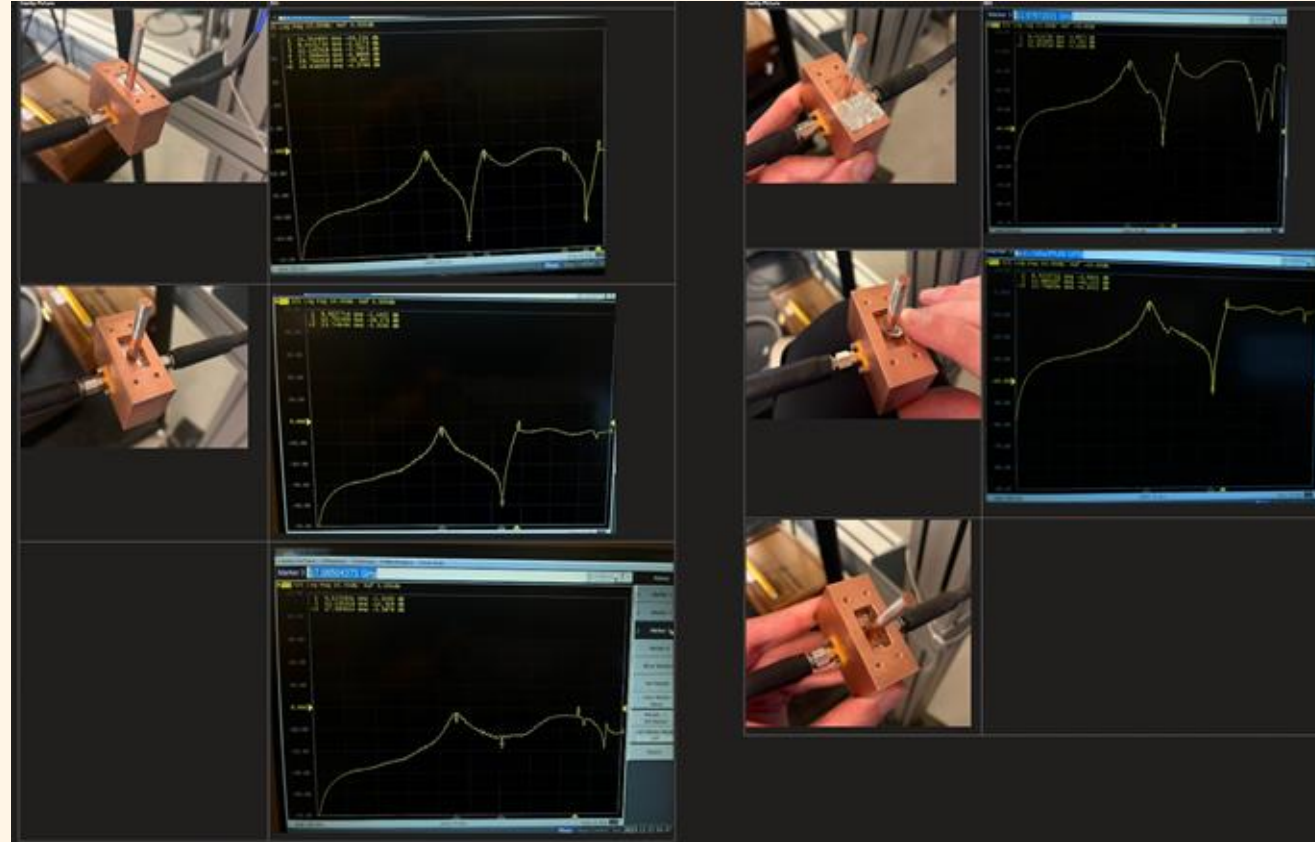




Let's Look at Different Types of Pins

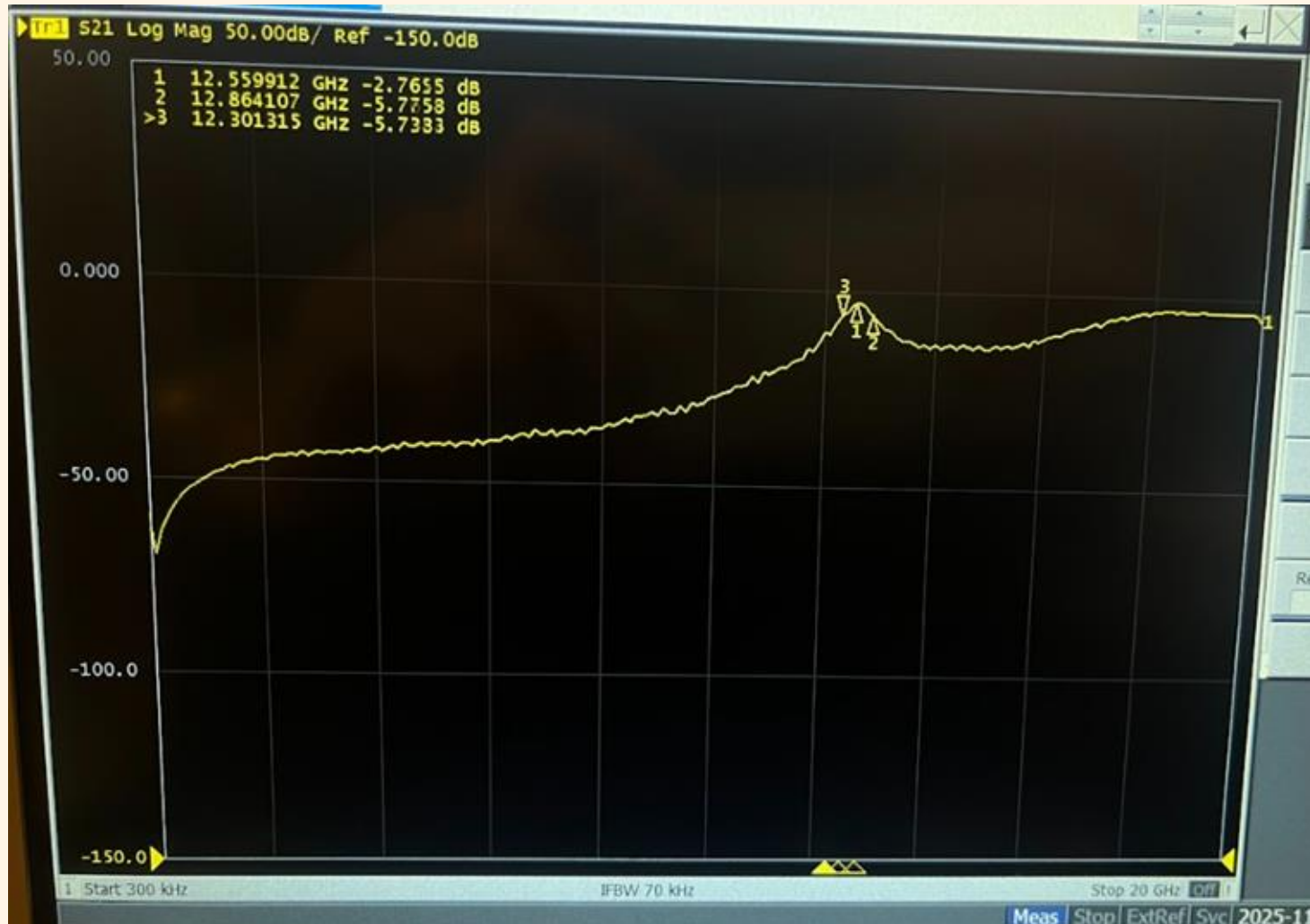


Long SMA
(12.56mm)





Meeting Requirement of PReSPA





Meeting Requirement of PReSPA

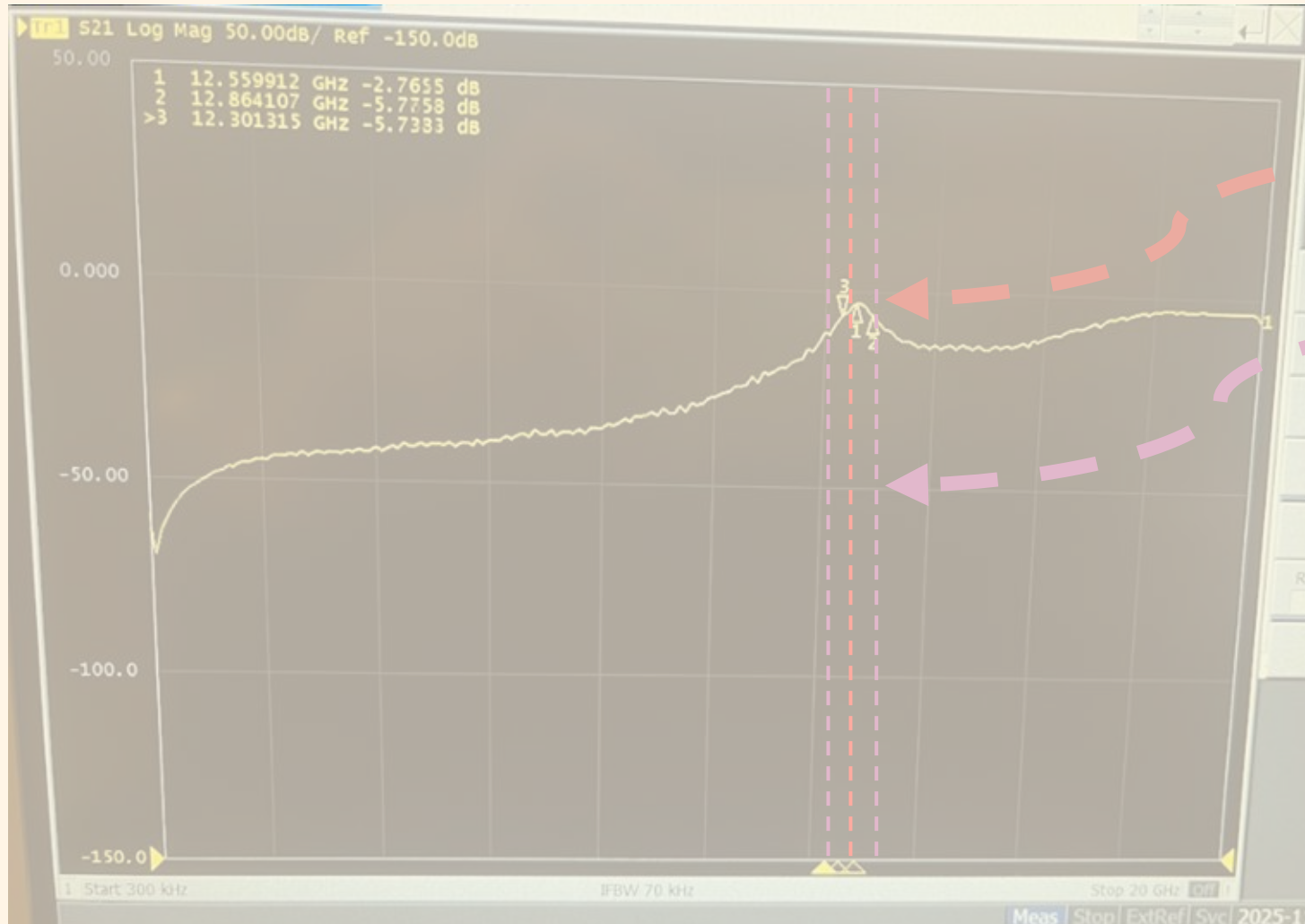


Resonance: 12.55 Ghz

$\kappa = 560$ MHz

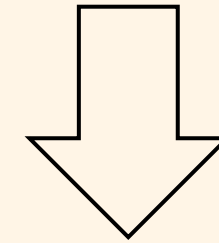


Meeting Requirement of PReSPA



Resonance: 12.55 Ghz

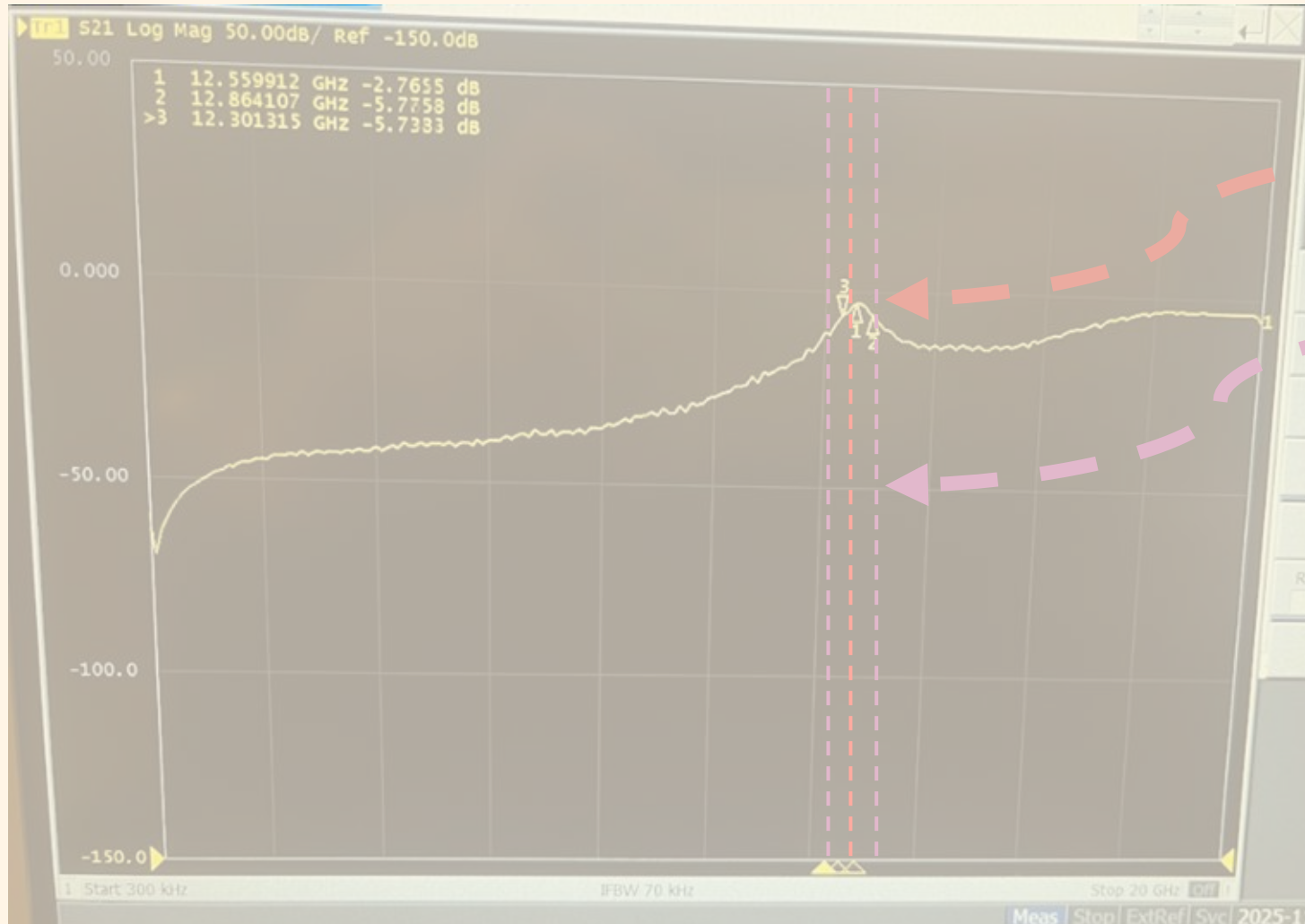
$\kappa = 560$ MHz



Can We Improve κ ?

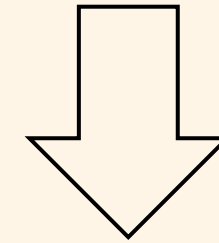


Meeting Requirement of PReSPA



Resonance: 12.55 Ghz

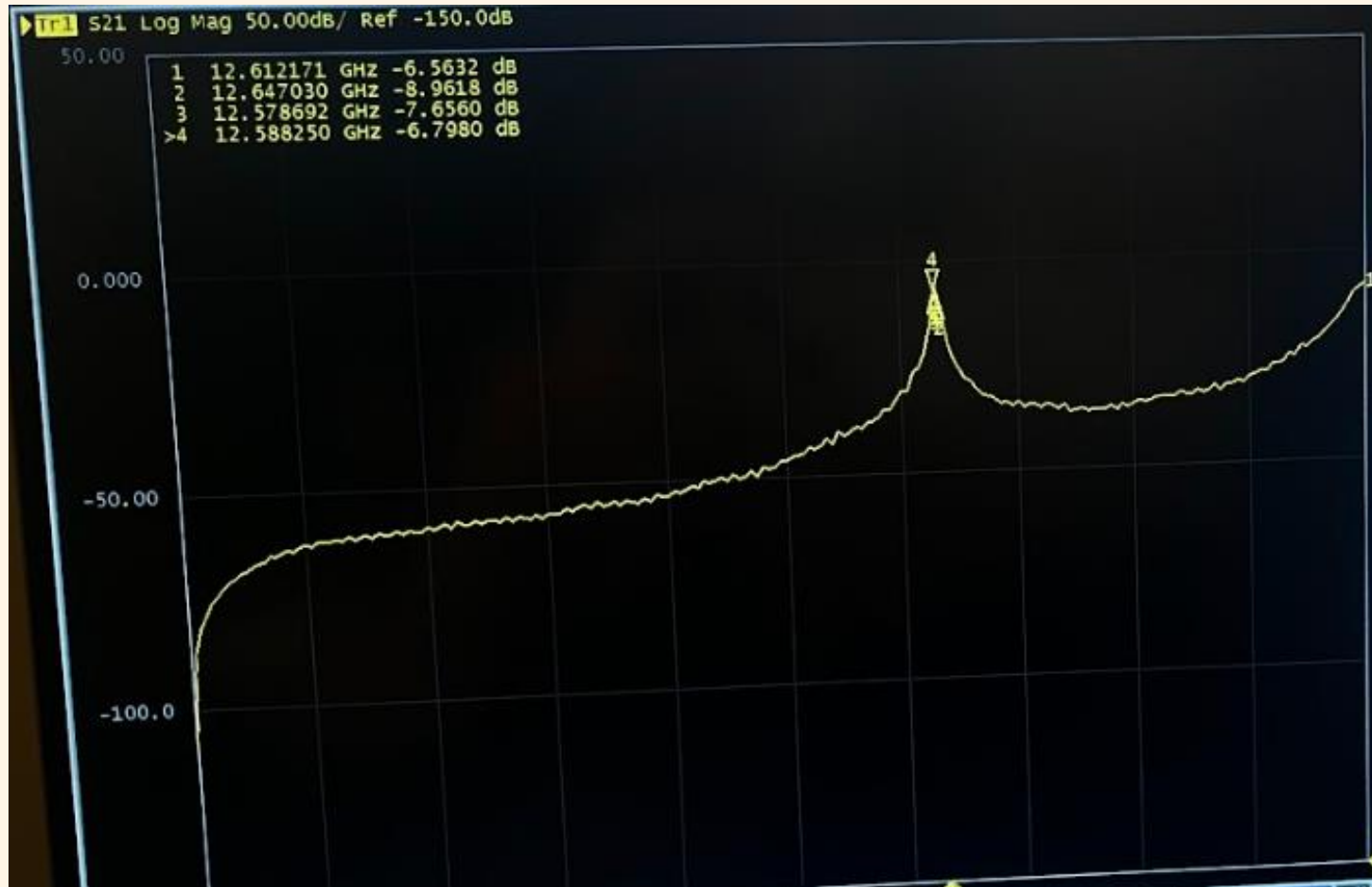
$\kappa = 560$ MHz



Change Coupling = Change Pin

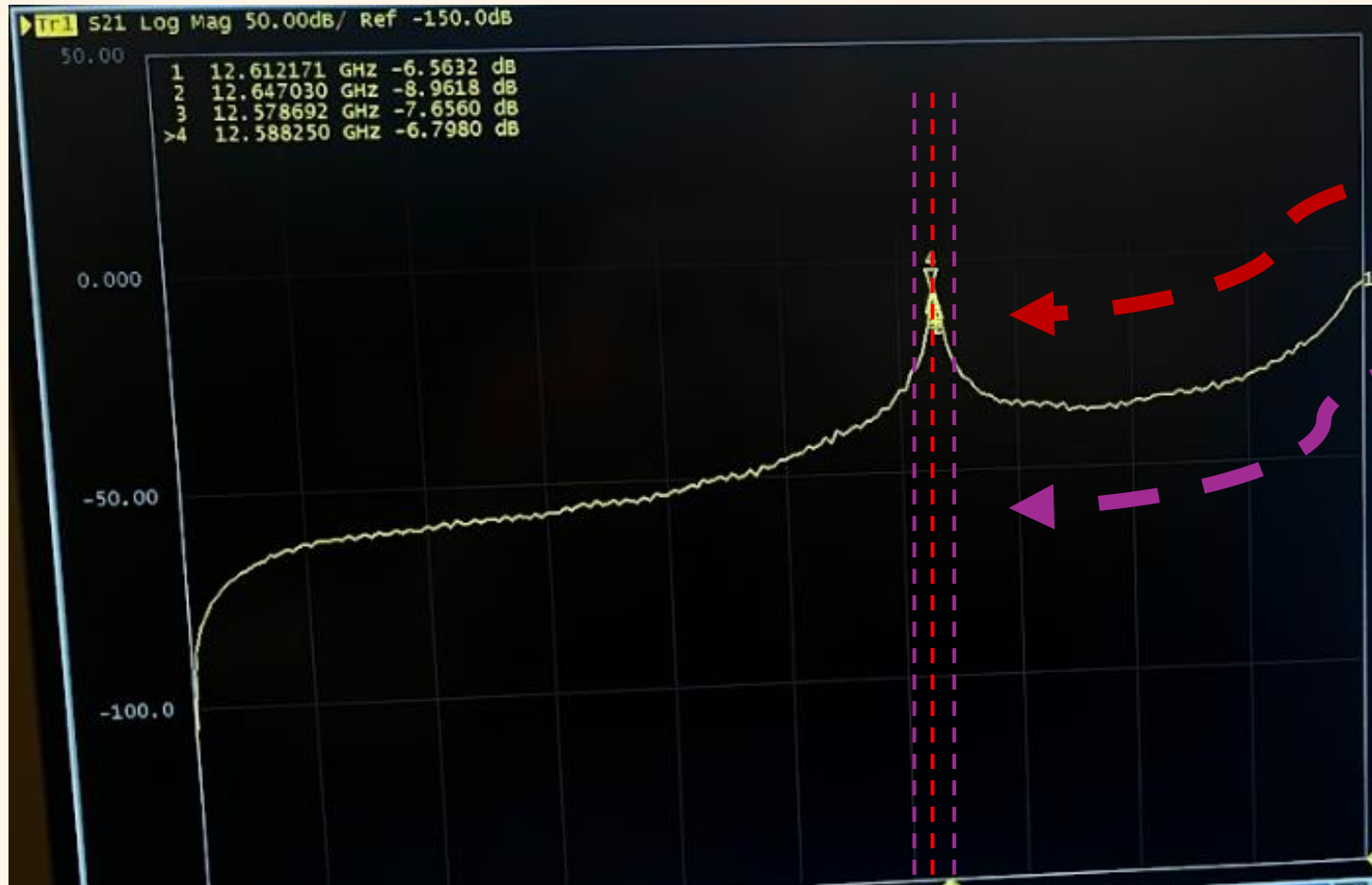


Meeting Requirement of PReSPA





Meeting Requirement of PReSPA

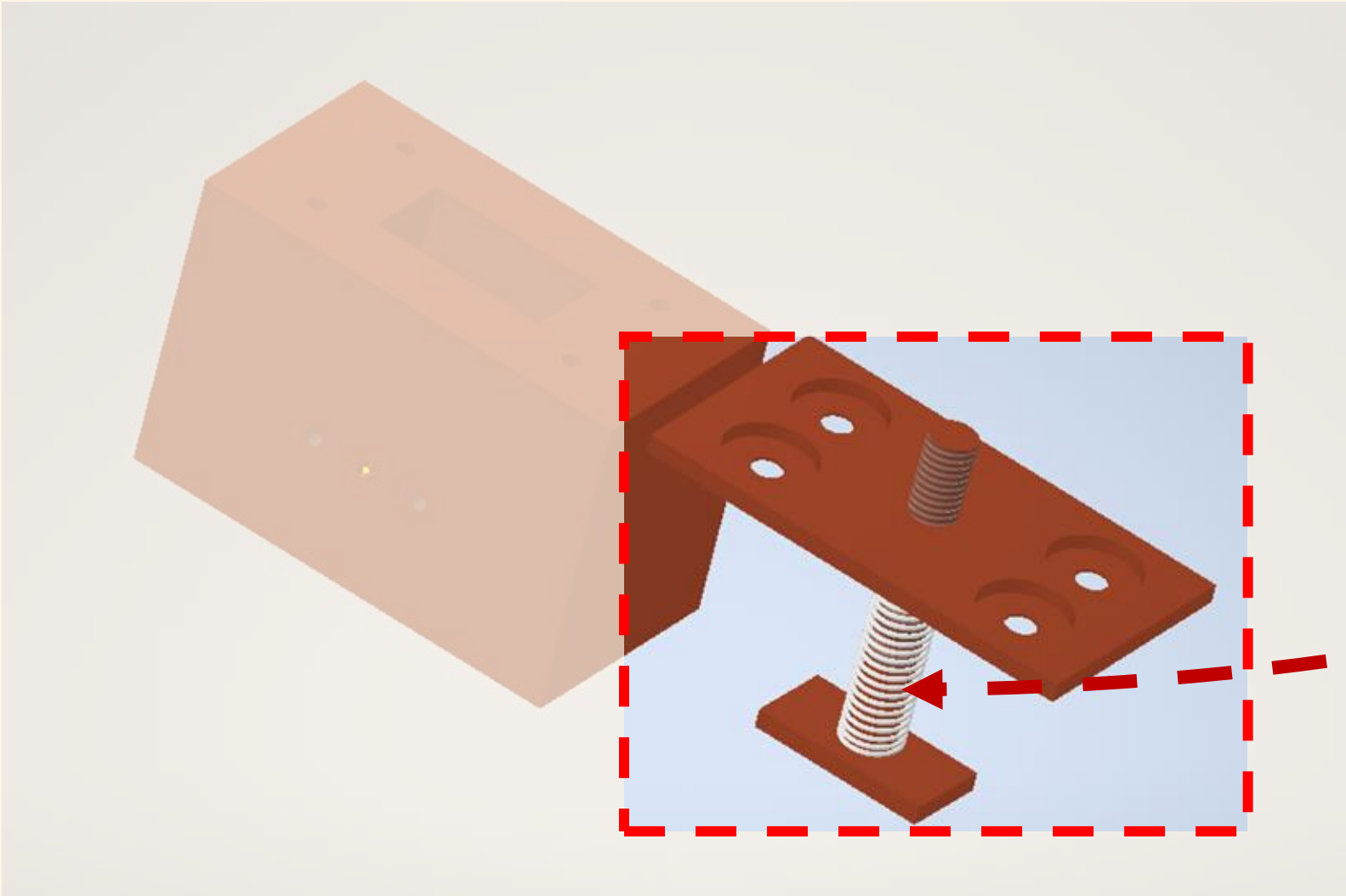


Resonance: 12.61 Ghz

$\kappa = 69$ MHz



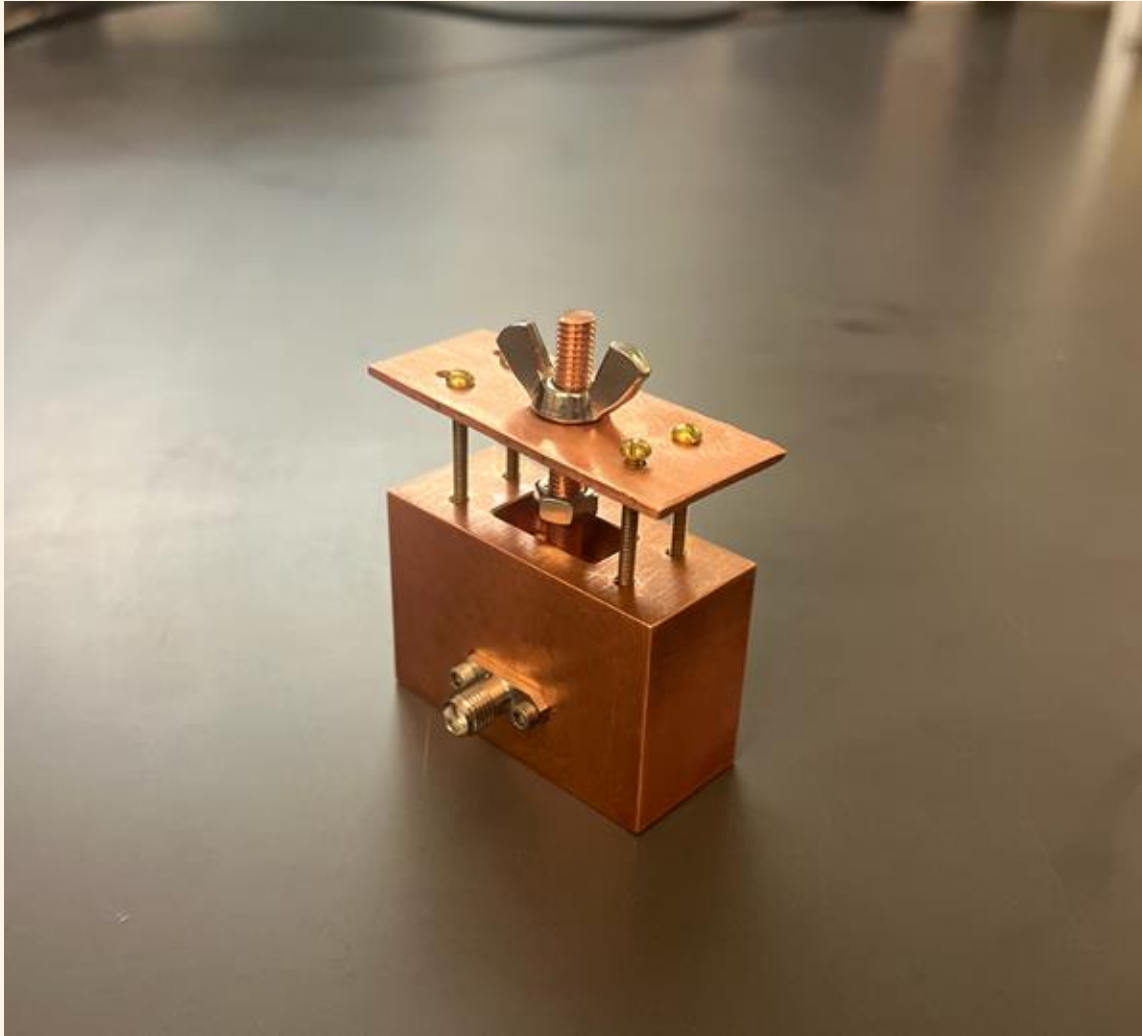
We Still Need to Get Around This...



Spring Did Not Function As Expected...



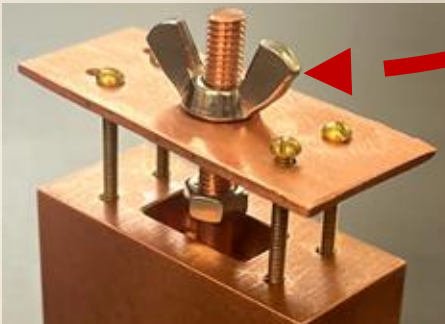
Hello Engineering.





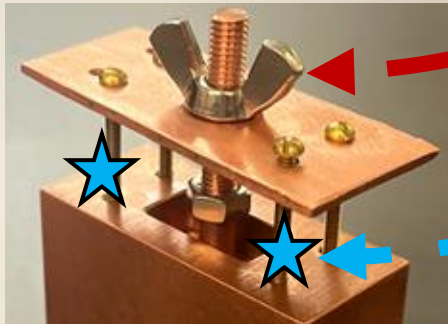
Hello Engineering.

Fine-tune the Cavity Frequency





Hello Engineering.

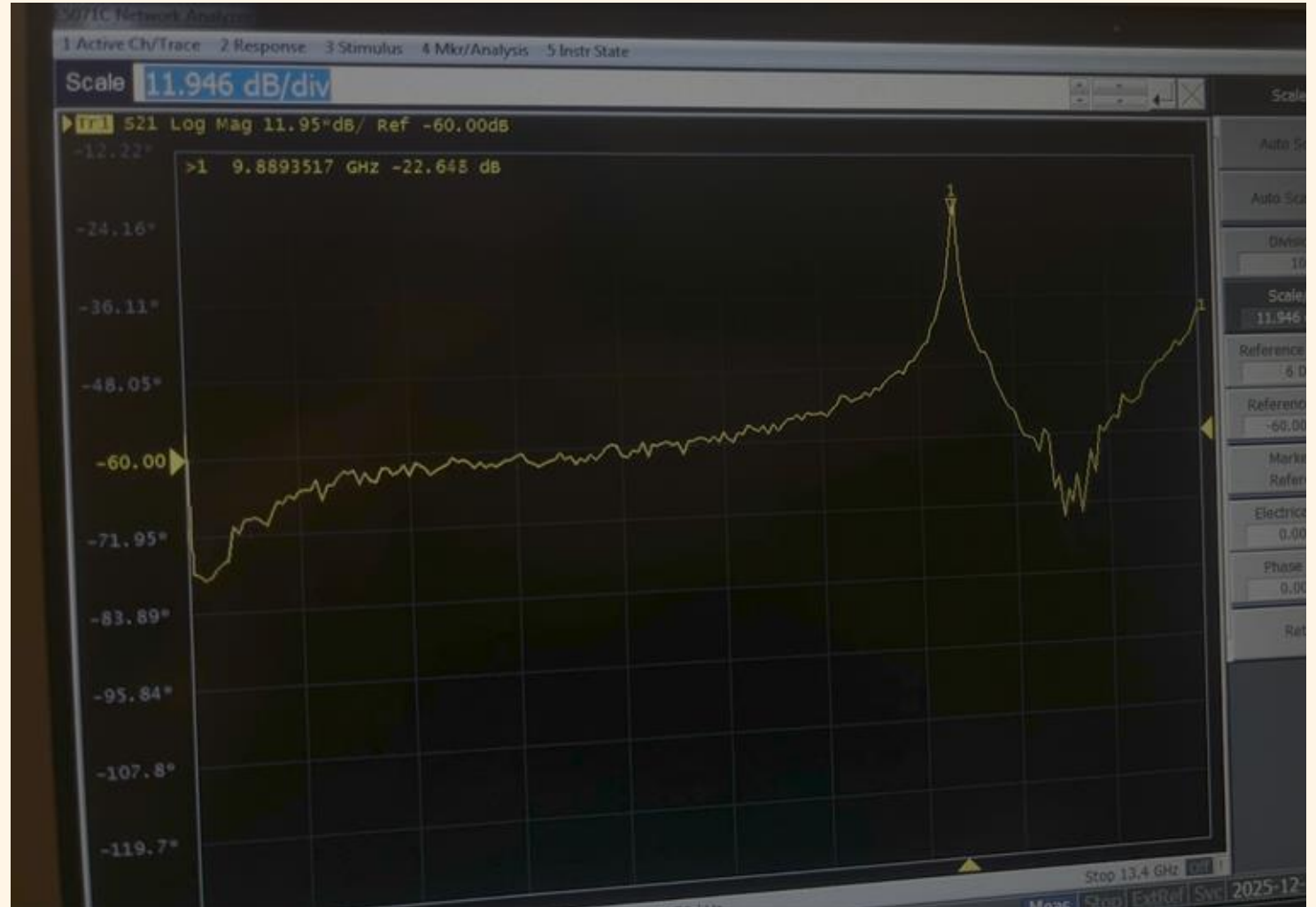
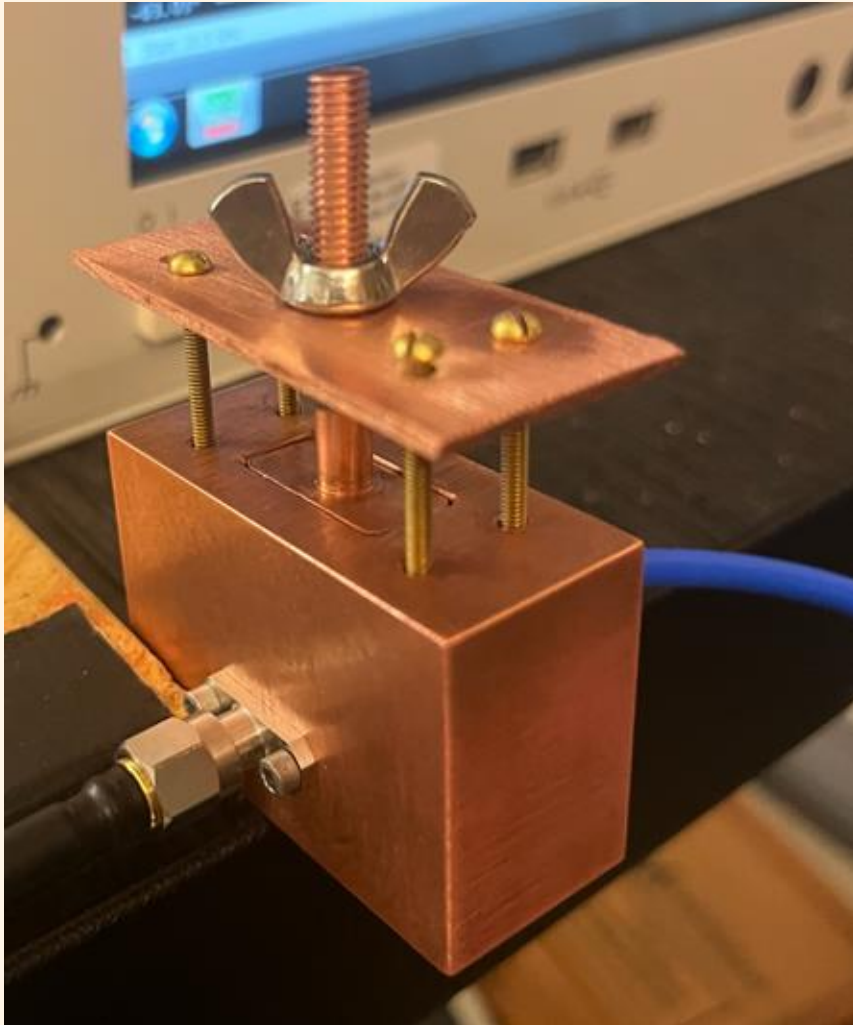


Fine-tune the Cavity Frequency

Posts Provide Structural Integrity

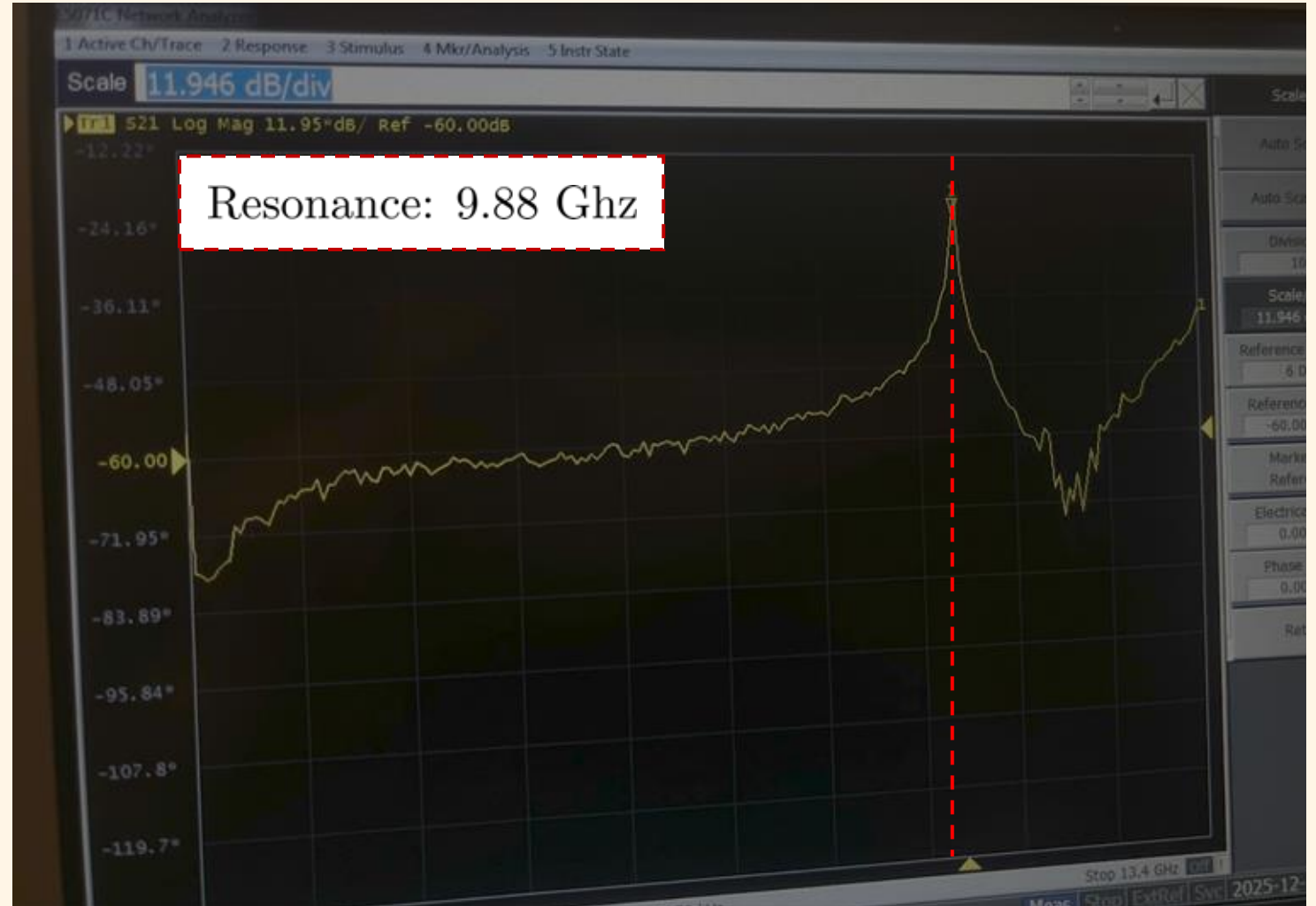
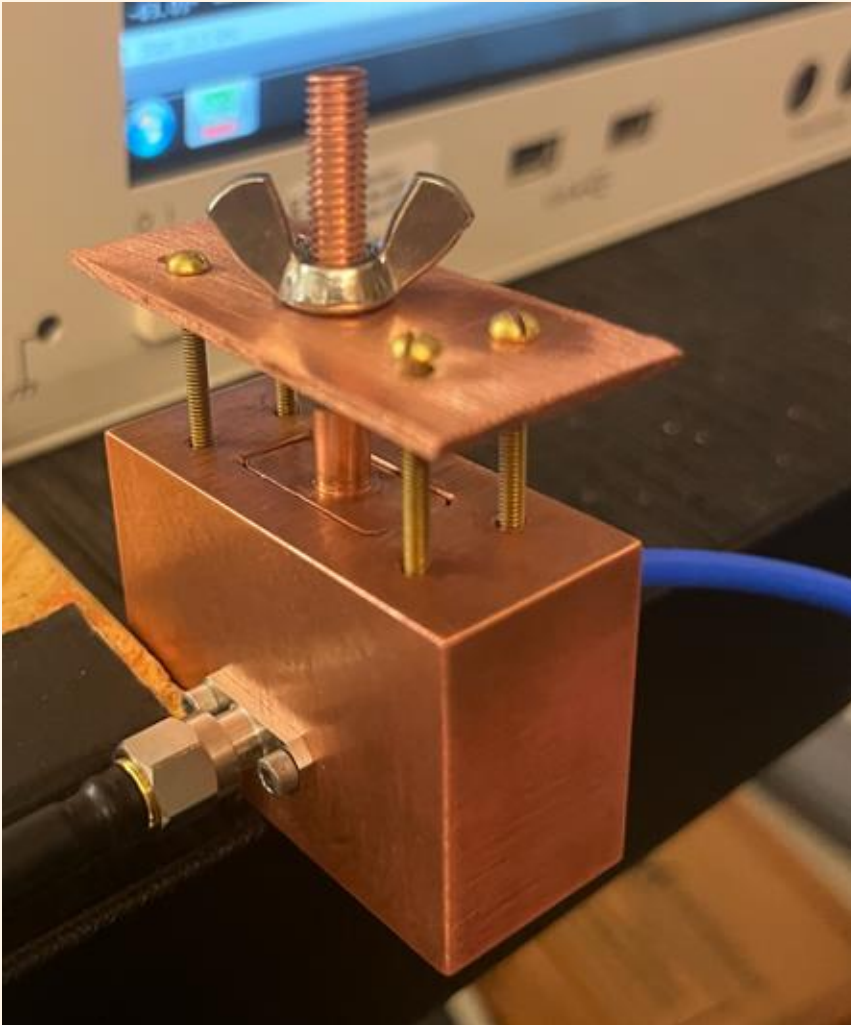


Seeing This Design in Action.



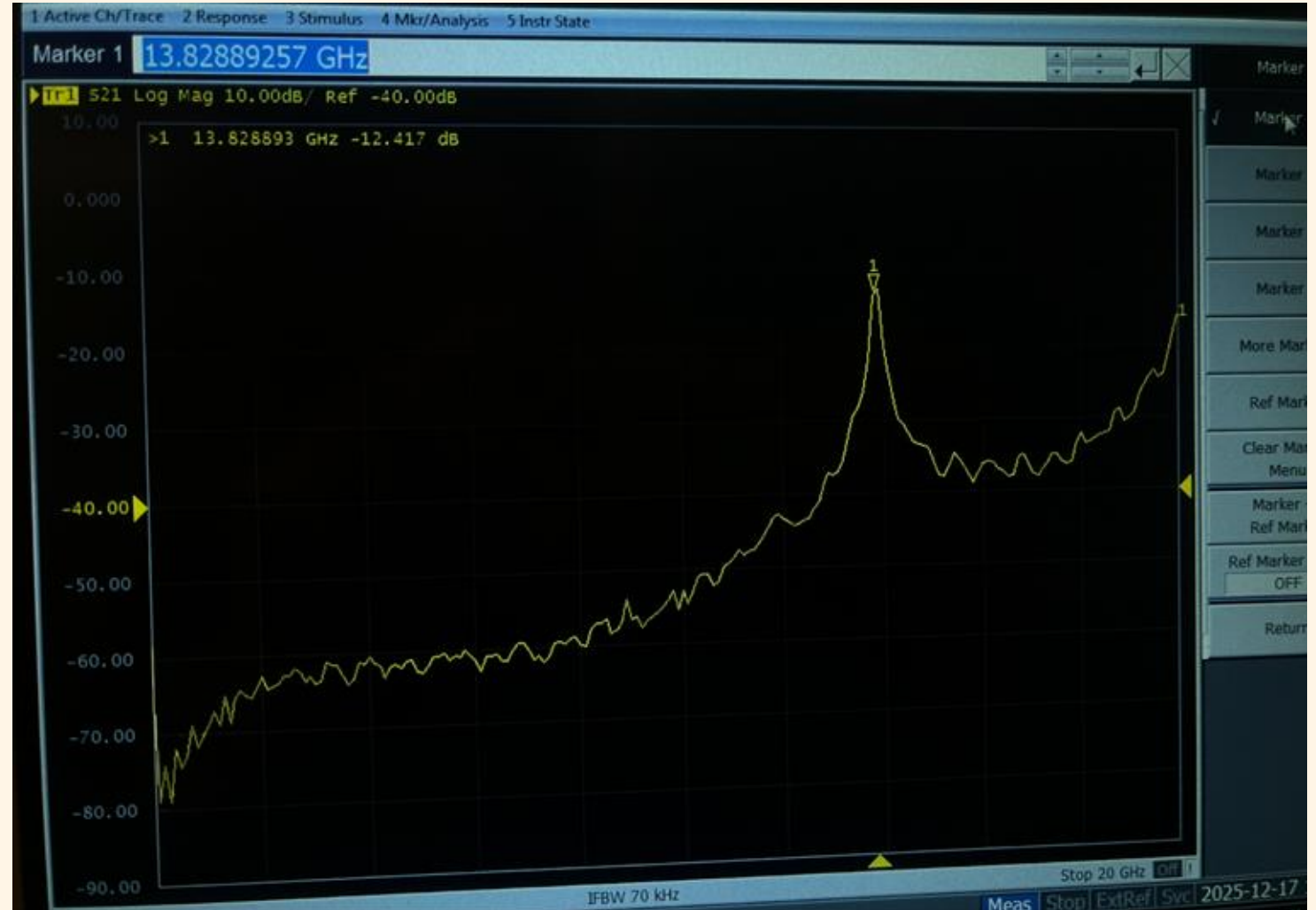


Seeing This Design in Action.



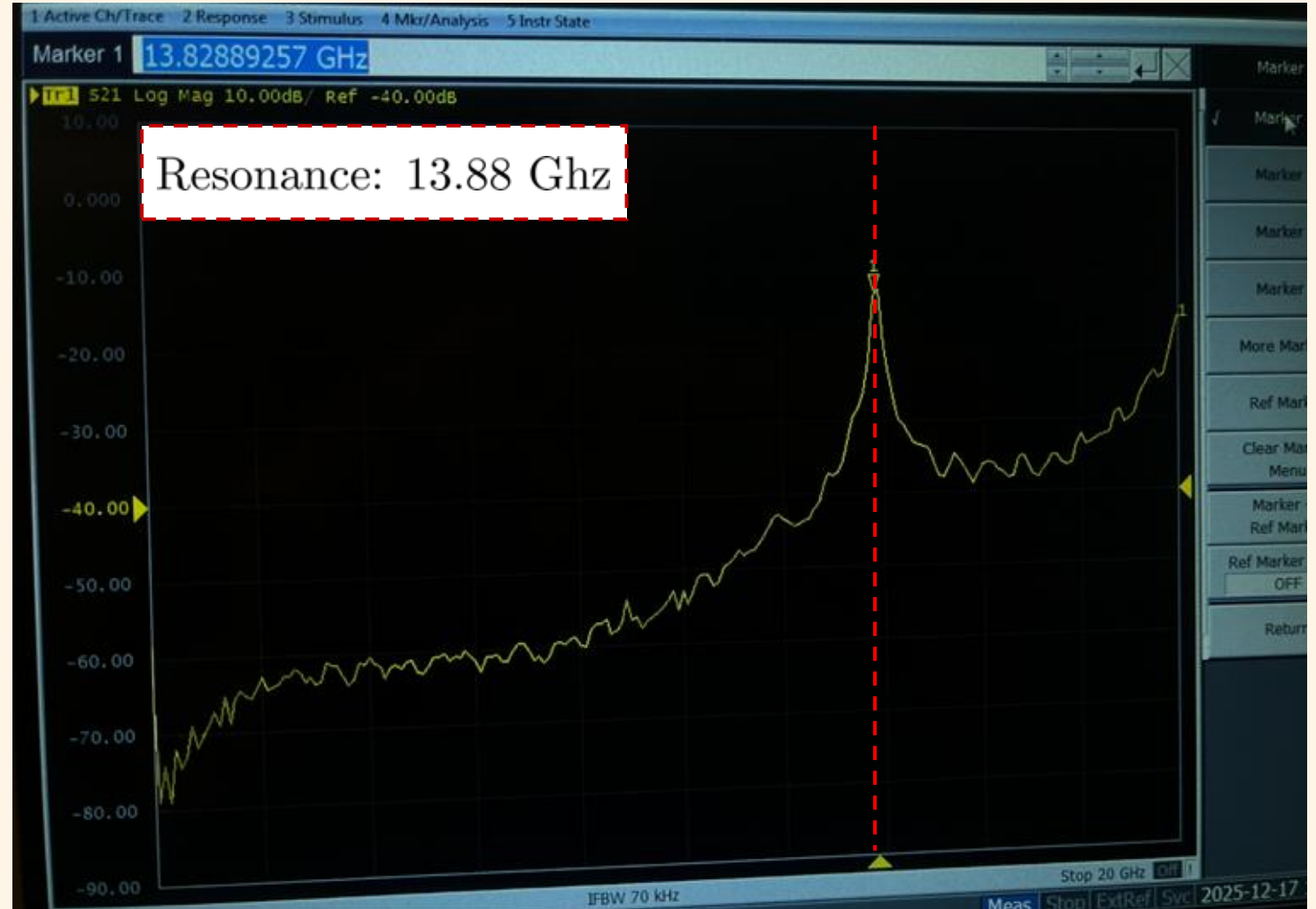


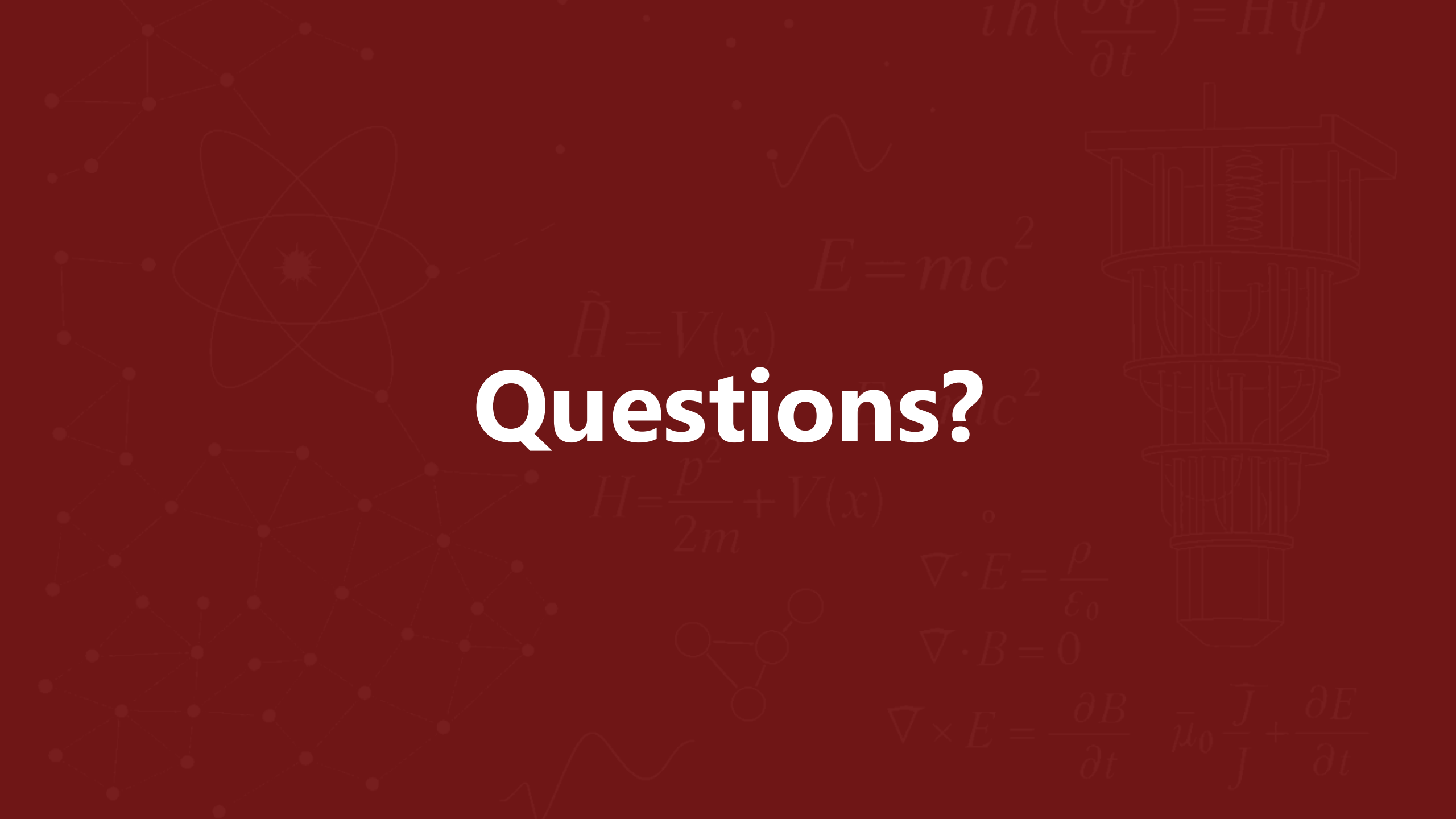
Seeing This Design in Action.





Seeing This Design in Action.



The background is a dark red field filled with faint, light-colored line art and mathematical symbols. On the left, there's a complex network of dots and lines resembling a molecular or crystal structure. In the upper left, a Bohr-style atomic model with a central nucleus and three elliptical orbits is visible. To the right, a vertical stack of four cylindrical capacitors with vertical plates is shown. Scattered throughout are various mathematical expressions: $i\hbar \left(\frac{\partial \Psi}{\partial t} \right) = H\Psi$ at the top right, $E=mc^2$ in the upper middle, $\hat{H}=V(x)$ below it, $H=\frac{p^2}{2m}+V(x)$ in the lower middle, $\nabla \cdot E = \frac{\rho}{\epsilon_0}$ and $\nabla \cdot B = 0$ on the right, and $\nabla \times E = -\frac{\partial B}{\partial t}$ at the bottom right. A small sine wave is also present near the bottom center.

Questions?